

The *fabB* gene, encoding β -ketoacyl-ACP synthase I, possesses a nucleotide sequence in its -40 region that matches perfectly with the highest affinity FadR operator sites. Accordingly, *fabB* is also positively regulated by FadR; however, the changes in *fabB* mRNA levels due to FadR regulation are much lower than for *fabA*.

10.4. Regulation of phospholipid headgroup composition

Within a given strain of *E. coli*, the phospholipid ratio (PE:PG:CL) is maintained under a variety of growth rates and conditions. The exception to this is the increased conversion of PG to CL during the stationary phase (Section 5.4). Thus, a mechanism must exist to maintain phospholipid homeostasis. Regulatory mutants resulting in the overexpression of PS synthase (*pssR*) and diacylglycerol kinase (*dgkR*) have been identified, suggesting the existence of *trans* acting factors that control the expression of these key enzymes, but their significance is unclear. The hypothetical 32-kDa PssR protein is similar to the LysR family of transcriptional regulators. Overexpression of PS synthase, PGP synthase or CL synthase in plasmid-based systems do not lead to dramatic changes in the membrane phospholipid composition. Thus, modulation of protein level is unlikely to have a role in the regulatory scheme.

So how is phospholipid homeostasis maintained? Control of the individual enzymes at the level of activity by feedback-regulation appears a more probable mechanism. Perturbations in the ratio of phospholipids were attempted experimentally by the activation of phosphoglycerol transferase I (*mdoB*). This enzyme, involved in MDO synthesis, catalyzes the transfer of glycerol phosphate from PG to the extracellular arbutin (4-hydroxyphenyl-*O*- β -D-glucoside). Treatment with arbutin (a MDO substrate analogue) causes a 7-fold increase in the rate of PG synthesis without a concomitant increase in PGP synthase proteins levels or significant changes in membrane phospholipid composition. Thus, PS synthase and PGP synthase are independently regulated by phospholipid composition. Similarly, purified CL synthase is strongly feedback inhibited by CL, and this inhibition is partially relieved by PE. Thus, the regulation of phospholipid content in *E. coli* appears to be an intrinsic property of the enzymes.

10.5. Coordinate regulation of fatty acid and phospholipid synthesis with macromolecular biosynthesis

Fatty acid biosynthesis is coordinately regulated with phospholipid synthesis since, in growing cultures of *E. coli*, there is no significant accumulation of any of the intermediates in fatty acid synthesis. Following inhibition of phospholipid biosynthesis, fatty acid biosynthesis carries on at about 10–20% of the uninhibited rate. Measurements of this rate were severely hampered by the requirement to first prevent β -oxidation of newly synthesized fatty acids, and second to use a strain in which acetate was solely channeled into fatty acid synthesis, so that incorporation of ^{14}C -label from the acetate could be measured.

Labeling the ACP moiety of the fatty acid intermediates by growth of a *panD* strain on medium containing tritiated β -alanine, a precursor of 4'-phosphopantetheine, also shows that long chain acyl-ACPs accumulate for a short period following the cessation

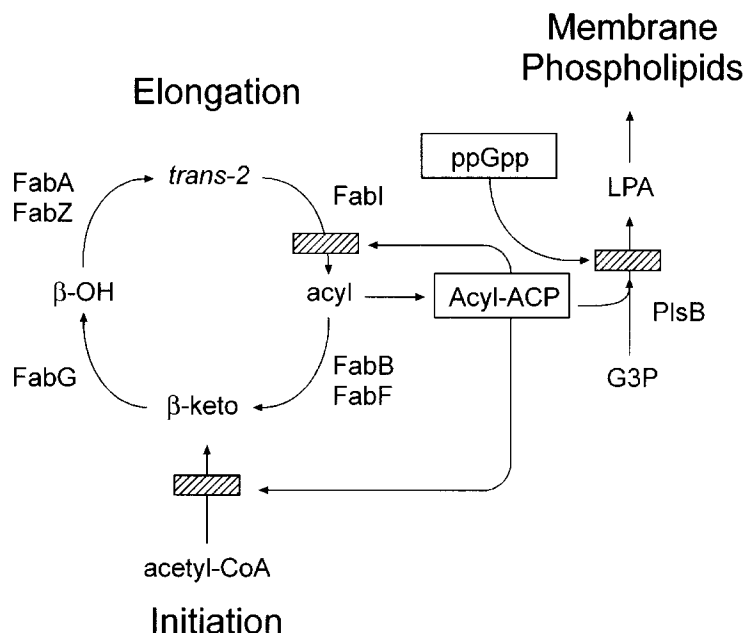


Fig. 10. Coordinate regulation of fatty acid and phospholipid metabolism. The pleiotropic regulator ppGpp regulates transfer of fatty acids to the membrane via inhibition of the PlsB acyltransferase step, coordinating phospholipid synthesis with macromolecular synthesis. Long chain acyl-ACPs accumulate for a period, then feedback inhibit their own synthesis at the point of initiation (inhibition of acetyl-CoA carboxylase and/or FabH) and elongation, by inhibition of FabI.

of phospholipid synthesis. This accumulation does not continue indefinitely, however, and reaches a plateau after about 20 min following inhibition of phospholipid synthesis. Thus, *de novo* fatty acid synthesis ceases, probably by a feedback inhibition mechanism involving long chain acyl-ACPs inhibiting early steps in the fatty acid biosynthesis pathway (Fig. 10). A significant finding in support of this idea is that overexpression of a thioesterase (which prevent the accumulation of acyl-ACP by cleavage of the thioester linkage and release of the acyl chain), allows continued fatty acid synthesis following cessation of phospholipid synthesis. This also further suggests that acyl-ACP and not free fatty acids mediate the inhibition. A reduction in total ACP is not responsible for the inhibition of fatty acid synthesis, since the free ACP pools of the glycerol-starved *plsB* mutants are not significantly depleted, and overproduction of ACP fails to relieve inhibition of fatty acid synthesis. A *fadD* mutant strain, which cannot produce acyl-CoA, overexpressing a thioesterase gave the same results as strains blocked elsewhere in β -oxidation or wild-type strains, thus ruling out a role for acyl-CoA.

As *E. coli* enter the stationary phase, levels of an unusual nucleotide, guanosine 5'-diphosphate-3'-diphosphate (ppGpp), rise [18]. Wild-type strains of *E. coli* undergo the so-called 'stringent response' following starvation for a required amino acid, an effect also mediated by increased intracellular ppGpp. Increased levels of ppGpp cause a strong inhibition of stable RNA synthesis, and inhibition of protein and

phospholipid synthesis. Mutant strains (*relA*) do not undergo the stringent response, due to the lack of ppGpp synthase I, a ribosomal protein that produces ppGpp in response to uncharged tRNA. The interaction of ppGpp with RNA polymerase mediates the inhibitory effects on stable RNA synthesis. ppGpp directly inhibits phospholipid biosynthesis by inhibition of the glycerol phosphate acyltransferase (PlsB) and causes an accumulation of long chain acyl-ACPs, which in turn lead to the inhibition of fatty acid biosynthesis. Overexpression of the acyltransferase relieves the inhibition on both fatty acid and phospholipid synthesis.

The target enzyme of the acyl-ACP feedback inhibition is not clear, and indeed several enzymes are still under consideration (Fig. 10). An obvious target is acetyl-CoA carboxylase, the first enzyme of the pathway. Inhibition of malonyl-CoA production would prevent fatty acid synthesis and elongation. Cronan has shown that acetyl-CoA carboxylase appears to be a rate-limiting step for fatty acid synthesis and that the enzyme is inhibited by acyl-ACP in vitro. However, β -ketoacyl-ACP synthases I and II (FabB and FabF) are potential regulators due to their ability to degrade malonyl-ACP to acetyl-ACP, an activity stimulated by the presence of long chain acyl-ACP, and hence attenuate cycles of fatty acid elongation. FabH catalyzes the first step in the pathway, and inhibition of this enzyme would halt initiation of new acyl chains, but would allow the elongation of existing fatty acid intermediates. Inhibition of FabH by physiologically relevant concentrations of long chain acyl-ACPs has been demonstrated in vitro. Finally, enoyl-ACP reductase (FabI) is a potential target since the activity of this enzyme is a determining factor in completing rounds of fatty acid elongation and acyl-ACP could act as product inhibitors. Accumulation of the precursors of the enoyl-ACP step can be seen in vivo following inhibition of the acyltransferase. It remains possible that multiple of steps may be targeted to different degrees. The identification of the inhibited enzyme(s) and their relative contribution to the regulation will require an in vitro system including acetyl-CoA carboxylase that accurately reflects in vivo metabolism, and the isolation of mutants refractory to inhibition.

The inhibition of fatty acid biosynthesis triggers the stringent response. The accumulation of ppGpp is in this case dependent on the activity of the *spoT* gene product (ppGpp synthase II). How this regulatory system operates is unknown and it will be important to determine whether intermediates, such as acyl-ACP or malonyl-CoA, are the intracellular metabolites that mediate ppGpp synthase II activity.

11. Lipid metabolism in other bacteria

11.1. Analysis of lipid metabolism by genomic inference

The availability of genomic sequences for a variety of bacteria [8] allows for the rapid assessment of the lipid metabolic pathways present. Open reading frames encoding species specific isoforms of known genes can be amplified by the polymerase chain reaction, the protein expressed in *E. coli*, and the properties of the enzyme compared to the *E. coli* or other known proteins. This approach has been used extensively in Rock's laboratory for the genes of fatty acid synthesis, with isoforms from pathogenic

bacteria being isolated to assess their unique biochemical characteristics and for use in drug-screening programs. For example, the enoyl-ACP reductase isoforms from *E. coli*, *Staphylococcus aureus* and *B. subtilis* have been compared. Subtle differences in cofactor specificity could be detected in this way that are not obvious from the primary sequences.

Novel proteins have been identified using genomic information [19]. The *fabI*-encoded enzyme is the sole reductase present in *E. coli*, and it was assumed that this was probably true for all bacteria. However, analysis of the genome of *Streptococcus pneumoniae* did not reveal a *fabI* homologue, while genes for all of the other enzymes required for saturated fatty acid biosynthesis could be identified in a gene cluster. Thus, a novel enoyl-ACP reductase isoform was sought, and the *fabK*-encoded enoyl-ACP reductase II identified. The *fabK*-encoded protein possesses no homology to the FabI protein, and utilizes flavin mononucleotide as a cofactor, and is resistant to inhibition by the antibiotic triclosan (Section 12.3).

A third enoyl-ACP reductase isoform was discovered in *B. subtilis*, which also possesses *fabI*. A gene, *fabL*, was identified with low overall homology to *fabI*, although it contained lysine and tyrosine residues in the distinct motif of the catalytic residues in the enoyl-ACP reductase I. Heath and coworkers cloned the gene and demonstrated that the product possesses enoyl-ACP reductase activity in vitro. Insertional inactivation of either *fabI* or *fabL* in *B. subtilis* results in no obvious growth phenotype, but the double null strain could not be constructed. Thus, *fabL* encodes enoyl-ACP reductase III.

11.2. Branched chain fatty acid biosynthesis

Not all bacteria regulate membrane fluidity through the production of straight chain unsaturated fatty acids. In fact, Gram-positive bacteria often use branched chain fatty acids to modulate membrane fluidity [20]. The branch is a methyl group in the iso- or anteiso-position in the chain (i.e. the second or third carbon from the distal end of the chain). Based on the concept of acetyl-CoA as a primer for straight chain fatty acid synthesis in *E. coli*, it can be seen that the methyl group could be introduced using a 'branched-chain' primer. Indeed, isotope labeling and biochemical analysis identifies precursors of the branched chain amino acids, valine and isoleucine (isobutyryl-CoA or 2-methylvaleryl-CoA, respectively), as the primers for branched chain fatty acid synthesis. As in straight chain synthesis, the primer is condensed with malonyl-ACP by the action of FabH. The substrate specificities of the FabH enzyme(s) present in the bacteria determine the relative amounts of the respective fatty acids produced. *B. subtilis* contains a high proportion of branched chain fatty acids in its membranes, and has two FabH enzymes, each of which prefer the branched chain substrates over acetyl-CoA. *E. coli* FabH cannot use branched chain primers. Why *B. subtilis* possesses two FabH enzymes, with only minor differences in substrate specificity, is not understood.

11.3. Other ways to make unsaturated fatty acids

The *fabA* gene encodes the dehydratase/isomerase specifically required for the production of unsaturated fatty acids in Gram-negative bacteria. Gram-positive bacteria do not

contain an identifiable *fabA* homologue in their genome, but do possess unsaturated fatty acids. In *Bacillus*, it has been shown that a cold-shock induced gene (*desA*) encodes a desaturase that is active on the existing fatty acids present in membrane phospholipids. Desaturase activities are dependent on oxygen, and thus an aerobic lifestyle. This mechanism is analogous to that observed in plants (Chapter 4). *Streptococcus pneumoniae*, on the other hand, has a fatty acid profile similar to *E. coli*, but possesses neither an identifiable isomerase nor a desaturase. Thus, these facultatively anaerobic bacteria must contain a novel gene that encodes an isomerase with no homology to *fabA*. Gram-positive bacteria possess a *fabZ* homologue that performs all of the dehydration steps.

11.4. Bacteria with other phospholipid headgroups

E. coli have a very simple phospholipid composition with just three major forms, PE, PG and CL. However, the prokaryotic kingdom possesses a wide array of headgroups that defy adequate description in this short space; hence the reader is referred to Goldfine's review [21] for a comprehensive treatment of bacterial phospholipid structures. The phosphocholine headgroup stands worthy of mention for its uniqueness and distinct mechanisms of synthesis. Phosphatidylcholine (PC) had long been considered a eukaryotic phospholipid, where it is synthesized by transfer of the choline from CDP-choline to diacylglycerol, or by methylation of PE (Chapter 8).

Rhodospseudomonas spheroides, *Bradyrhizobium japonicum* and a few other specialized photosynthetic or nitrogen-fixing bacteria synthesize PC by three subsequent methylations of PE. The first methylase, encoded by *pmtA*, has been disrupted in *Bradyrhizobium japonicum*, and the mutants, which contain significantly reduced PC content, are less able to fix nitrogen in colonization assays. Thus, PC seems to be involved in host:bacteria interactions to establish symbiosis. The prokaryotic PE methyltransferases share weak homology to other bacterial methyltransferases, but no homology with their eukaryotic counterparts. *Rhodospseudomonas spheroides* are also somewhat unique amongst bacteria in that they contain intracellular membranes that hold the photosynthetic machinery. The amount of the intracellular membrane correlates to the amount of incident light, indicating a light-specific regulation of phospholipid synthesis in these organisms.

Sinorhizobium meliloti has been shown to synthesize PC by direct condensation of choline with CDP-diacylglycerol, as well as by the methyltransferase pathway. The *pcs* gene was identified and expression in *E. coli* demonstrates that it does code for a PC synthase [22]. The genomes of *Pseudomonas aeruginosa* and *Borrelia burgdorferi* contain similar genes, and have been reported to possess PC in their membranes. The PC synthase protein shares weak homology with PS synthase (a CDP-diacylglycerol:serine O-phosphatidyltransferase) from other bacteria, but not to any eukaryotic proteins.

A genus of bacteria, termed the Sphingobacterium, produce sphingolipids by a pathway similar to that in mammals. However, little is known about the enzymes involved at this time. Clostridia produce plasmalogens (1-alk-1'-enyl lipids) by an anaerobic pathway clearly different to the O₂-dependent pathway in mammals (Chapter 9). Branched chains fatty acids are also found in which the methyl group is inserted

post-synthetically into the middle of the chain, in a manner analogous to cyclopropane fatty acid synthesis. *S*-Adenosylmethionine is also the methyl donor for these reactions.

11.5. Bacteria with a type I fatty acid synthase

A general distinction between prokaryotic and eukaryotic fatty acids synthases is that bacteria possess the dissociated enzymes described above (type II), while higher organisms have a single, multifunctional, protein (type I) that catalyses all of the reactions. There are exceptions to this rule, however. Mycobacteria, for example, possess a type I fatty acid synthase for the production of their membrane fatty acids. This enzyme is a homohexamer of 290-kDa subunits. Each subunit possesses the six different active sites required to generate a fatty acid. Unlike the type II system, the products of a type I enzyme are acyl-CoAs. For the mycobacterial enzyme, the saturated acyl chains produced are between 16 and 24 carbons in length. Unsaturations must be added post-synthetically by a desaturase. Even more unusual is that the *Mycobacterium* possesses a type II synthase system for the further elongation of the fatty acyl products of the type I system into the 70–80 carbon mycolic acids. *Brevibacterium ammoniagenes*, a highly developed bacteria thought to be a progenitor of the fungi, possesses a type I fatty acids synthase that is capable of producing both saturated and unsaturated fatty acids anaerobically.

11.6. Lipid synthesis in Archea

Archea are a group of organisms, previously classified as bacteria, from which eubacteria and other life may have evolved. A melavonic acid (6 carbon) building block is used for synthesis instead of acetic acid. The generated phytanyl chains are attached to glycerol moieties of complex lipids by ether linkages. Thus, these lipids are unlike anything found in eubacteria or eukaryotes today.

11.7. Other organisms with a bacterial-like fatty acid synthase system

The dissociated enzymes that form the fatty acid synthesis system of most bacteria are not limited to the prokaryotic kingdom [23]. Plants utilize a homologous series of enzymes for synthesis of their fatty acids (Chapter 4). Although the genes are present on the nuclear chromosomes, fatty acid biosynthesis occurs in the chloroplasts of plants. It is thus hypothesized that the pathway evolved from the endosymbiont bacteria that became the chloroplast in plants. Recently, it has also been shown that the apicomplexans, a group of intracellular parasites including *Plasmodium falciparum*, the causative organism of malaria, possess a bacterial-like fatty acid synthase system [24]. These eukaryotic organisms possess an organelle described as a vestigial chloroplast (the apiplast), and thus are presumably evolved from chloroplast-containing algae. *Plasmodium* have been shown to be sensitive to the antibiotics thiolactomycin and triclosan, indicating that this pathway could be exploited for the development of novel antimalaria drugs.

12. Inhibitors of lipid metabolism

Much of the work on prokaryotic membrane biosynthesis in recent years has been directed towards analysis of the pathway enzymes as potential targets for the discovery on new antimicrobial agents [9]. It should be remembered that bacterial fatty acid biosynthesis is catalyzed on a series of small, discrete proteins and utilizes the soluble ACP to shuttle the soluble intermediates within the cytosol. This setup is quite different from mammalian fatty acid synthesis, and thus should allow for specific inhibitors to be designed with minimal mammalian toxicity.

12.1. β -Decynoyl-*N*-acetylcysteamine

The β -hydroxydecanoyl-ACP dehydrase (the *fabA* gene product) is specifically and irreversibly inhibited by the synthetic acetylenic substrate analog β -decynoyl-*N*-acetylcysteamine. A covalent adduct is formed with the active site histidine and results in the loss of both dehydrase and isomerase reactions. The description of this mode of inhibition was the first report of a suicide or mechanism-based inhibitor [25]. β -Decynoyl-*N*-acetylcysteamine concentrations between 10 and 50 μ M are sufficient to completely inhibit unsaturated fatty acid synthesis and bacterial growth. Growth inhibition is relieved by the addition of unsaturated fatty acids to the medium. Saturated fatty acid synthesis continues normally in the presence of β -decenoyl-*N*-acetylcysteamine due to the activity of the *fabZ*-encoded dehydrase. Since most Gram-positive bacteria do not contain unsaturated fatty acids, inhibitors of this step are not being intensively developed.

12.2. Cerulenin and thiolactomycin

Cerulenin, (2*R*)(3*S*)-2, β -epoxy-4-oxo-7,10-dodecadienolyamide, is a fungal product that irreversibly inhibits β -ketoacyl-ACP synthases I and II. It is extremely effective in blocking the growth of a large spectrum of bacteria. Cerulenin forms a covalent bond with the active site cysteine of the synthases. Cerulenin can be seen in the hydrophobic acyl binding tunnel in the crystal structures of both FabB and FabF complexed with this drug. Although cerulenin is a versatile biochemical tool, it is not a suitable antibiotic for clinical use because it is also a potent inhibitor of the multifunctional mammalian fatty acid synthase, which contains similar active site residues and catalyzes the same chemical reaction.

Thiolactomycin, (4*S*)(2*E*,5*E*)-2,4,6-trimethyl- β -hydroxy-2,5,7-octatriene-4-thiolide, inhibits all three β -ketoacyl-ACP synthases of bacterial fatty acid synthesis in vivo and in vitro, but not the multifunctional fatty acid synthases. Malonyl-ACP protects the synthases from thiolactomycin inhibition, indicating that this antibiotic targets a different site on the condensing enzyme from that targeted by cerulenin. The crystal structure of FabB in complex with thiolactomycin reveals that the drug does in fact bind in the malonyl-ACP binding pocket [26], which is presumably distinct from the malonyl-pantotheine site of mammalian type I fatty acid synthase. The structure of the mammalian enzyme has not been solved to date, due in part at least to the large size of the enzyme. Overproduction of FabB but not FabH imparts thiolactomycin resistance.

Since FabF is not essential, these data suggest that FabB is the relevant thiolactomycin target *in vivo*. Thiolactomycin resistance phenotype maps to the *emrR* locus of the *E. coli* chromosome and results in the inactivation of a repressor that governs the expression of the *emrAB* multidrug resistance pump. The subtly different active site architecture of FabH compared to FabB and FabF makes it resistant to both cerulenin and thiolactomycin [26].

12.3. Diazaborines, isoniazid and triclosan

Diazaborines, a group of boron-containing antibacterial heterocyclic compounds, inhibit fatty acid synthesis in *E. coli* by inhibition of the enoyl-ACP reductase I. Inhibition by diazaborines requires the presence of NAD or NADH, and the drug mimics the binding of the substrate. The discovery that the FabI analog in *M. tuberculosis* (InhA) is the target for isoniazid and ethionamide, drugs used to treat tuberculosis, illustrates the potential importance of enoyl-ACP reductase I as an antibiotic target. Isoniazid is first activated by the KatG catalase/peroxidase, and then forms a covalent adduct with the nicotinamide cofactor [27]. It was recently discovered that the enoyl-ACP reductase was also targeted by another antimicrobial compound in widespread use. Triclosan, the active component in a plethora of household items including toothpaste, hand soaps, cutting boards, mattress pads and undergarments, is a very effective inhibitor of the enoyl-ACP reductase I. Its ubiquitous use has been justified by the claim, based on improperly analyzed data, that it caused non-specific membrane disruption, a delayed leakage of cytoplasmic material and finally cell death. This phenotype is essentially identical to that observed by Egan and Russell in 1973 [28] for the original temperature-sensitive *fabI* mutant at the non-permissive temperature. Mutants of *E. coli* can be generated that are resistant to triclosan, and contain a Gly to Val substitution in the active site of enoyl-ACP reductase I. The mutant protein is resistant to triclosan *in vivo* and *in vitro*. Definitive proof that triclosan does not disrupt the membranes of Gram-negative bacteria in a non-specific manner comes from the overexpression of the triclosan-resistant enoyl reductase II (from a plasmid-borne copy of the *fabK* gene) in *E. coli*, which increases resistance by over 10,000-fold.

A key feature of triclosan's mode of action is the slow but complete inactivation of the enoyl reductase. The drug binds to the active site of the enzyme and forms very strong, non-covalent interactions with active site residues and the nicotinamide ring of the cofactor. The half life of the ternary enzyme–NAD⁺–triclosan complex is over 1 h. The mutation that leads to triclosan resistance is the same as that for diazaborine resistance, and adjacent to the mutation required for isoniazid resistance in *M. tuberculosis*. There is some concern that the unregulated use of triclosan in the household setting, where it has not been proven to be effective in promoting public health, will lead to the generation of bacteria resistant to all enoyl-ACP reductase I-directed inhibitors.

12.4. Lipid A biosynthesis inhibitors

A new class of antibiotics was recently discovered that inhibit lipid A biosynthesis [29]. These compounds are still in their infancy, but could prove useful additions to

our antibacterial arsenal. The first generation of compounds are effective inhibitors of bacterial growth *in vivo*, and potent inhibitors of the LpxC deacetylase, the second step in the pathway, *in vitro*. Optimization of this family of compounds may lead to effective agents against Gram-negative bacteria.

13. Future directions

Much progress has been made over the last several years in elucidating the details of the enzymes of fatty acid biosynthesis. This pathway has become a major target for therapeutic intervention in bacterial-mediated disease. Significant inroads have also been made into the molecular mechanisms that regulate fatty acid synthesis. The availability of genomic data has also allowed for the facile translation of many of the findings made in *E. coli* to other bacteria facilitating the discovery of novel genes involved in lipid biosynthesis.

Many of the details of the regulation of fatty acid biosynthesis are still to be worked out, including the probable discovery of new transcription factors and new effector molecules. The fine details of the enzymatic mechanisms, and the comparison of the biochemical properties and functions of different isoforms, will continue in an effort, in part, to elucidate probable activity spectrums of next generation antibiotics that will surely be generated against this pathway. The rush to study fatty acid synthesis has somewhat overshadowed bacterial lipid synthesis in recent years, and discoveries in this area are more difficult since the enzymes and substrates involved are membrane-associated. However, techniques to study the structure and function of membrane proteins are rapidly evolving and will certainly be applied to resolving outstanding issues in bacterial lipid biogenesis.

Abbreviations

ACP	acyl carrier protein
CoA	coenzyme A
<i>fabA</i> , FabA	lowercase italics indicates gene, while uppercase Roman type indicates the protein product of the gene
PE	phosphatidylethanolamine
PG	phosphatidylglycerol
PS	phosphatidylserine
PGP	phosphatidylglycerolphosphate
CL	cardiolipin
LPS	lipopolysaccharides
ABC	ATP-binding cassette
MDO	membrane-derived oligosaccharides

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Lipid metabolism in plants

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1. Introduction

Plants produce the majority of the world's lipids, and most animals, including humans, depend on these lipids as a major source of calories and essential fatty acids. Like other eukaryotes, plants require lipids for membrane biogenesis, as signal molecules, and as a form of stored carbon and energy. In addition, soft tissues and bark each have distinctive protective lipids that help prevent desiccation and infection. To what extent does the biochemistry of plant lipid metabolism resemble that in other organisms? This chapter mentions a number of similarities, but emphasizes aspects unique to plants. Major differences between lipid metabolism in plants and other organisms are summarized in Table 1.

The presence of chloroplasts and related organelles in plants has a profound effect on both gross lipid composition and the flow of lipid within the cell. Fatty acid

Table 1
Comparison of plant, mammalian, and bacterial lipid metabolism

	Higher plants	Mammals	<i>E. coli</i>
<i>Fatty acid synthase</i>			
Structure	Type II (multicomponent)	Type I (multifunctional)	Type II (multicomponent)
Location	Plastids	Cytosol	Cytosol
Acetyl-CoA carboxylase(s)	Multisubunit and multifunctional	Multifunctional	Multisubunit
<i>Primary desaturase substrates</i>			
Δ^9	18:0-ACP	18:1-CoA	none
ω -6	18:1 on glycerolipids	none	none
ω -3	18:2 on glycerolipids	none	none
Primary substrate(s) for phosphatidic acid synthesis	acyl-ACP and acyl-CoA	acyl-CoA	acyl-ACP
Prominent bilayer lipids	Galactolipid > phospholipid	Phospholipid	Phospholipid
Main β -oxidation function	Provides acetyl-CoA for glyoxylate cycle	Provides acetyl-CoA for TCA cycle	Provides acetyl-CoA for TCA cycle

synthesis occurs not in the cytosol as in animals and fungi, but in the chloroplast and other plastids. Acyl groups must then be distributed to multiple compartments, and the complex interactions between competing pathways are a major focus of plant lipid biochemists. It is also significant that the lipid bilayers of chloroplasts are largely composed of galactolipids rather than phospholipids. As a result, galactolipids are the predominant acyl lipids in green tissues and probably on earth.

Plant lipids also have a substantial impact on the world economy and human nutrition. More than three-quarters of the edible and industrial oils marketed annually are derived from seed and fruit triacylglycerols. These figures are particularly impressive given that, on a whole organism basis, plants store more carbon as carbohydrate than as lipid. Since plants are not mobile, and since photosynthesis provides fixed carbon on a regular basis, plant requirements for storage lipid as an efficient, light weight energy reserve are less acute than those of animals.

Finally, hundreds of genes required for plant lipid biosynthesis, utilization and turnover have now been cloned. In addition to providing valuable information on enzyme structure and function, these genes are being exploited to design new, more valuable plant oils. The coordination of lipid metabolic genes with each other and with their potential regulators may also become better understood, as DNA microarray and other genomic technologies mature.

2. *Plant lipid geography*

2.1. *Plastids*

Although all eukaryotic cells have much in common, the ultrastructure of a plant cell differs from that of the typical mammalian cell in three major ways. The plasma membrane of plant cells is shielded by the cellulosic cell wall, preventing lysis in the naturally hypotonic environment but making preparation of cell fractions more difficult. The nucleus, cytosol and organelles are pressed against the cell wall by the tonoplast, the membrane of the large, central vacuole that can occupy 80% or more of the cell's volume. Finally, all living plant cells contain one or more types of plastid.

The plastids are a family of organelles containing the same genetic material, a circular chromosome present in multiple copies. Young or undifferentiated cells contain tiny proplastids that, depending on the tissue, may differentiate into photosynthetic chloroplasts, carotenoid-rich chromoplasts, or any of several varieties of colorless leucoplasts, including plastids specialized for starch storage [1]. These different types of plastids, which may be interconverted *in vivo*, have varying amounts of internal membrane but invariably are bounded by two membranes. The internal structure of chloroplasts is dominated by the flattened green membrane sacks known as thylakoids. The thylakoid membranes contain chlorophyll and are the site of the light reactions of photosynthesis.

As noted above, chloroplasts and other plastids are enriched in galactolipids (Fig. 1). They also contain a unique sulfolipid, sulfoquinovosyldiacylglycerol, whose head group is a modified galactose. The phospholipid components of plastids are less

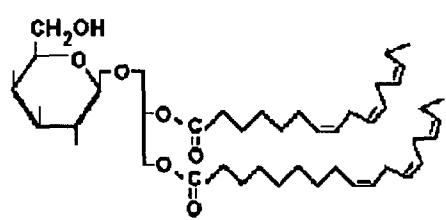
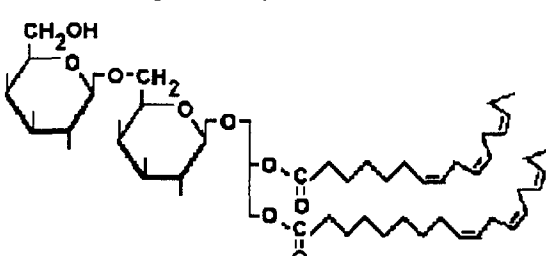
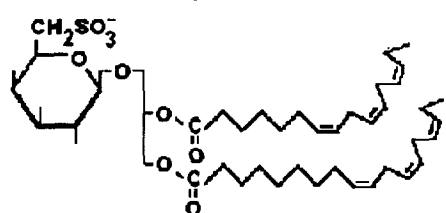
Monogalactosyldiacylglycerol		Pea Chloroplasts	%
		Thylakoids	56
		Inner membrane	64
		Outer membrane	1
		Daffodil chromoplasts	63
		Cauliflower proplastids	31
		Potato leucoplasts	14
Digalactosyldiacylglycerol		Pea Chloroplasts	%
		Thylakoids	32
		Inner membrane	31
		Outer membrane	40
		Daffodil chromoplasts	18
		Cauliflower proplastids	29
		Potato leucoplasts	45
Sulfoquinorosyldiacylglycerol		Pea Chloroplasts	%
		Thylakoids	4
		Inner membrane	4
		Outer membrane	4
		Daffodil chromoplasts	5
		Cauliflower proplastids	6
		Potato leucoplasts	5

Fig. 1. Composition of plastid membranes. Figures given are percentages (as % of total lipid) of the pictured lipid in the membranes specified. Data from Harwood, J.L. (1980) Plant acyl lipids: structure, distribution and analysis. In: P.K. Stumpf (Ed.) The Biochemistry of Plants, Vol. 4., Academic Press, pp. 2–56 and Sparace, S.A., Kleppinger-Sparace, K.F. (1993) Metabolism in non-photosynthetic, non-oilseed tissues. In: T.S. Moore Jr. (Ed.) Lipid Metabolism in Plants, Boca Raton, FL, CRC Press, pp. 569–589.

abundant. Phosphatidylglycerol is the most prominent phospholipid contributor to the thylakoid membrane system of chloroplasts (but < 10% of the glycerolipids), whereas most of the limited phosphatidylcholine of chloroplasts is associated with their outer membrane.

The synthetic capabilities of plastids and other plant organelles are summarized in Table 2. Representatives of each type of plastid have been isolated and found to incorporate acetate into long-chain fatty acids, to desaturate 18:0 to 18:1, and to assemble phosphatidic acid and galactolipids. Chloroplasts have also been shown to synthesize phosphatidylglycerol, including molecular species containing the unusual *trans*-3-hexadecenoic acid at the 2-position. In addition to the components normally retained within the plastids, large quantities of fatty acids, particularly 18:1 and 16:0,

Table 2
Compartmentation of lipids and lipid biosynthesis in plant cells^a

Membrane or organelle	Activities	Prominent lipids (not listed if <5%)
Plastids	Acetyl-CoA carboxylase, fatty acid synthase, 18:0-ACP desaturase, ω 3 and ω 6 desaturases, glycerol-3-phosphate acyltransferase, lysophosphatidic acid acyltransferase, phosphatidic acid phosphohydrolase, CTP: phosphatidate cytidyltransferase, phosphatidylglycerol phosphate synthase and phosphatase, galactolipid and sulfolipid synthesis, diacylglycerol acyltransferase (minor), acyl-CoA synthetase, CTP-dependent lipid kinase	Digalactosyldiacylglycerol, monogalactosyldiacylglycerol, phosphatidylglycerol, sulfoquinovosyldiacylglycerol, phosphatidylcholine
Endoplasmic reticulum	Fatty acid elongase, ω 3 and ω 6 desaturases, other fatty acid modifying reactions, acyl-CoA thioesterase, glycerol-3-phosphate acyltransferase, lysophosphatidic acid acyltransferase, phosphatidic acid phosphatase, CTP: phosphatidate cytidyltransferase, phosphatidylglycerol phosphate synthase and phosphatase, CDP-choline: diacylglycerol cholinephosphotransferase, phosphatidylmono- and dimethylethanolamine methyltransferases, CDP-ethanolamine: diacylglycerol ethanolaminephosphotransferase, phosphatidylserine decarboxylase, phosphatidylethanolamine: serine phosphatidyltransferase, CDP-diacylglycerol: serine phosphatidyltransferase, CDP-diacylglycerol: myo-inositol phosphatidyltransferase, diacylglycerol acyltransferase, diacylglycerol: diacylglycerol acyltransferase, some enzymes of sterol and sphingolipid synthesis, <i>N</i> -acylphosphatidylethanolamine synthase	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, phosphatidylglycerol
Golgi bodies	Glycerol-3-phosphate acyltransferase, lysophosphatidic acid acyltransferase, CDP-choline: diacylglycerol cholinephosphotransferase, CDP-ethanolamine: diacylglycerol ethanolaminephosphotransferase	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine
Lipid bodies	Fatty acid elongase, diacylglycerol acyltransferase, lipase	Triacylglycerol
Mitochondria	Fatty acid synthesis (minor), glycerol-3-phosphate acyltransferase, lysophosphatidic acid acyltransferase, CTP: phosphatidate cytidyltransferase, phosphatidylglycerol phosphate synthase and phosphatase, phosphatidylglycerol: CDP diacylglycerol phosphatidyltransferase, β -oxidation	Phosphatidylcholine, phosphatidylethanolamine, cardiolipin, phosphatidylinositol
Nuclear membranes	None reported	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, phosphatidylglycerol

Plasma membrane	Phosphatidylinositol and phosphatidylinositol phosphate kinases, serine exchange enzyme, sterol glucosyltransferase, glucosylceramide synthase	Phosphatidylethanolamine, phosphatidylcholine, phosphatidylinositol, sphingolipids, sterols and derivatives
Protein body membranes	CTP:phosphocholine cytidyltransferase, diacylglycerol kinase, phosphatidylinositol and phosphatidylinositol phosphate kinases	Phosphatidic acid, phosphatidylcholine, phosphatidylethanolamine, phosphatidylglycerol, phosphatidylinositol
Tonoplast	None reported	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, sphingolipids, sterols and derivatives
Glyoxysomes	Lipase, β -oxidation, glyoxylate cycle	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol
Peroxisomes	β -Oxidation, acyl-CoA thioesterase	Phosphatidylcholine, phosphatidylethanolamine, phosphatidylglycerol, phosphatidylinositol

^a Information collated primarily from T.S. Moore Jr., (Ed.) (1993) *Lipid Metabolism in Plants*, CRC Press, Boca Raton, FL; J.L. Harwood (1989) *Crit. Rev. Plant Sci.* 8, 1–43. Lipids that comprise less than 5% are not listed as prominent lipids. Note that many of the fractions cited were not analyzed for sterols, sterol esters and glycosides, and sphingolipids.

are produced for export to the rest of the cell. An acyl-CoA synthetase identified on the outer membrane of plastids is thought to facilitate release of acyl groups into the cytosol. It should also be noted that, although net lipid traffic is from the plastids, this organelle can likewise be on the receiving end. In addition to small quantities of plastidial phospholipids whose head groups are not known to arise in that compartment, there may be considerable flow of extraplastidially constructed diacylglycerol backbones into the galactolipid synthesis pathway. The quantitative significance of this backflow depends on the plant species, as will be discussed in Section 5.3.

2.2. Endoplasmic reticulum and lipid bodies

The endoplasmic reticulum has traditionally been viewed as the primary source of phospholipids in plant cells. With the exception of cardiolipin, all of the common phospholipids can be produced by microsomal fractions. The endoplasmic reticulum also serves as the major site of fatty acid diversification. Although plastids do have the ability to synthesize polyunsaturated fatty acids, they are formed on acyl lipid substrates and are not typically exported. Thus, the endoplasmic reticulum desaturation pathways are of particular importance for developing seeds that store large quantities of 18:2 and 18:3. Pathways for the production of unusual fatty acids found primarily in seed oils have likewise been described in microsomes. Not surprisingly, the endoplasmic reticulum also appears to be instrumental in the formation of the triacylglycerols themselves and the lipid bodies in which they are stored (Section 7).

2.3. Mitochondria

Next to plastids and the endoplasmic reticulum, the plant mitochondrion is probably the organelle investigated the most thoroughly with respect to lipid metabolism. Its ability to synthesize phosphatidylglycerol and cardiolipin is well established. Although most fatty acids for mitochondrial membranes are imported from the plastids or the ER, recently mitochondria have been shown to synthesize low levels of fatty acids from malonate. Octanoate is a major product of this pathway and serves as a precursor for the lipoic acid cofactor needed by glycine decarboxylase and pyruvate dehydrogenase [2].

2.4. Glyoxysomes and peroxisomes

A discussion of the compartmentation of lipids and their metabolism would be incomplete without reference to the organelles responsible for fatty acid oxidation. As in mammals, there is evidence for both mitochondrial and peroxisomal β -oxidation systems. In plants, the peroxisomal system appears to be the more significant [3]. Unlike mammals, plants can use the peroxisomal enzymes to catabolize long-chain fatty acids all the way to acetyl-CoA. Under certain conditions, such as oilseed germination, plants also differentiate specialized peroxisomes called glyoxysomes. In addition to the β -oxidation pathway, glyoxysomes contain the enzymes of the glyoxylate cycle, a pathway absent from animals. Plants are able to use the glyoxylate cycle to feed the

acetyl-CoA produced by β -oxidation into carbohydrate synthesis. Since plants cannot transport fatty acids over long distances, only this conversion of acetate to sucrose, which can be transported by the plant's vascular system, makes lipid a practical carbon reserve for the growing shoots and roots of seedlings.

3. *Acyl-ACP synthesis in plants*

Fatty acid synthases may be classified into two groups. 'Type I' fatty acid synthases are characterized by the large, multifunctional proteins typical of yeast and mammals (Chapter 6), while 'Type II' synthases of most prokaryotes are dissociable into components that catalyze individual reactions (Chapter 3). Plants, while certainly themselves eukaryotic, appear to have inherited a Type II fatty acid synthase from the photosynthetic prokaryotes from which plastids originated.

The ground-breaking studies of Overath and Stumpf in 1964 [4] established not only that the constituents of the avocado fatty acid synthesis system could be dissociated and reconstituted, but also that the heat stable fraction from *E. coli* we now know as acyl carrier protein (ACP) could replace the corresponding fraction from avocado. Plant ACPs share both extensive sequence homology and significant elements of three-dimensional structure with their bacterial counterparts. In plants, this small, acidic protein not only holds the growing acyl chain during fatty acid synthesis, but also is required for synthesis of monounsaturated fatty acids and plastidial glycerolipids.

3.1. *Components of plant fatty acid synthase*

Fatty acid synthase is generally defined as including all polypeptides required for the conversion of acetyl and malonyl-CoA to the corresponding ACP derivatives, the acyl-ACP elongation cycle diagrammed in Chapter 3, and the cleavage of ACP from completed fatty acids by enzymes termed thioesterases or acyl-ACP hydrolases [5]. All components of fatty acid synthase occur in plastids, although they are encoded in the nuclear genome and synthesized on cytosolic ribosomes. Most of the 8–10 enzymes of the pathway are soluble when isolated from homogenates. Nevertheless, some evidence suggests that at least ACP and some subunits of acetyl-CoA carboxylase may be associated with the plastid membranes.

Despite the presence of acetyl-CoA : ACP acyltransferase activity in plant fatty acid synthase preparations, acetyl-ACP does not appear to play a major role in plant fatty acid synthesis (Jaworski, 1992). Instead, the first condensation takes place between acetyl-CoA and malonyl-ACP. This reaction is catalyzed by β -ketoacyl-ACP synthase III, one of three ketoacyl synthases in plant systems (Fig. 2). The acetoacetyl-ACP product then undergoes the standard reduction–dehydration–reduction sequence to produce 4 : 0-ACP, the initial substrate of ketoacyl-ACP synthase I. KAS I is responsible for the condensations in each elongation cycle up through that producing 16 : 0-ACP. The third ketoacyl synthase, KAS II, is dedicated to the final plastidial elongation, that of 16 : 0-ACP to 18 : 0-ACP.

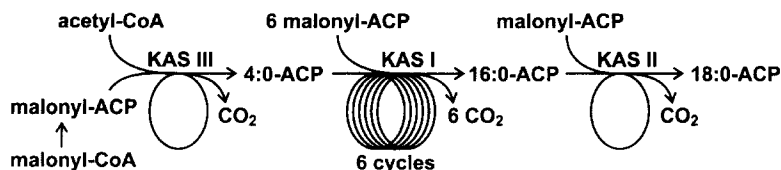


Fig. 2. Contribution of the three ketoacyl synthases (KAS I, II and III) to fatty acid elongation. Each circle represents one round of the elongation cycle catalyzed by ketoacyl-ACP synthase, enoyl-ACP reductase, hydroxyacyl-ACP dehydrase, and acyl-ACP reductase.

3.2. Desaturation of acyl-ACPs

The major components of the long-chain acyl-ACP pool in most plant tissues are 16:0-ACP, 18:0-ACP and 18:1-ACP. This finding highlights the importance of stearoyl-ACP desaturase, the plastidial enzyme responsible for Δ^9 -desaturation in plants. In contrast to the desaturation system of *Escherichia coli* (Chapter 3), the plant enzyme introduces the double bond directly to the Δ^9 -position. Unlike yeast and mammalian Δ^9 -desaturases, it is a soluble enzyme and is specific for acyl-ACPs rather than acyl-CoAs (McKeon, 1982). In recent years, work on stearoyl-ACP desaturase has progressed rapidly and is now providing a more detailed understanding of the fundamental mechanisms of oxygenic fatty acid desaturation. Genes for the enzyme have been cloned from a number of species, and the structure of the castor bean Δ^9 -desaturase has been determined to 2.4 Å resolution. A combination of the crystal structure and spectroscopic methods has revealed two identical monomers, each with an active site containing a diiron-oxo cluster. Reduction of the iron by ferredoxin leads to its binding of molecular oxygen. The resulting complex ultimately removes electrons at the Δ^9 -position, resulting in double bond formation [6].

Although the most common unsaturated fatty acids in plants are derived from oleic acid, a wide range of unusual fatty acids are found in the seed oils of different species. Divergent plastid acyl-ACP desaturases have been shown to account for some of this diversity. For example, *Coriandrum sativum* achieves seed oils rich in Δ^6 -18:1 (petroselinic acid) by desaturation of 16:0 at the Δ^4 -position followed by elongation, while *Thunbergia alata* attains a similar oil by direct Δ^6 -desaturation of 18:0. Several structure–function relationships suggested by unusual desaturases have been tested in the acyl-ACP desaturase system. Shortening the acyl binding pocket of the Δ^9 -18:0 desaturase by altering a single amino acid as in *Doxantha unguis-cati* shifts substrate specificity in favor of 16:0-ACP (Cahoon, 1998). In addition, a set of five specific amino acids suggested by the *Thunbergia* gene transforms the castor bean Δ^9 -desaturase to a Δ^6 -desaturase, while enzymes with certain subsets of the five amino acids can desaturate at either position [6].

3.3. Acyl-ACP thioesterases

Among prokaryotes, all acyl groups exiting the dissociable fatty acid synthase are transferred directly from ACP to polar lipids. However, plants must also release sufficient fatty acid from ACP to supply the extraplastidial compartments. Since the

typical chloroplast exports primarily 18:1 and 16:0, the same fatty acids that comprise the greatest fraction of long-chain acyl-ACPs, it might be assumed that a relatively non-specific thioesterase releases 16- and 18-carbon fatty acids from ACP. However, molecular and biochemical analyses of cloned plant thioesterases suggest that plants possess individual thioesterases with specificity either for 18:1 or for one or more saturated fatty acids [7]. The most prominent thioesterase in most plants has a strong preference for 18:1-ACP, making 18:1 the fatty acid most available for extraplastidial glycerolipid synthesis. In contrast, mangosteen, a plant with seed oil particularly high in 18:0, contains an 18:0-ACP thioesterase gene that has been used to engineer rapeseed with high 18:0 content (Hawkins, 1998).

Plants that synthesize certain unusual fatty acids have additional or modified thioesterases. For example, several plant species that produce storage oils containing large amounts of 8- to 14-carbon acyl chains contain thioesterases specific for those chain lengths. By removing acyl groups from ACP prematurely, the medium-chain thioesterases simultaneously prevent their further elongation and release them for triacylglycerol synthesis outside the plastids. In addition, both the standard Δ^9 -18:1 thioesterase and a Δ^6 -18:1 thioesterase have been purified from the Δ^6 -18:1 accumulating coriander plant. Thus plants, by regulating expression of different thioesterases, can both fine tune and radically modify the exported fatty acid pool.

4. *Acetyl-CoA carboxylase and control of fatty acid synthesis*

4.1. *Two forms of acetyl-CoA carboxylase*

The malonyl-CoA that supplies all but two carbons per fatty acid is produced from acetyl-CoA and carbon dioxide by acetyl-CoA carboxylase (ACC). In plants, malonyl-CoA for fatty acid synthesis is apparently provided by a plastid ACC, while a cytosolic ACC contributes malonyl units for fatty acid elongation as well as synthesis of flavonoids, polyketides, and other metabolites. As with fatty acid synthase, ACC forms may be categorized either as 'eukaryotic' enzymes, which are dimers of a multifunctional polypeptide (Chapter 6), or 'prokaryotic' enzymes, which are heteromers of biotin carboxyl carrier protein, biotin carboxylase, and two subunits of carboxyltransferase (Chapter 3). In the grass family, both plastids and cytosol house 'eukaryotic' enzymes. However, dicots and monocots other than grasses appear to have both forms, with the 'eukaryotic' form limited primarily to the cytosol, and 'prokaryotic' enzymes dominating in the plastids [8]. Assembly of the 'prokaryotic' form requires participation of both the nuclear genome, which encodes biotin carboxyl carrier protein, biotin carboxylase, and the α -subunit of carboxyltransferase, and the plastid genome, which has retained the gene for the carboxyltransferase β -subunit, perhaps due to a requirement for RNA editing (Sasaki, 2000).

4.2. *Acetyl-CoA carboxylase as control point*

In other kingdoms, ACC is a major control point for fatty acid biosynthesis. Although the mechanisms acting in plants are incompletely characterized, there is evidence that

plant ACCs are also tightly regulated [9]. For example, both redox regulation via thioredoxin and phosphorylation of the carboxyltransferase have been implicated in up-regulation of the 'prokaryotic' ACC by light. Conversely, feedback inhibition is observed at the level of ACC when tobacco cell cultures are given exogenous fatty acids. Due to its impact on the rate of fatty acid synthesis, ACC is considered a promising target in oilseed improvement programs, and some increases in oil content have been obtained by engineering a cytosolic ACC gene to be expressed in rapeseed plastids.

5. Phosphatidic acid synthesis: 'prokaryotic' and 'eukaryotic' acyltransferases

Since phosphatidic acid serves as a precursor of phospholipids, galactolipids and triacylglycerols, it is not surprising that its own synthesis has been reported in four plant compartments: plastids, endoplasmic reticulum, mitochondria, and Golgi bodies. In each case, esterification of the first acyl group to the *sn*-1 position of glycerol-3-phosphate is catalyzed by glycerol-3-phosphate acyltransferase. Lysophosphatidic acid acyltransferase then completes the synthesis by acylating the *sn*-2 position. However, plastidial and extraplastidial acyltransferases show distinct differences in structure and specificity.

5.1. Plastidial acyltransferases

In the plastids, acyltransferases provide a direct route for acyl groups from ACP to enter membrane lipids. Since this is the standard pathway in *E. coli* and cyanobacteria, both the enzymes of phosphatidic acid synthesis in plastids and the glycerolipid backbones they produce are termed 'prokaryotic.' In both chloroplasts and non-green plastids, the glycerol-3-phosphate acyltransferase is a soluble enzyme that, unlike the *E. coli* enzyme, shows preference for 18:1-ACP over 16:0-ACP. The lysophosphatidic acid acyltransferase, which is a component of the inner envelope of plastids, is extremely selective for 16:0-ACP. The presence of a 16-carbon fatty acid at the 2-position is therefore considered diagnostic for lipids synthesized in the plastids.

5.2. Extraplastidial acyltransferases

At least superficially, the mitochondrial and Golgi acyltransferase activities resemble those of the better studied endoplasmic reticulum system. All three compartments have membrane-bound glycerol-3-phosphate and lysophosphatidic acid acyltransferases that utilize acyl-CoA substrates. In the endoplasmic reticulum, which is quantitatively the most significant of the extraplastidial sites for phosphatidic acid synthesis, saturated fatty acids are almost entirely excluded from the *sn*-2 position. The glycerol-3-phosphate acyltransferase is less selective, but, due to substrate availability, more often fills the *sn*-1 position with 18:1 than with 16:0. It is therefore possible to judge the relative contributions of the prokaryotic and eukaryotic pathways by comparing the proportions of eukaryotic 18/18 or 16/18 glycerolipids with prokaryotic 18/16 or

16/16 glycerolipids [5]. Mitochondrial lysophosphatidic acid acyltransferase, which shows little selectivity for chain length or unsaturation, can usually be ignored in discussions of lipid flow.

5.3. 16:3 and 18:3 plants

Relative fluxes through the prokaryotic and eukaryotic pathways vary between organisms and among tissues. Plastids have the potential to use phosphatidic acid from the prokaryotic pathway for all of their glycerolipid syntheses. However, not all plants do so; in some cases, the prokaryotic acyl chain arrangement is found only in plastidial phosphatidylglycerol, whereas galactolipids are derived from diacylglycerol imported to the plastids from the ER. As indicated above, the eukaryotic acyltransferases of the endoplasmic reticulum produce substantially more 18/18 than 16/18 lipids, and it is chiefly the 18/18 units that are assembled into galactolipids by plants with a minor prokaryotic pathway. Because galactolipids become highly unsaturated, plants that import diacylglycerol for galactolipids are rich in 18:3 and are called 18:3 plants. Species in which most galactolipid is derived from the prokaryotic 18/16 or 16/16 diacylglycerol contain substantial 16:3 and are known as 16:3 plants.

Kunst et al. [10] have demonstrated that a 16:3 plant, *Arabidopsis thaliana*, may be converted to a de facto 18:3 plant by a single mutation in plastidial glycerol-3-phosphate acyltransferase. Under these conditions, 16:3 content is reduced dramatically, and when isolated chloroplasts are labeled with glycerol-3-phosphate, only phosphatidylglycerol is labeled. Nevertheless, the percentage of galactolipids in mutant plants is practically identical to that in wild-type plants, emphasizing the ability of plants to compensate for reduction of the prokaryotic pathway. Other studies in mutants have confirmed that plants have an amazing capacity to adapt to many, but not all, perturbations of lipid metabolism (Section 11).

6. Glycerolipid synthesis pathways

In plants, glycerolipid biosynthesis involves a complex web of reactions distributed among multiple compartments [11,12]. As in mammals, the synthesis of individual glycerolipids is initiated either by the formation of CDP-diacylglycerol from phosphatidic acid and CTP, or by cleavage of phosphate from phosphatidic acid to produce diacylglycerol.

CTP:phosphatidate cytidyltransferase has been observed in plastids, mitochondria and endoplasmic reticulum. In all three compartments, the CDP-diacylglycerol derived from phosphatidic acid is used in the synthesis of phosphatidylglycerol; in mitochondria, the reaction of phosphatidylglycerol with a second CDP-diacylglycerol then produces cardiolipin. The endoplasmic reticulum can also incorporate CDP-diacylglycerol into phosphatidylinositol and phosphatidylserine.

Phosphatidic acid phosphatase is present in the same three compartments. In the endoplasmic reticulum and mitochondria, diacylglycerol combines with CDP-ethanolamine or CDP-choline to produce phosphatidylethanolamine or phosphatidylcholine,

respectively. Although separate enzymes catalyze ethanolamine and choline transfer in animals and yeast, there are indications that a single aminoalcoholphosphotransferase may be responsible in plants. Flux into phosphatidylcholine is at least partially determined by regulation of the phosphocholine cytidyltransferase that generates CDP-choline. The production of CDP-ethanolamine from phosphoethanolamine is less well studied, but is also considered a probable regulatory step. In addition, there is clear evidence that phosphoethanolamine can be methylated to monomethylphosphoethanolamine, dimethylphosphoethanolamine, and phosphocholine, and that this pathway is inhibited by exogenous choline at the initial methylation step. The methylation of phosphoethanolamine in plants is frequently contrasted with the methylation of phosphatidylethanolamine to phosphatidylcholine in animals and yeast. In general, no significant methylation of phosphatidylethanolamine itself occurs in plants. However, phosphatidylmonomethylethanolamine is synthesized and converted to phosphatidylcholine [11].

The diacylglycerol released in plastids reacts either with UDP-galactose or with UDP-sulfoquinovose to generate sulfolipid or monogalactosyldiacylglycerol. Synthesis of digalactosyldiacylglycerol from the latter may use either UDP-galactose or a second monogalactosyldiacylglycerol as the galactose donor [12] (Kelly, 2002). There is also evidence for some extraplastidial galactolipid synthesis in plants suffering from phospholipid deficiency (Hartel, 2000).

6.1. Glycerolipids as substrates for desaturation

In addition to the soluble acyl-ACP desaturases, plants contain a number of membrane-bound enzymes that desaturate fatty acids while they are esterified within glycerolipids [5,6]. The recent cloning and characterization of these desaturases is of great interest to the scientific community because the products of the membrane-bound systems include $\Delta^{9,12}$ -18:2 and $\Delta^{9,12,15}$ -18:3, both of which are essential to the human diet, and thought to play a major role in human health and disease.

Once again, separate pathways occur in plastids and endoplasmic reticulum, although, as should be evident from the discussion above, fatty acids from the endoplasmic reticulum may make their way back to the plastids. Clarification of the number of desaturases involved in plant lipid metabolism and isolation of their genes has been greatly assisted by the isolation of a large number of mutants in *A. thaliana*, a small weed of the mustard family used as a model organism by plant geneticists and molecular biologists. Briefly, three membrane-bound desaturation sequences are evident in *Arabidopsis* [5,6].

- (1) In chloroplasts, 16:0 at the 2-position of phosphatidylglycerol is desaturated to *trans*-3-16:1. This desaturase is most likely encoded by the *FAD4* gene.
- (2) Plastids are able to convert 18:1 to 18:3 and 16:0 to 16:3 using a combination of three membrane-bound desaturases. One of them, encoded by *FAD5*, is relatively specific for the conversion of 16:0 on monogalactosyldiacylglycerol to Δ^7 -16:1. This 16:1 and Δ^9 -18:1 may then be given a second and third double bond by the *FAD6* and the *FAD7* or *FAD8* gene products, respectively. The latter two desaturases are less selective in their choice of glycerolipid substrate, and will

accept appropriate fatty acids on phosphatidylglycerol, sulfolipid, or either of the major galactolipids.

- (3) In the endoplasmic reticulum, 18 : 1 esterified to phosphatidylcholine or occasionally phosphatidylethanolamine may be desaturated to 18 : 2 by *FAD2* and to 18 : 3 by *FAD3*.

It should be noted that fatty acids entering one of the multistep desaturation pathways listed above are not necessarily committed to completing that set of reactions. It is particularly common for 18 : 2 to be an end product of endoplasmic reticulum desaturation. This 18 : 2 may remain in phospholipid, be incorporated into triacylglycerol, or enter the galactolipid pathway and receive a third double bond in the chloroplast.

7. Lipid storage in plants

A plant stores reserve material in its seeds in order to allow seedling growth of the next generation until photosynthetic capacity can be established. The three major storage materials are oil, protein and carbohydrate, and almost all seeds contain some of each. However, their proportions vary greatly. For example, the amount of oil in different species may range from as little as 1–2% in grasses such as wheat, to as much as 60% of the total dry weight of the castor seed. With the exception of the jojoba plant, which accumulates wax esters in seeds, plants store oil as triacylglycerol (TAG).

7.1. Lipid body structure and biogenesis

In the mature seed, TAG is stored in densely packed lipid bodies, which are roughly spherical in shape with an average diameter of 1 μm (Fig. 3) [13]. This size does not change during seed development, and accumulation of oil is accompanied by an increase in the number of lipid bodies. The very large number of lipid bodies in an oilseed cell (often >1000) contrasts strikingly with animal adipose tissue where oil droplets produced in the cytosol can coalesce into a few or only one droplet. The plant lipid bodies appear to be surrounded by a lipid monolayer in which the polar headgroups face the cytosol, while the non-polar acyl groups are associated with the non-polar TAG within. The membranes of isolated lipid bodies, which comprise less than 5% of a lipid body's weight, contain both phospholipids and characteristic proteins known as caleosins and oleosins. The recently discovered caleosins are calcium binding proteins whose function is still unknown. Oleosins are small (15–26 kDa) proteins that are believed to preserve individual lipid bodies as discrete entities. Desiccated seeds lacking oleosins undergo lipid body fusion and cell disruption when rehydrated (Leprince, 1998). The cDNAs encoding many oleosins have been cloned and each has a sequence encoding a totally hydrophobic domain of 68–74 amino acids which is likely to be the longest hydrophobic sequence found in any organism. Structurally, oleosins are roughly analogous to the animal apolipoproteins which coat the surface of lipid droplets during their transport between tissues.

When a seed germinates, the TAG stored in the lipid bodies becomes the substrate for lipases. In at least some cases, peroxidation of polyunsaturated fatty acids by a lipid

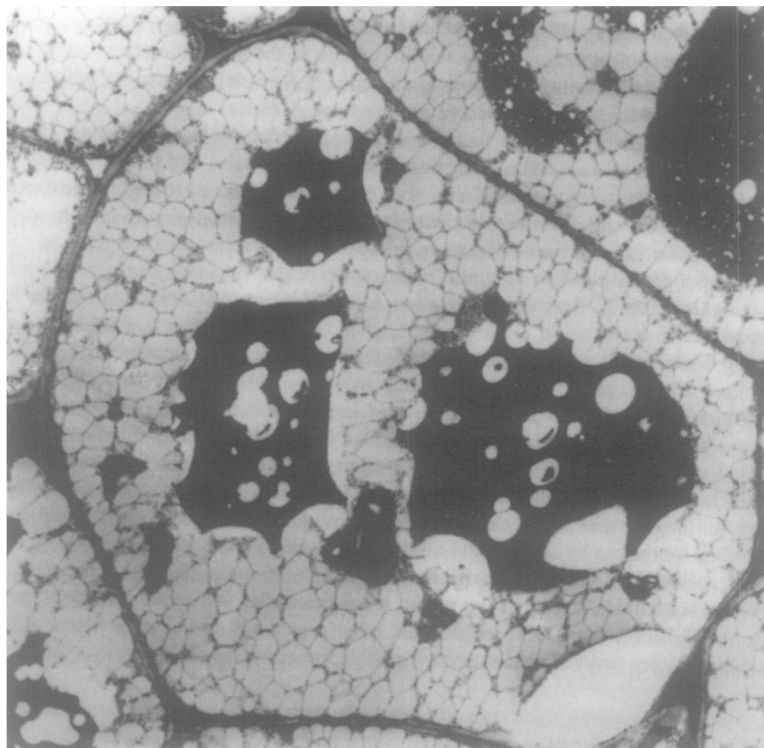


Fig. 3. Thin-sectional view of cells in a cotyledon of a developing cotton embryo harvested 42 days after anthesis. The cells are densely packed with lipid bodies and several large storage protein bodies (dark). Magnification $\times 9000$. Photo courtesy of Richard Trelease, Arizona State University.

body lipoxygenase precedes the release of fatty acids from TAG [14]. Typically lipases and lipid body lipoxygenase are active only after germination is triggered by imbibition and other environmental signals. Fatty acids released by the lipid bodies are further metabolized through the β -oxidation pathway and glyoxylate cycle in the glyoxysomes (Section 2.4).

7.2. Seed triacylglycerols often contain unusual fatty acids

The structural glycerolipids of all plant membranes contain predominantly 5 fatty acids (18:1, 18:2, 18:3, 16:0, and in some species, 16:3). However, the fatty acid composition of storage oils varies far more than in membrane glycerolipids. Altogether more than 300 different fatty acids are known to occur in seed TAG [15]. Chain length may range from less than 8 to over 22 carbons. The position and number of double bonds may also be unusual, and hydroxy, epoxy or other functional groups can modify the acyl chain. Many of the different fatty acid structures, including hydroxy, epoxy, acetylenic and conjugated varieties, are now known to originate from minor modifications in the amino acid sequence of the oleate desaturase. For example, only

Table 3
Some unusual fatty acids produced in plant seeds

Fatty acid type	Example	Major sources	Major or potential uses	Approx. US market size, 10 ⁶
Medium chain	Lauric acid (12:0)	Coconut, palm kernel	Soaps, detergents, surfactants	350
Long chain	Erucic acid (22:1)	Rapeseed	Lubricants, anti-slip agents	100
Epoxy	Vernolic acid 18:1 Δ^9 epoxy12,13	Epoxidized soybean oil, <i>Vernonia</i>	Plasticizers	70
Hydroxy	Ricinoleic acid 18:1 Δ^9 ,12OH	Castor bean	Coatings, lubricants	80
Acetylenic	Crepennic acid 18:2 Δ^9 ,12yne	<i>Crepis foetida</i>	Polymers	—
Cyclopropene	Sterculic acid 19:1	<i>Sterculia foetida</i>	Lubricants, polymers	—
Conjugated	Parinaric 18:4 Δ^9 c11t13t15c	<i>Impatiens balsamina</i>	Coatings	—
Trienoic	Linolenic acid (α 18:3)	Flax	Paints, varnishes, coatings	45
Wax esters	Jojoba oil	Jojoba	Lubricants, cosmetics	10

4 amino acid changes have been shown to convert a desaturase into a hydroxylase [16].

The reason for the great diversity in plant storage oils is unknown. However, the special physical or chemical properties of the 'unusual' plant fatty acids have been exploited for centuries. In fact, approximately one-third of all vegetable oil is used for non-food purposes (Table 3). Reading the ingredients of a soap or shampoo container reveals one of the major end uses of high lauric acid specialty plant oils. Other major applications include the use of erucic acid (22:1) derivatives to provide lubricants and as a coating for plastic films. Hydroxy fatty acids from the castor bean have over 100 industrial applications including plastic and lubricant manufacture. As discussed further below, the ability of genetic engineering to transfer genes for some unusual fatty acid production from exotic wild species to high yielding oil crops is now providing the ability to produce new renewable agricultural products and to replace feedstocks derived from petroleum.

7.3. The pathway of triacylglycerol biosynthesis

As in animal tissues, it has been suggested that TAGs are produced by a relatively simple four reaction pathway. According to this model, phosphatidic acid is synthesized by the extraplastidial pathway (Section 5) and dephosphorylated to diacylglycerol. A third fatty acid is then transferred from CoA to the vacant third hydroxyl of the diacylglycerol, producing TAG. This last and single committed step is catalyzed by diacylglycerol acyltransferase (Fig. 4, reaction 6). Although plants possess all of the enzymes for the reactions above, the assembly of three fatty acids onto a glycerol backbone is not always

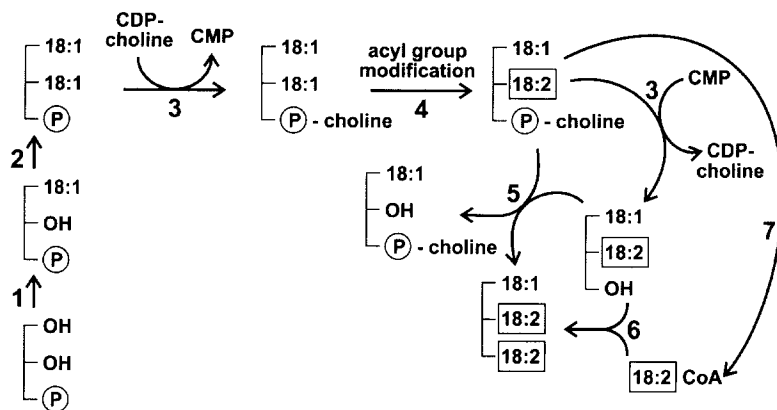


Fig. 4. Pathway depicting how flux through phosphatidylcholine (product of reaction 3) can promote acyl group diversity in plant triacylglycerols. Production of 18:2 (boxed) at the *sn*-2 position and its transfer to TAG is used as a sample modification. Other fatty acid alterations may be substituted. Enzymes: 1 = glycerol-3-phosphate:acyl-CoA acyltransferase; 2 = lysophosphatidic acid:acyl-CoA acyltransferase; 3 = CTP:phosphatidate cytidyltransferase; 4 = T6 18:1-desaturase or other fatty acid modifying enzyme; 5 = phospholipid:diacylglycerol acyltransferase; 6 = diacylglycerol acyltransferase; 7 = acyl-CoA:phosphatidylcholine acyltransferase or phospholipase plus acyl-CoA synthetase.

as straightforward as suggested by the above pathway. In many oilseeds, pulse-chase labeling has revealed that fatty acids reach TAG only after passing through phosphatidylcholine, (or phosphatidylethanolamine to a lesser extent). Given the range of desaturation and other modification reactions that can take place on phosphatidylcholine, transit through this phospholipid helps to explain some of the fatty acid diversity in TAG.

Fatty acids from phosphatidylcholine may become available for TAG synthesis in several ways [17]. In some plants, a phospholipid:diacylglycerol acyltransferase produces TAG by direct transfer of a fatty acid from the 2-position of phospholipid to diacylglycerol (Fig. 4, reaction 5) [18]. The second mechanism by which phosphatidylcholine can participate in TAG synthesis is by donation of its entire diacylglycerol unit. In many plants, the synthesis of phosphatidylcholine from diacylglycerol and CDP-choline appears to be rapidly reversible. As shown in Fig. 4 (reaction 3), this activity of the choline phosphotransferase can allow diacylglycerol moieties modified on phosphatidylcholine to be incorporated into TAG. Finally, a fatty acid removed from phosphatidylcholine may subsequently be used for TAG synthesis. Such an 'acyl exchange' may provide acyl-CoA either by the combined reverse and forward reactions of an acyl-CoA:phosphatidylcholine acyltransferase, or by a combination of phospholipase and acyl-CoA synthase.

7.4. Challenges in understanding triacylglycerol synthesis

Although the basic reactions of TAG biosynthesis have been determined, several fundamental and potentially related questions persist. As highlighted above, TAG and membrane lipids frequently have radically different fatty acid compositions. How do

plants control which fatty acids are stored in TAG as opposed to which fatty acids are restricted to membranes? Are unusual fatty acids excluded from membranes because their physical and chemical idiosyncracies would perturb membrane fluidity or other physical characteristics? Is TAG synthesis spatially distinct from the synthesis of membrane lipids, or do enzyme specificities dictate the partitioning of fatty acid species among glycerolipids? Although all of these factors may be significant, selectivity by enzymes such as phospholipid:diacylglycerol acyltransferase for unusual fatty acids, and editing of unusual fatty acids from phospholipids, are currently the best documented [17,18].

8. *Protective lipids*

In plants, tissues are protected against desiccation and pathogens by both a cuticle and epicuticular wax. The cuticle itself contains some wax, but is anchored to the plant cell wall by cutin, a complex polyester of fatty acid derivatives with a wide range of oxygen-containing functional groups (Fig. 5). 16- and 18-carbon dicarboxylic acids with one or more hydroxyl groups are particularly common [19]. *Arabidopsis* transformed with cutinase from a fungal plant pathogen not only develops a leaky cuticle, but also suffers from fusions between leaves and flower parts (Sieber, 2000).

Surface waxes are complex mixtures including a range of very long-chain alkanes, aldehydes and ketones as well as wax esters and their building blocks. Although only one mutation in cutin formation has been identified (Wellesen, 2001), visual screening of plant surfaces has allowed isolation of several wax mutants blocked in malonyl-CoA-dependent elongation of fatty acids, decarbonylation and reduction (Post-Beittanmiller, 1996). The cDNA of an elongase required for extension of wax acyl units beyond 24 carbons has been cloned, and resembles the condensing enzymes involved in synthesis of erucic acid in seed oil and wax ester precursors in jojoba seeds (Millar, 1999). A cDNA encoding the wax synthase of jojoba seeds has been cloned and successfully expressed in *Arabidopsis* (Lardizabal, 2000), and may provide clues to the corresponding genes for surface waxes.

Bark, wound callus, and specialized tissues such as the endodermis that controls entry into the root vascular system, have walls lined with suberin. Suberin, like cutin, is a polyester incorporating fatty acids enriched in carboxyl and hydroxyl groups. In addition to placement on the inner surface of cell walls rather than outside, the tough, waterproof suberin differs from cutin in its preference for longer fatty acids and in its incorporation of large amounts of phenylpropanoids [19].

9. *Sterol, isoprenoid and sphingolipid biosynthesis*

In the plant kingdom, isoprenoids represent the most diverse range of natural products with over 25,000 lipophilic structures known, ranging from small, volatile compounds to rubber. Quantitatively, the photosynthetic apparatus is probably the primary consumer of isoprenoids, since carotenoids, plastoquinone and the phytol tail of chlorophyll

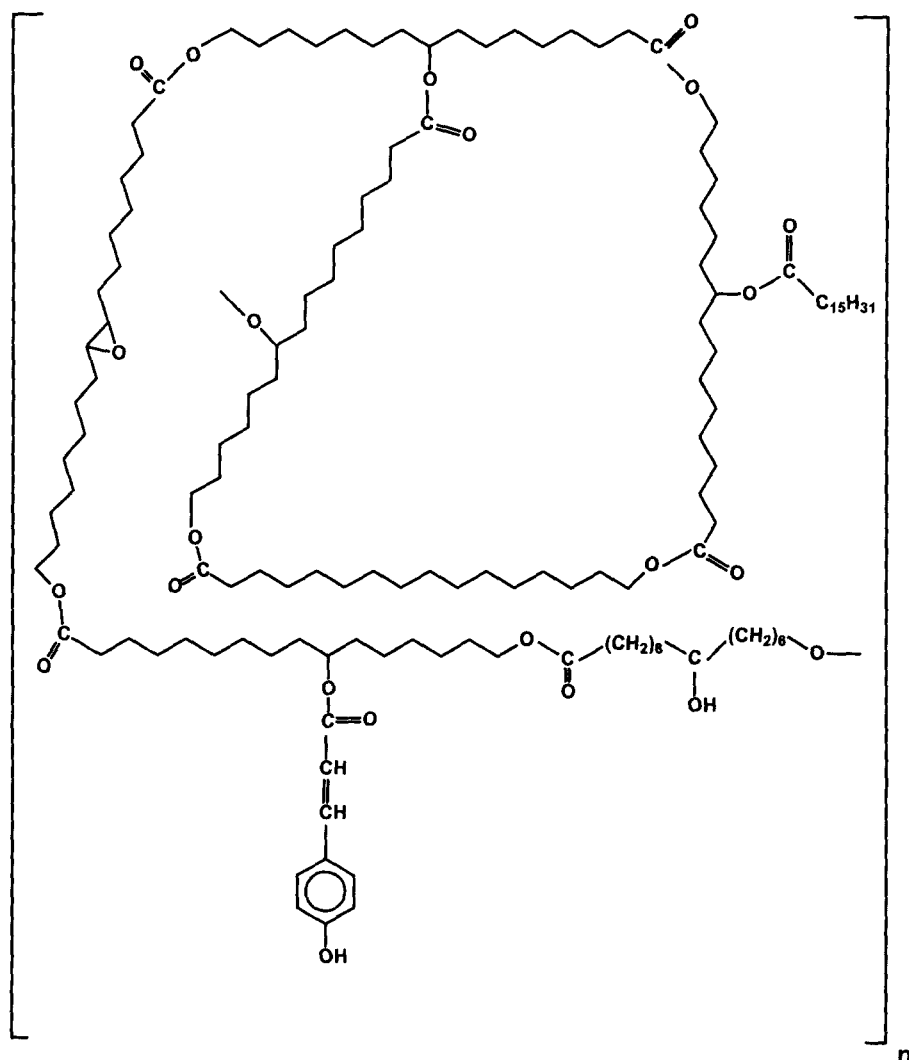


Fig. 5. Model showing some of the linkages in cutin, after Kolattukudy [19]. Note the cross-linking made possible by mono- and dihydroxy-fatty acids.

all belong to this group. Given that vital plant hormones such as gibberellin and abscisic acid, plus many defensive compounds, are isoprenoids, the early steps of this pathway have been studied intensely. However, surprisingly, it was not until the late 1990s that researchers realized that plants have two very different pathways for production of isopentenyl pyrophosphate, the five-carbon central precursor of all isoprenoids [20]. For several decades it was known that, as in other organisms, plants join three molecules of acetyl-CoA to form hydroxymethylglutaryl-CoA followed by the highly regulated reduction of that compound to mevalonic acid. Furthermore, plants contain multiple well-studied hydroxymethylglutaryl-CoA reductase genes that

are differentially expressed during development and in response to such stimuli as light, wounding and infection. It was incorrectly suspected that this 'mevalonate' pathway was localized in both cytosol and plastids and produced all classes of isoprenoids. The story has now been clarified [20] with the discovery that plastids produce isopentenyl pyrophosphate by a 'non-mevalonate' pathway that begins with the condensation of pyruvate with glyceraldehyde-3 phosphate to produce 1-deoxy-D-xylulose-5-P. At least three additional enzymes are required to produce isopentenyl pyrophosphate in the plastids. In parallel to work in plants, this pathway has also been demonstrated in bacteria and algae. The non-mevalonate pathway in plastids is responsible for production of the classic plant photosynthetic isoprenoids such as phytols and carotenoids, as well as mono- and diterpenes.

The mevalonate pathway in the cytosol is responsible for biosynthesis of sterols, sesquiterpenes and triterpenoids. After conversion of mevalonic acid to isopentenyl pyrophosphate, three C5 units can be joined head to tail to produce a C15 compound, farnesyl pyrophosphate. Two farnesyl pyrophosphates are then united head to head to form squalene, the progenitor of the C30 isoprenoids from which sterols are derived. The plant squalene synthetase, like its mammalian homologue, is found in the endoplasmic reticulum and the reaction proceeds via a presqualene pyrophosphate intermediate. In the last step prior to cyclization, squalene is converted to squalene 2,3-epoxide.

It is also in the cyclization step that photosynthetic and non-photosynthetic organisms diverge. Whereas animals and fungi produce lanosterol, organisms with a photosynthetic heritage produce cycloartenol. Despite the differences in the cyclization product, there is substantial conservation between the enzymes responsible, with 34% identity between an *Arabidopsis* cycloartenol synthase and lanosterol synthase.

A complex series of reactions including opening of the cyclopropane ring, double bond formation and isomerization, demethylation of ring carbons, and methylation of the side chain result in formation of a number of different plant sterols. Sitosterol is the most common plant sterol (Fig. 6); however, plants normally contain mixtures of sterols whose proportions differ from tissue to tissue. In addition, sterol esters, sterol glycosides, and acylated sterol glycosides are common plant constituents whose physiological significance is under scrutiny. Both cold adaptation and pathogenesis drastically alter free and derivatized sterol pools. Plants also produce a steroid hormone, brassinolide, required for both light-induced development and fertility. Interestingly, the gene for a 5α -reductase in the brassinolide pathway can complement the corresponding reductase in the testosterone pathway [21].

Sphingolipids are usually considered minor constituents of plant lipids, accounting for 5% or less of most lipid extracts. This fact, and the more complex methods needed for their identification and characterization have resulted in a comparative lack of information on plant sphingolipid biosynthesis and function. Nevertheless, sphingolipids make up a substantial proportion (25% or more) of the composition of plasma and tonoplast membranes, with the glucosylceramides constituting the largest fraction. In addition, difficulties in extraction and analysis may have led to underestimates of sphingolipid contents. As in animals (Chapter 14), sphingolipid biosynthesis begins with condensation of palmitoyl-CoA with serine to form 3-keto-sphinganine, with the enzyme from *A. thaliana* showing 68% similarity to the human homologue (Tamura,

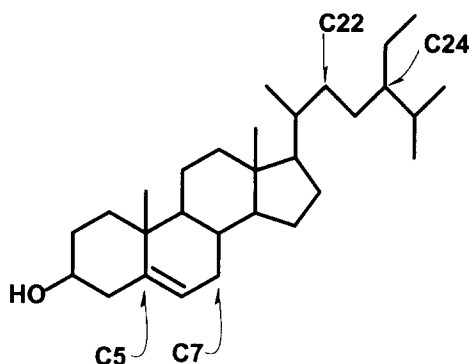


Fig. 6. Sitosterol (24 α -ethylcholesterol), shown here, is the most common plant sterol, but plants generally contain complex mixtures of sterols. Other prominent phytosterols differ from sitosterol as follows. Campesterol, 24 α -methyl; stigmasterol, C22 double bond; dihydrospinasterol, move double bond from C5 to C7; spinasterol, move C5 double bond to C7, add C22 double bond; dihydrobrassicasterol, 24 β -methyl; brassicasterol, 24 β -methyl, add C22 double bond.

2000). Reduction of 3-keto-sphinganine by an NADPH-dependent reaction yields sphinganine, and cDNAs encoding enzymes for the subsequent C4-hydroxylation and Δ^8 -desaturation of sphinganine have recently been cloned [22]. Less information is available on the synthesis of ceramides and glucosylceramides, although there is evidence that either sterol glucoside or UDP-glucose could serve as the glucose donor for the latter [23]. The finding that sphingolipids serve as signal molecules in plants, particularly in the defensive apoptosis induced by some plant pathogens, has begun to stimulate interest in this relatively neglected area of plant lipid biochemistry.

10. Oxylipins as plant hormones

Jasmonate is one of several lipid-derived plant growth regulators referred to as oxylipins [24]. The structure and biosynthesis of jasmonate have intrigued plant biologists because of the parallels to some eicosanoids (Chapter 13), which are central to inflammatory responses and other physiological processes in mammals. In plants, jasmonate derives from α -linolenic, presumably released from membrane lipids by a phospholipase A. The linolenic acid is oxidized by lipoxygenase and the resulting products, 9-hydroperoxylinolenic acid or 13-hydroperoxylinolenic acid may be further metabolized by one of three routes to produce a wide variety of oxylipin (Fig. 7). The pathways by which 13-hydroperoxylinoleic acid may be metabolized include hydroperoxide lyase catalyzed scission of the *trans*-11,12-double bond to produce a C6 aldehyde, *cis*-3-hexenal, and the C12 compound, 12-oxo-*cis*-9-dodecenoic acid. The acid is subsequently metabolized to 12-oxo-*trans*-10-dodecenoic acid, the wound hormone traumatin. The enzyme hydroperoxide dehydratase (allene oxide synthase) catalyzes the dehydration of hydroperoxides to unstable allene oxides that readily decompose to form a 9,12-ketol or a 12,13-ketol. The allene oxide of 13-hydroperoxylinolenic acid may also be converted by allene oxide cyclase to 12-oxo-phytodienoic acid which can

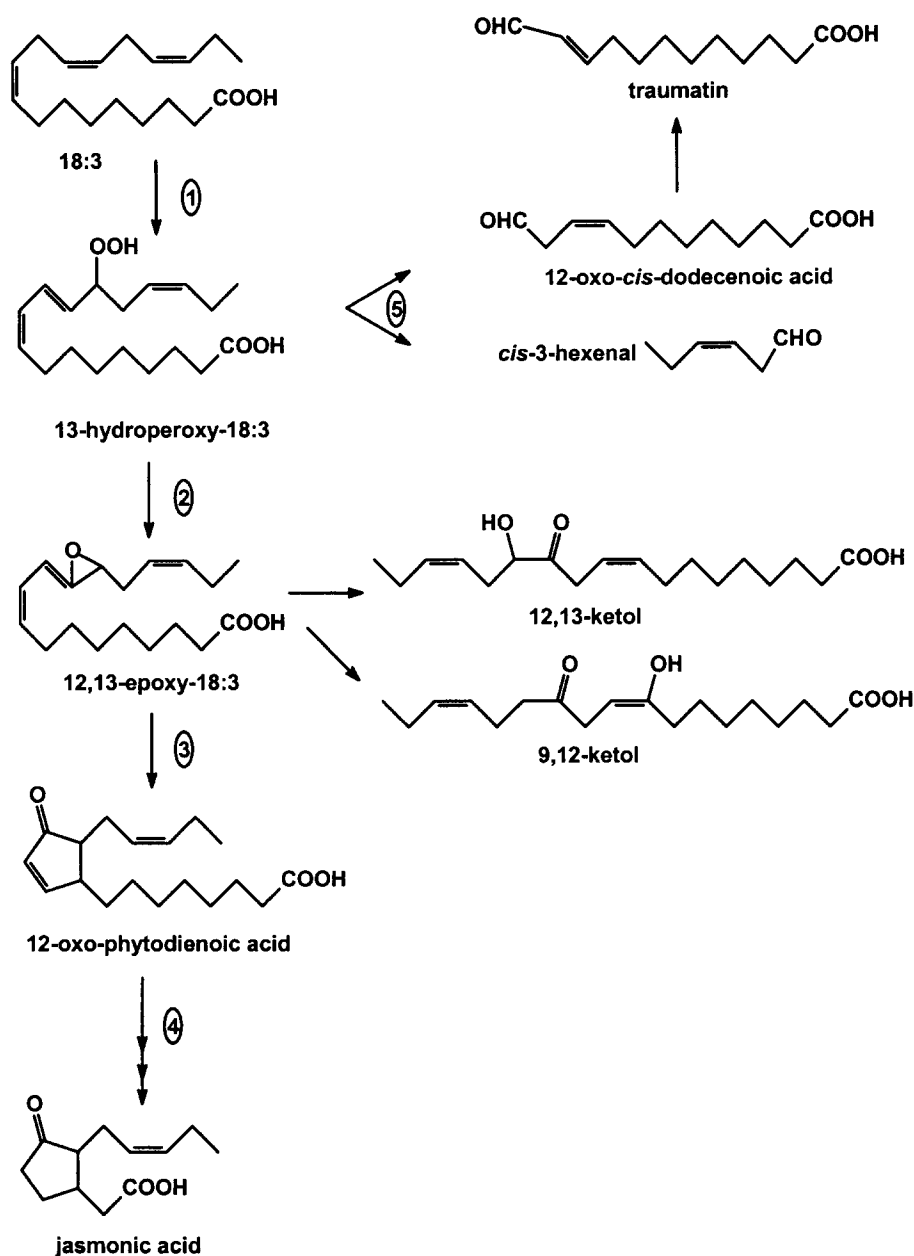


Fig. 7. Metabolism of 18:3 to oxylipins. 1, lipoxygenase; 2, allene oxide synthase; 3, allene oxide cyclase; 4, 12-oxo-phytodienoic acid reductase, β -oxidation; 5, hydroperoxide lyase.

be further metabolized to 7-iso-jasmonic acid. In the last few years, it has become clear that jasmonate is a key component of a wound-signaling pathway that allows plants to protect themselves against insect attack. When experimentally applied to

plants at low concentrations, jasmonate leads to the induction of protease inhibitors and other defense compounds. Furthermore, mutants of tomato and *Arabidopsis* that are deficient in jasmonate synthesis are much more susceptible to insect damage. In addition to jasmonate, a number of the other oxylipins have been reported to function as signal molecules. In particular, the oxylipin traumatin has been suggested to trigger cell division at the site of wounds, leading to the development of a protective callus. The lipoxygenase product 13-hydroxylinolenic acid triggers phytoalexin production. Similarly, C6–C10 alkenals act as volatile elicitors of a defense response in cotton.

11. Progress in plant lipid research: the value of mutants

Biochemical approaches toward understanding plant lipid biosynthesis and function provided much of the information summarized above. However, in recent years, the isolation of mutants in plant lipid metabolism has been extremely fruitful in providing new insights and new methods for gene isolation. Much of the progress in the genetic dissection of plant lipid metabolism has come from the extensive studies of Somerville, Browse and coworkers with *A. thaliana*, which has one of the smallest genomes (113 megabases) of higher plants. By using gas chromatography to screen several thousand randomly selected plants from a mutagenized population, Somerville and Browse were able to obtain an extensive collection of mutants showing altered leaf or seed fatty acid compositions. As described above, these mutants included isolates instrumental in confirming the relationships between prokaryotic and eukaryotic phosphatidic acid synthesis and in the analysis and cloning of membrane-bound desaturases.

11.1. Mutants in lipid metabolism have helped link lipid structure and function

Two other major benefits have been derived from the *Arabidopsis* lipid mutants. First, the physiological effects of the mutations have provided the opportunity to evaluate the relationships between lipid structure and function. There has been a long-term assumption, based on the strong association of high levels of polyunsaturated fatty acids with photosynthetic membranes and the conservation of this property among higher and lower plant species, that these fatty acids must be essential to photosynthesis. However, many attempts to understand the relationships between membrane fatty acid composition and cell physiology or photosynthesis have led to equivocal results. The isolation of mutants totally lacking certain unsaturated fatty acids has now provided much more convincing evaluations of their function and indeed, the results have forced re-evaluation of several previous hypotheses. For example, Δ^3 -*trans*-hexadecenoate is an unusual plant fatty acid which is associated with phosphatidylglycerol of chloroplast membranes, is evolutionarily conserved, and is synthesized in coordination with the assembly of the photosynthetic apparatus. These observations led to the suggestion that Δ^3 -*trans*-hexadecenoate is a highly essential component of photosynthesis. However, mutants which contain no detectable Δ^3 -*trans*-hexadecenoate grow as well as wild type plants, and all photosynthetic parameters examined appear normal (Browse, 1985). A minor difference in stability of some components of the photosystem can be detected by polyacrylamide

gel electrophoresis. It has been concluded from such analyses that, although Δ^3 -*trans*-hexadecenoate may facilitate assembly of the light harvesting complex into thylakoids, a more obvious phenotype could be restricted to certain unusual environmental conditions.

As mentioned above, a number of mutants blocked in the production of polyunsaturated fatty acid biosynthesis have also been isolated (Table 4). Because leaves have desaturases both in chloroplasts and in the endoplasmic reticulum, single mutations lead only to partial reduction of polyunsaturated fatty acid levels. Again, these mutants grow normally under most conditions and have normal photosynthetic parameters. However, several alterations in physiology are observed including changes in chloroplast ultrastructure, a reduction in the cross-sectional area of chloroplasts, and increased stability to thermal disruption of photosynthesis. Moreover, whereas wild-type *Arabidopsis* plants are chilling resistant and can reproduce normally at temperatures as low as 6°C, the mutants blocked in plastidial Δ^7 (*fad5*) and $\omega 6$ (*fad6*) desaturation become chlorotic at 6°C and show a 20–30% reduction in growth rate relative to the wild-type plant. The *fad2* mutants, in which the endoplasmic reticulum ω -6 desaturase is blocked, are even more sensitive to 6°C and die if left at this temperature for several days. These results demonstrate that polyunsaturated fatty acids are essential for maintaining cellular function and plant viability at low temperatures.

While most mutants which are reduced only in polyunsaturated fatty acid synthesis grow and develop normally at 22°C, a high-stearate mutant with 14% 18:0 is strikingly abnormal (Fig. 8) [26]. Many cell types fail to expand, resulting in mutant plants growing to less than one-tenth the size of wild-type. At higher growth temperatures (36°C), the effects are less dramatic, suggesting that the physical properties or fluidity of highly saturated membranes are less impaired under these conditions.

Other large scale alterations in membrane fatty acid composition and phenotypes have been obtained by creation of multiple-mutant lines. When mutants defective in endoplasmic reticulum $\omega 6$ -desaturase were crossed with plants defective in the plastid $\omega 6$ -desaturase, double mutants could be recovered only on sucrose-supplemented media. The sucrose grown plants, which contained less than 6% polyunsaturated fatty acids, were chlorotic and unable to carry out photosynthesis but otherwise remarkably normal. These results, while confirming the significance of polyunsaturated fatty acids to photosynthesis, indicate that the vast majority of membrane functions can proceed despite drastically reduced levels of polyunsaturates [27].

Triunsaturated fatty acids normally dominate chloroplast membranes and thus are the most abundant fatty acids on the planet. By constructing a triple mutant of *fad3*, *fad7* and *fad8*, it has been possible to eliminate triunsaturated fatty acids from *Arabidopsis* without affecting 16:2 and 18:2 production (McConn, 1996). Surprisingly, these plants are able to grow, photosynthesize, and even flower. However, they are male sterile and therefore cannot produce seeds. This observation led to the discovery of a very different role for jasmonate. This mutant cannot synthesize jasmonate because it lacks the 18:3 precursor, and the plants are male-sterile because pollen does not mature properly and is not released from the anthers. Application of jasmonate or linolenic acid to the anthers restores fertility, demonstrating that jasmonate is a key signal in pollen development. This result is a dramatic example of a change in fatty acid composition having a very specific effect on an essential developmental and tissue specific reproductive process.

Table 4
Biochemical and physiological responses of selected *Arabidopsis* lipid mutants.^a

Mutant	Enzyme blocked ^a	Fatty acid or lipid phenotype	Physiological response
<i>fab1</i>	3-Ketoacyl-ACP synthase II	16:0↑	Death of plants after prolonged exposure to 2°C
<i>fab2</i>	18:0-ACP Δ ⁹ -desaturase	18:0↑	Dwarf at 22°C
<i>fab4</i>	tΔ ³ , 16:0 desaturase?	tΔ ³ , 16:1↓	Altered stability of photosystem?
<i>fab5</i>	16:0 Δ ⁷ -desaturase	16:0↑; 16:3↓	Enhanced growth rate at high temperatures. Leaf chlorosis, reduced growth rate and impaired chloroplast development at low temperature.
<i>fab6</i>	Plastid ω-6 desaturase	16:1↑; 16:3↓	Leaf chlorosis, reduced growth rate and impaired chloroplast development at low temperature. Enhanced thermotolerance of photosynthetic electron transport at high temperatures.
<i>fab7</i>	Plastid ω-3 desaturase	16:2↑; 18:2↑; 16:3↓; 18:3↓	Reduced chloroplast size and altered chloroplast ultrastructure.
<i>fab2</i>	Cytosolic ω-6 desaturase	18:1↑; 18:2↓	Greatly reduced stem elongation at 12°C. Death at 6°C.
<i>fab2/fad6</i>	Plastid and cytosolic ω-6 desaturase	<6% Polyunsaturated	Loss of photosynthesis
<i>fab3/fad7/fad8</i>	Plastid and cytosolic ω-3 desaturase	<1% Trienoic	Male sterile, insect resistance decreased
<i>dgd1</i>	Digalactosyldiacylglycerol synthase	Digalactosyldiacylglycerol↓	Dwarfism, abnormal chloroplast size
<i>act1</i>	Plastid acyl-ACP: G3P acyltransferase	Phosphatidylglycerol↓, 16:3↓	Altered chloroplast structure
<i>mgd1</i>	Diacylglycerol glycosyltransferase	Monogalactosyldiacylglycerol↓	Abnormal chloroplast development
<i>AS11</i>	Diacylglycerol acyltransferase	50% Reduction in seed TAG	Slow germination
<i>writ1</i>	Unknown; glycolysis impaired	80% Reduction in seed TAG	Slow germination

^a In some cases, the enzyme defect in the mutation is not known and this table lists the most likely possibility.

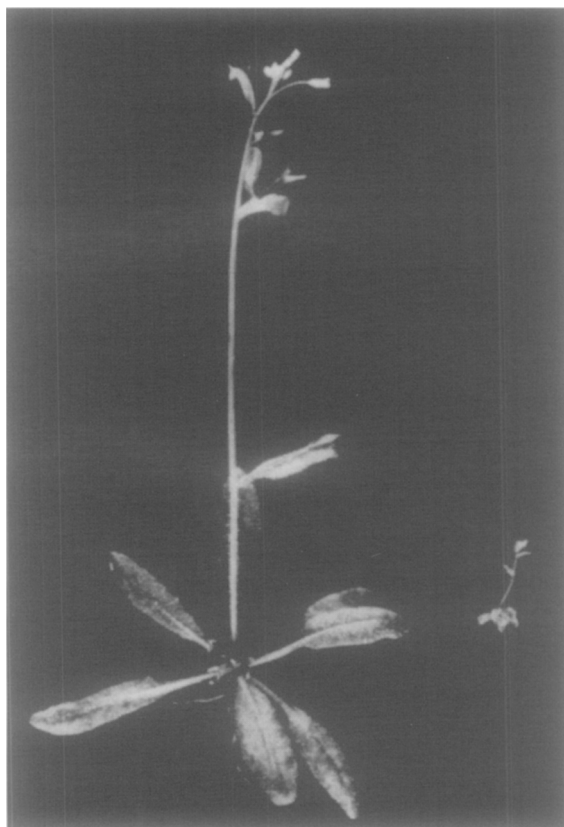


Fig. 8. Increased stearic acid causes severe dwarfing of *Arabidopsis*. A wild-type *Arabidopsis* plant (left) is compared to the *fab2* mutant (right) in which leaf stearic acid content has increased from 1 to 14%. Photo courtesy of John Browse, Washington State University.

11.2. *Arabidopsis* mutants have allowed cloning of desaturases and elongases

A. thaliana mutants have also provided a means of cloning genes difficult to isolate by other methods. As in other kingdoms, the membrane bound enzymes of plants have been notoriously difficult to purify and characterize. However, cDNAs or genes encoding a number of these enzymes have now been isolated using molecular genetic strategies based on mutations. Several approaches have been successful. A cDNA encoding the ω -3 desaturase which converts linoleic to linolenic acid was isolated in 1992 by Arondel et al. [28] after a mutation leading to the loss of function was genetically mapped and a yeast artificial clone corresponding to this region was selected. The genomic region was then used to screen a cDNA library, and some of the clones which hybridized to the yeast artificial chromosome had sequence similarity to cyanobacterial desaturases. These clones subsequently were shown to complement the loss of 18:3.

Gene 'tagging' strategies have also proved enormously valuable in identifying clones of membrane bound enzymes. *Arabidopsis* genes can be 'tagged' by insertional

inactivation when T-DNA from *Agrobacterium tumefaciens* inserts randomly in the genome. When a promising phenotype is observed, the inactivated gene can be identified by probing with the T-DNA sequence. This method was used to identify the ω -6 desaturase required for the 18:1 to 18:2 conversion in endoplasmic reticulum [25]. In addition, transposon tagging led to the cloning of a gene which controls elongation of oleic acid to 20:1 and 22:1 in developing *Arabidopsis* seeds [29]. Since no membrane-bound fatty acid elongase had ever been completely purified or cloned from a eukaryotic organism, the finding that this gene encodes a 60-kDa condensing enzyme provided the first direct evidence that membrane fatty acid elongation is not catalyzed by a type I multifunctional fatty acid synthase.

12. Design of new plant oils

In recent years, tremendous progress has occurred not only in the isolation of many plant lipid biosynthetic genes, but also in the use of these genes to manipulate plant oil composition. As shown in Table 5, both substantial changes in seed oil composition and introduction of unusual fatty acids to heterologous species have been achieved. Progress in this area has been accelerated by several industrial laboratories whose goal has been the production of higher value oilseeds.

Table 5
Some examples of genetic engineering of plant lipid metabolism

Modification achieved	Enzyme engineered	Source of gene	Plant transformed
Lauric acid production	Acyl-ACP thioesterase	California Bay	Rapeseed
Increased stearic acid	Antisense of stearoyl-ACP desaturase	Rapeseed	Rapeseed
Reduced saturated fatty acids	Stearoyl-CoA desaturase	Rat, yeast	Tobacco
Reduced 18:3	Acyl-ACP thioesterase	Soybean	Soybean
Altered lauric acid distribution in TAG	ω -3 Desaturase	Soybean, canola	Soybean, rapeseed
Altered cold tolerance	1-Acyl-glycerol-3-phosphate acyltransferase	Coconut	Rapeseed
Petroselinic acid production	Acyl-ACP: glycerol-3-phosphate acyltransferase	<i>E. coli</i> , squash, <i>Arabidopsis</i>	Tobacco, <i>Arabidopsis</i>
Increased linolenic acid	Acyl-ACP desaturase	Coriander	Tobacco
Cyclopropane fatty acid production	ω -3 Desaturase	<i>Arabidopsis</i>	<i>Arabidopsis</i>
γ -Linolenic acid production	Cyclopropane synthase	<i>E. coli</i> , <i>Sterculia</i>	Tobacco
Increased long-chain fatty acids	Linolenic Δ^6 -desaturase	<i>Synechocystis</i>	Tobacco
Wax ester synthesis	Fatty acid elongase	Jojoba	Rapeseed
	Wax ester synthase, fatty acid reductase, fatty acid elongase	Jojoba	<i>Arabidopsis</i>

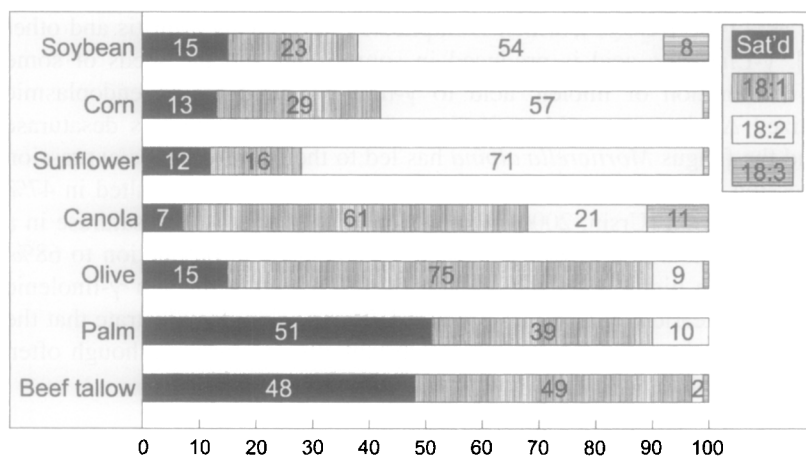


Fig. 9. Fatty acid composition of dietary vegetable oils and beef tallow. The values shown represent typical compositions of varieties grown commercially. Lines modified substantially through breeding or genetic engineering are available for soybean, canola, corn and sunflower.

12.1. Design of new edible oils

12.1.1. Improvements in nutritional value and stability of vegetable oils

Vegetable oils have gradually replaced animal fats as the major source of lipids in human diets and now constitute 15–25% of total caloric intake by industrialized nations. As shown in Fig. 9, vegetable oils display a wide range in the relative proportions of saturated and unsaturated fatty acid acids although in the United States, up to 80% of vegetable oil consumed is soybean oil making it the single largest source of calories in American diets. Most nutritionists recommend a reduction in saturated fat content in diets, and genetic engineering of plant oils can substantially help achieve this goal. Most of the saturated fatty acid in common plant oils is palmitic acid, and its occurrence is largely related to the action of a palmitoyl-ACP thioesterase. Reduction of the expression of this activity in transgenic soybean using co-suppression has led to decreases of the palmitic acid content from 15 to 6%.

A very high stability liquid soybean oil with low saturated fatty acids was also produced in soybean by suppression of the oleoyl-desaturase (Kinney, 1996). Oleic acid content was increased up to 89%, 18:2 content reduced from 57% to less than 3%, and saturated fatty acids reduced to less than 7%. This oil has been produced commercially and is extremely stable for high-temperature frying applications. In addition, its stability matches that of mineral oil derived lubricants, and therefore such non-food uses as biodegradable lubricants are underway. One added consumer benefit to wide future use of the engineered high-oleic acid oils may be reduction in the pathologies associated with high ω -6 consumption.

Besides reducing the amounts of *trans* and ω -6 unsaturated fatty acids in vegetable oils, efforts directed at increasing the amount of health beneficial fatty acids in vegetable oils appear to be promising. Two potential health-promoting fatty acids are stearidonic, (18:4- $\Delta^{6,9,12,15}$), an ω -3 fatty acid precursor of the ω -3 structures found in fish oils,

and γ -linolenic acid ($18:3-\Delta^{6,9,12}$), which is implicated in relieving arthritis and other conditions [30]. γ -Linolenic acid is produced in some fungi and the seeds of some plants by the desaturation of linoleic acid to γ -linolenic acid by an endoplasmic reticulum-localized Δ^6 -desaturase. Identification of cDNAs encoding this desaturase from borage and the fungus *Mortierella alpina* has led to their heterologous expression in plants. Expression of the *Mortierella* gene in *Brassica napus* seed resulted in 47% γ -linolenic acid production (Ursin, 2000). Expression of the fungal Δ^6 -desaturase in a low α -linolenic acid canola line further enhanced γ -linolenic acid production to 68%. When canola lines with high α -linolenic content were crossed with high γ -linolenic acid lines, up to 17% stearidonic acid was produced. These examples illustrate that the level of desired end-product can sometimes be substantially increased, although often unpredictably, by the choice of host and/or enzyme source.

12.1.2. Alternatives to hydrogenated margarines and shortenings

Since most vegetable oils are liquids at room temperature, the production of margarines and shortenings from such oils requires alteration of their physical properties. This is most frequently achieved by catalytic hydrogenation of the oil, a process which reduces the double bonds and thereby increases the melting point of the oil. However, hydrogenation substantially increases the saturated fat content of the oil. An additional side reaction which occurs during hydrogenation is the conversion of much of the naturally occurring *cis* double bonds to the *trans* configuration. Typically, hydrogenated oils used in margarine or shortening contain up to 25–40% saturated or *trans* fatty acids. Although convincing evidence for a deleterious effect of *trans*-isomers in the diet is lacking, some nutritionists consider reduction of *trans*-double bonds in the diet advantageous. An added disadvantage to hydrogenation is the 2–3 cents per pound cost which it adds to the price of the oil. Thus, for several reasons, an alternative to vegetable oil hydrogenation is desirable for manufacture of margarines and shortenings.

Further progress toward reducing the need for hydrogenation has been made using a molecular genetic approach to increase the melting point of rapeseed oil. As described in Section 3.2, the introduction of the first double bond into plant fatty acids occurs by the action of stearoyl-ACP desaturase. An obvious route to alter the activity of this enzyme in oilseeds is the use of antisense RNA. This objective was achieved in *Brassica napus* and other plants, by suppression of stearoyl-ACP desaturase messenger RNA, which reduced enzyme activity and desaturase protein to barely detectable levels (Knutzon, 1992). As a result, the content of stearic acid in the seed oil was increased from 2 to 40%. Due to its high saturated fatty acid content, the oil from these plants has a high melting point and may be directly suitable for margarine or shortening manufacture without added hydrogenation.

An alternative fatty acid modification that might simultaneously increase unsaturation in diets and reduce the need for hydrogenation is the production of petroselinic acid rich vegetable oils. Petroselinic acid is an isomer of oleic acid which has a *cis* double bond at the 6th carbon from the carboxyl end of the molecule rather than the 9th carbon. This minor modification of the structure alters the melting point of the fatty acid such that petroselinic acid melts at 33°C whereas oleic acid melts at 5°C. Thus petroselinic acid might provide the means to produce an unsaturated vegetable oil which

is also a solid at room temperatures and therefore ideal for margarine and shortening manufacture. Although petroselinic acid is a major component of seed oils in species such as coriander and carrot, these crops have a low yield of oil per hectare. It is therefore hoped that an oilseed crop can be engineered to become a commercially viable source of petroselinic acid. When a cDNA encoding the acyl-ACP desaturase involved in petroselinic acid synthesis was cloned from coriander and used to transform other plants, petroselinic acid was produced but only at low levels. Therefore, to achieve economic production of petroselinic acid, it may be necessary to introduce additional genes. It is now known that coriander and its relatives possess several proteins devoted to production of this special fatty acid, including ACP isoforms, a modified condensing enzyme (3-ketoacyl-ACP synthase) specific for the elongation of *cis*-4-16:1-ACP to *cis*-6-18:1-ACP and a novel acyl-ACP thioesterase specific for petroselinoyl-ACP.

12.2. Design of new industrial oils

As described above, plants have evolved the ability to produce a diverse range of fatty acid structures. A number of specialty fatty acids have already been extensively exploited for industrial uses such as lubricants, plasticizers and surfactants (Table 3). In fact, approximately one-third of all vegetable oil is now used for non-food products, and this figure is expected to increase as petroleum reserves are depleted. Thus, in addition to providing food, oilseed crops can be considered efficient, low polluting chemical factories which are able to harness energy from sunlight and transform it into a variety of valuable chemical structures with a multitude of non-food uses.

12.2.1. High laurate and caprate oils

One of the major non-food uses of vegetable oils, consuming approximately 500 million pounds of oil per annum in the US, is the production of soaps, detergents and other surfactants. The solubility and other physical properties of medium-chain fatty acids and their derivatives make them especially suited for surfactant manufacture. Coconut and palm kernel oils, which contain 40–60% lauric acid (12:0), are the current major feedstocks for the surfactant industry. The mechanism for synthesis of lauric and other medium-chain fatty acids in plants involves the action of a medium-chain acyl-ACP thioesterase which terminates fatty acid synthesis after a 10 or 12 carbon chain has been assembled (Pollard, 1991). A cDNA encoding such a thioesterase was isolated from seeds of the California Bay tree and transformed into rapeseed. As shown in Fig. 10, the introduction of this specialized thioesterase resulted in transgenic seeds that produce up to 60 mol% lauric acid. The plants grow normally and oil yields are very similar to those of the untransformed cultivars. Commercial production of high lauric rapeseed oil began in 1995, and this crop has the potential to provide a new, non-tropical source of lauric oils for the surfactant industry.

Since current oil crops do not produce significant amounts of caprate (10:0) and caprylate (8:0), the cloning of thioesterases from plants accumulating these species raised hopes for development of commercial sources of 8:0 and 10:0. Surprisingly, medium-chain thioesterase from elm, a 10:0 accumulator, recognized either 10:0 or 16:0, and a thioesterase from a C_{8–10}-rich *Cuphea* species, although showing strong

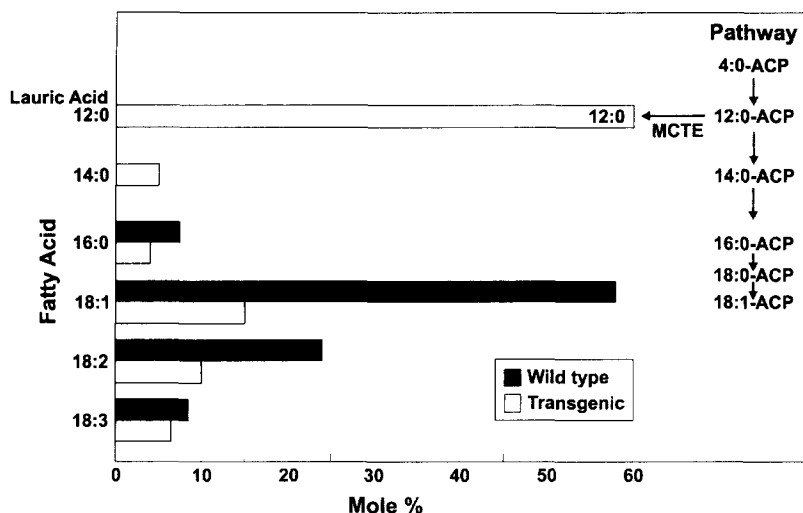


Fig. 10. Genetic engineering of rapeseed oil to produce lauric acid. mol% of major fatty acids in a typical canola cultivar are compared to the composition achieved through genetic engineering using a California Bay medium chain acyl-ACP thioesterase (MCTE) under control of a napin promoter.

selectivity for medium-chain fatty acids, gave disappointing results in transformed plants. Part of this discrepancy was resolved with the discovery that medium-chain production in *Cuphea* also involves a specific keto-acyl synthase. Cotransformation with the medium-chain thioesterase and the keto-acyl synthase IV give plants with substantial caprate and caprylate [31], and further improvements may be possible with addition of acyltransferases better able to introduce 8:0 and 10:0 to underrepresented positions in triacylglycerol [32].

12.2.2. Production of waxes

Long-chain wax esters were once harvested from sperm whales and were a major ingredient of industrial lubricants and transmission fluids. Banning of whale harvests led to searches for alternative biological sources of such structures. Jojoba, a desert shrub found in the American southwest, is the only plant species known to accumulate waxes (up to 60% dry weight) rather than TAG as a seed storage reserve. These waxes are mostly derived from C20–C24 monounsaturated fatty acids and alcohols and are synthesized by the elongation of oleate followed by reduction to alcohols by a fatty acid reductase. The wax storage lipid is formed by a fatty acyl-CoA: fatty alcohol acyltransferase, also referred to as wax synthase. The reductase and acyltransferase were purified from jojoba and the cognate cDNAs cloned. Coordinated expression of three genes, a *Lunaria annua* long-chain acyl-CoA elongase, and the jojoba reductase and acyltransferase in *Arabidopsis*, resulted in wax production at up to 70% of the oil present in mature seeds (Lardizabal, 2000). The high levels of accumulation indicated that all the genes necessary for this trait were identified. If this trait can be successfully transferred to commercial crops this would represent a large potential source of waxes used for a variety of applications including cosmetics and industrial lubricants.

12.2.3. Other industrial oils

Many of the unusual fatty acids produced by plants would have substantial value as industrial feedstocks if they were available in sufficient quantity at low prices. Examples in this category include fatty acids with hydroxy, epoxy, cyclopropane or branched chains. These specialty fatty acids are often produced in wild species which have not been optimized for high agronomic and oil yields, and therefore such specialty oils are expensive to produce. An alternative to the long-term effort required for domestication of such plants is the introduction of genes relevant to unusual fatty acid production into existing high-yielding oil crops. A step in this direction was made by the isolation of a cDNA for a fatty acid hydroxylase from the castor oil plant. When this gene was introduced into transgenic plants, approximately 20% hydroxy fatty acids were produced (Broun, 1997). Genes for specialty fatty acid production need not be isolated only from plants. As mentioned above, the stearyl-CoA desaturases from animals and yeast are active in plants. Furthermore, the cyclopropane synthase of *E. coli* (Schmid, 1995) and desaturases from cyanobacteria and fungi have been successfully expressed in transgenic plants. Thus, in principle there are no fundamental barriers to producing a wide range of oil compositions using genes borrowed from diverse organisms. Furthermore, protein engineering offers even more possibilities to tailor the substrates and products of plant enzymes to produce 'designer oil crops' for specific end uses [6].

13. Future prospects

It is now possible to say that the biosynthetic pathways have been determined for all major plant lipids and most of the genes identified for enzymes in these pathways have been cloned. Clearly there has been great progress, although the biosynthetic pathways of sphingolipids, *trans*- Δ^3 -16:1 and some details of several pathways remain elusive. In addition, the enzymes involved in the production of many unusual fatty acids found in seed oils are largely unexplored.

One area of expanding interest is the production of lipid hormones and signal molecules. Several lipids including phosphatidylinositol phosphates, diacylglycerol and *N*-acylphosphatidylethanolamine have been implicated in signal transduction in plants. Another intriguing similarity between plants and animals is their use of oxygenated fatty acids in response to wounding. As mentioned in Section 10, jasmonate, a plant growth regulator derived from 18:3, is able at femtomolar concentrations to induce proteinase inhibitors and other plant defense genes. Like leukotriene synthesis in animals, jasmonate biosynthesis begins with the generation of a hydroperoxide by lipoxygenase. Jasmonate itself contains a cyclopentane ring comparable to those of prostaglandins. The common roles and origins of oxygenated fatty acids in plants and animals suggest a very early common ancestor for these pathways.

Application of molecular genetics and genomics to problems in lipid biochemistry should continue to expand. Particularly stimulating has been the complete sequencing of the *Arabidopsis* genome and the availability of over 1 million expressed sequence tag (EST) sequences from a variety of plants. A recent survey of *Arabidopsis* genes involved in plant acyl lipid metabolism identified over 450 genes

Table 6
Internet resources related to plant lipid metabolism

Organization	Web site content	URL
National Plant Lipid Consortium (NPLC)	Directory of scientists involved in plant lipid research. E-mail newsgroup for information on plant lipids. Abstracts of NPLC meetings	http://www.msu.edu/user/ohlrogge/
Michigan State University	Survey and catalog of genes for plant lipid metabolism. Gene expression profiles based on plant lipid ESTs and microarrays	http://www.canr.msu.edu/lgc/index.html http://www.plantbiology.msu.edu/gene_survey/front_page.htm http://www.bpp.msu.edu/Seed/SeedArray.htm
Kathy Schmid, Butler University	Links to many oilseed research laboratories and web sites	http://blue.butler.edu/~kschmid/lipids.html
Benning and Ohlrogge Laboratories	Database and analysis of > 10,000 ESTs from developing <i>Arabidopsis</i> seeds	http://benningnt.bch.msu.edu
USDA Oilseed Database	Data on fatty acid composition of seeds of thousands of species	www.ncaur.usda.gov/nc/ncdb/search.html-ssi

(http://www.plantbiology.msu.edu/gene_survey/front_page). Although most of these could be assigned a tentative function based on sequence similarity to previously characterized genes, only a handful have been examined individually at an experimental level, and therefore the precise function of hundreds of genes awaits further work. For example, there are 8 genes which are similar to acyl-CoA desaturases whose function has not yet been identified and there are 19 genes for lipid transfer proteins and 51 additional lipid transfer protein-like genes. The physiological reasons underlying the existence of large gene families for lipid transfer proteins, plastid ACPs, and stearoyl desaturases, but only one gene for keto acyl-ACP synthase III and for most fatty acid synthase and ACC subunits, etc. remain to be elucidated. The availability of large collections of T-DNA insertion mutants means that the impact of gene disruptions can be tested. However, because 60% of *Arabidopsis* genes are present as duplicates, such gene disruptions must be supplemented by other strategies of functional genomics. Some websites related to these efforts in the plant lipid field are presented in Table 6.

Much of past lipid research has focused on a reductionist approach in which cells are taken apart and their pieces analyzed. The overall success of this approach and the wealth of new clones and sequence information have given us an unprecedented knowledge of the pieces of the puzzle which represent lipid metabolism. However, as in any puzzle, it is not just complete knowledge of the pieces, but an understanding

of how (and when) they fit together that defines the challenge. Microarrays that permit simultaneous monitoring of expression of many genes have begun to provide a more global overview of how genes work together to control seed metabolism (Girke, 2000). Together with the ability to rapidly over- and under-express genes in transgenic plants and the strengths of classical biochemistry, such recent advances in analytical techniques should allow us to enter a new stage of lipid research emphasizing the interplay between metabolic compartments and the control of lipid synthesis during the plant life cycle.

Abbreviations

ACC	acetyl-CoA carboxylase
ACP	acyl carrier protein
EST	expressed sequence tag
CoA	coenzyme A
TAG	triacylglycerol

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Oxidation of fatty acids in eukaryotes

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1. The pathway of β -oxidation: a historical account

Fatty acids are a major source of energy in animals. The study of their biological degradation began in 1904 when Knoop [1] performed the classical experiments that led him to formulate the theory of β -oxidation. In his experiments, Knoop used fatty acids with phenyl residues in place of the terminal methyl groups. The phenyl residue served as a *reporter group* because it was not metabolized, but instead was excreted in the urine. When Knoop fed phenyl substituted fatty acids with an odd number of carbon atoms, like phenylpropionic acid ($\text{C}_6\text{H}_5\text{—CH}_2\text{—CH}_2\text{—COOH}$) or phenylvaleric acid ($\text{C}_6\text{H}_5\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—COOH}$), to dogs, he isolated from their urine hippuric acid ($\text{C}_6\text{H}_5\text{—CO—NH—CH}_2\text{—COOH}$), the conjugate of benzoic acid and glycine. In contrast, phenyl-substituted fatty acids with an even number of carbon atoms, such as phenylbutyric acid ($\text{C}_6\text{H}_5\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—COOH}$), were degraded to phenylacetic acid ($\text{C}_6\text{H}_5\text{—CH}_2\text{—COOH}$) and excreted as phenylaceturic acid ($\text{C}_6\text{H}_5\text{—CH}_2\text{—CO—NH—CH}_2\text{—COOH}$). These observations led Knoop to propose that the oxidation of fatty acids begins at carbon atom 3, the β -carbon, and that the resulting β -keto acids are cleaved between the α -carbon and β -carbon to yield fatty acids shortened by two carbon atoms. Knoop's experiments prompted the idea that fatty acids are degraded in a stepwise manner by successive β -oxidation. In the years following Knoop's initial study, Dakin [2] performed similar experiments with phenylpropionic acid. Besides hippuric acid he isolated the glycine conjugates of the following β -oxidation intermediates: phenylacrylic acid ($\text{C}_6\text{H}_5\text{—CH=CH—COOH}$), β -phenyl- β -hydroxypropionic acid ($\text{C}_6\text{H}_5\text{—CHOH—CH}_2\text{—COOH}$), and benzoylacetic acid ($\text{C}_6\text{H}_5\text{—CO—CH}_2\text{—COOH}$). At the same time, Embden and coworkers demonstrated that in perfused livers unsubstituted fatty acids are degraded by β -oxidation and converted to ketone bodies. Thus, by 1910 the basic information necessary for formulating the pathway of β -oxidation was available.

After a 30-year period of little progress, Munoz and Leloir in 1943, and Lehninger in 1944, demonstrated the oxidation of fatty acids in cell-free preparations from liver. Their work set the stage for the complete elucidation of β -oxidation. Detailed investigations with cell-free systems, especially the studies of Lehninger, demonstrated the need for energy to 'spark' the oxidation of fatty acids. ATP was shown to meet this requirement and to be essential for the activation of fatty acids. Activated fatty acids were shown by Wakil and Mahler, as well as by Kornberg and Pricer, to be thioesters formed from fatty acids and coenzyme A. This advance was made possible by earlier studies of Lipmann and coworkers who isolated and characterized coenzyme A, and Lynen [3] and

coworkers who proved the structure of 'active acetate' to be acetyl-CoA. Acetyl-CoA was found to be identical with the two-carbon fragment removed from fatty acids during their degradation. The subcellular location of the β -oxidation system was finally established by Kennedy and Lehninger, who demonstrated that mitochondria were the cellular components most active in fatty acid oxidation. The mitochondrial location of this pathway agreed with the observed coupling of fatty acid oxidation to the citric acid cycle and to oxidative phosphorylation. The most direct evidence for the proposed β -oxidation cycle emerged from enzyme studies carried out in the fifties primarily in the laboratories of Green in Wisconsin, Lynen in Munich, and Ochoa in New York. Their studies were greatly facilitated by newly developed methods of protein purification and by the use of spectrophotometric enzyme assays with chemically synthesized intermediates of β -oxidation as substrates.

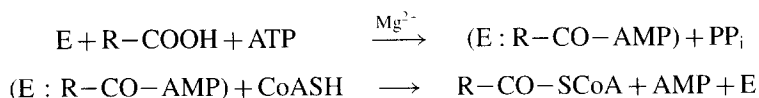
2. Uptake and activation of fatty acids in animal cells

Fatty acids are transported between organs either as unesterified fatty acids complexed to serum albumin or in the form of triacylglycerols associated with lipoproteins. Triacylglycerols are hydrolyzed outside of cells by lipoprotein lipase to yield free fatty acids. The mechanism by which free fatty acids enter cells remains poorly understood despite a number of studies performed with isolated cells from heart, liver, and adipose tissue [4]. Kinetic evidence has been obtained for both a saturable and non-saturable uptake of fatty acids. The saturable uptake, which predominates at nanomolar concentrations of free fatty acids, is presumed to be carrier-mediated, whereas the non-saturable uptake, which is significant only at higher concentrations of free fatty acids, has been attributed to non-specific diffusion of fatty acids across the membrane. Several suspected fatty acid transport proteins have been identified [5]. However, their specific function(s) in fatty acid uptake and their molecular mechanisms remain to be elucidated.

Once long-chain fatty acids have crossed the plasma membrane, they either diffuse or are transported to mitochondria, peroxisomes, and the endoplasmic reticulum where they are activated by conversion to their CoA thioesters. Whether this transfer of fatty acids between membranes is a facilitated process or occurs by simple diffusion is an unresolved issue. The identification of low-molecular-weight (14–15 kDa) fatty acid binding proteins (FABPs) in the cytosol of various animal tissues prompted the suggestion that these proteins may function as carriers of fatty acids in the cytosolic compartment [6]. FABPs may also be involved in the cellular uptake of fatty acids, their intracellular storage, or the delivery of fatty acids to sites of their utilization. The importance of FABPs in fatty acid metabolism is supported by the observation that the uptake and utilization of long-chain fatty acids are reduced in knock-out mice lacking heart FABP. These animals exhibit exercise intolerance and, at old age, develop cardiac hypertrophy. Nonetheless, the molecular mechanism of FABP function remains to be elucidated.

The metabolism of fatty acids requires their prior activation by conversion to fatty acyl-CoA thioesters. The activating enzymes are ATP-dependent acyl-CoA synthetases,

which catalyze the formation of acyl-CoA by the following two-step mechanism in which E represents the enzyme:



The evidence for this mechanism was primarily derived from a study of acetyl-CoA synthetase. Although the postulated intermediate, acetyl-AMP, does not accumulate in solution, and therefore only exists bound to the enzyme, the indirect evidence for this intermediate is very compelling. Other fatty acids are assumed to be activated by a similar mechanism, even though less evidence in support of this hypothesis has been obtained. The activation of fatty acids is catalyzed by a group of acyl-CoA synthetases that differ with respect to their subcellular locations and their specificities for fatty acids of different chain lengths [7]. Their chain-length specificities are the basis for classifying these enzymes as short-chain, medium-chain, long-chain and very-long-chain acyl-CoA synthetases.

A short-chain-specific acetyl-CoA synthetase that is present in mammalian mitochondria has been purified and its cDNA has been cloned. This 71-kDa enzyme, which is most active with acetate as a substrate but exhibits some activity towards propionate, has been detected in mitochondria of heart, skeletal muscle, kidney, adipose tissue and intestine, but not in those of liver. A cytosolic 78-kDa acetyl-CoA synthetase has been identified in liver, intestine, adipose tissue and mammary gland, all of which have high lipogenic activities. Expression studies support the hypothesis that the cytosolic enzyme synthesizes acetyl-CoA for lipogenesis, whereas the mitochondrial acetyl-CoA synthetase activates acetate headed for oxidation.

Medium-chain acyl-CoA synthetases are present in mitochondria of various mammalian tissues. The partially purified enzyme from beef heart mitochondria acts on fatty acids with 3–7 carbon atoms, but is most active with butyrate. In contrast, the 66-kDa enzyme purified from bovine liver mitochondria activates fatty acids with 3–12 carbon atoms with hexanoate being the best substrate. This enzyme also activates aromatic carboxylic acids like benzoic acid and its substituted derivatives. Overall, liver mitochondria, in contrast to heart mitochondria, contain medium-chain acyl-CoA synthetase activities with much broader substrate specificities toward fatty acids of varying chain lengths and structures.

Long-chain acyl-CoA synthetase is a membrane-bound enzyme that is associated with the endoplasmic reticulum, peroxisomes, and the outer mitochondrial membrane. The enzyme acts efficiently on saturated fatty acids with 10–20 carbon atoms and on common unsaturated fatty acids having 16–20 carbon atoms. Molecular cloning and expression studies of long-chain acyl-CoA synthetase have revealed the existence of five different isozymes (ACS 1–5) in the rat. A detailed investigation of the hepatic acyl-CoA synthetases ACS 1, 4 and 5 suggests that ACS 1 and 4, which are present in the endoplasmic reticulum and related subcellular structures, function in lipid biosynthesis while ACS 5 with a mitochondrial localization may catalyze the activation of fatty acids for β -oxidation. Such functional commitment of isozymes has previously been recognized. For example, long-chain acyl-CoA synthetases ACS I and ACS II of

Candida lipolytica are thought to activate long-chain fatty acids for complex lipid synthesis and peroxisomal β -oxidation, respectively.

Very long-chain acyl-CoA synthetase activates fatty acids with 22 or more carbon atoms and also acts on long-chain and branched-chain fatty acids. This membrane-bound enzyme is strongly expressed in liver where it is associated with the endoplasmic reticulum and peroxisomes but not with mitochondria. Purification of the 70-kDa very long-chain acyl-CoA synthetase enabled the cloning of the rat and human cDNAs coding for this enzyme. Sequence homologies with other cDNAs resulted in the identification of choloyl-CoA synthetase and in the demonstration that fatty acid transport protein 1 (FATP 1), a suggested transporter of fatty acids across the plasma membrane, exhibits very long-chain acyl-CoA synthetase activity.

In addition to ATP-dependent acyl-CoA synthetases, several GTP-dependent acyl-CoA synthetases have been described. The best known of these enzymes is succinyl-CoA synthetase, which cleaves GTP to GDP plus phosphate and functions in the tricarboxylic acid cycle. Although a mitochondrial GTP-dependent acyl-CoA synthetase activity was described, the existence of a distinct enzyme with such activity has been questioned.

3. Fatty acid oxidation in mitochondria

In animal cells, fatty acids are degraded in both mitochondria and peroxisomes, whereas in lower eukaryotes β -oxidation is confined to peroxisomes. Mitochondrial β -oxidation provides energy for oxidative phosphorylation and generates acetyl-CoA for ketogenesis in liver. The oxidation of fatty acids with odd numbers of carbon atoms proceeds by β -oxidation and yields, in addition to acetyl-CoA, 1 mole of propionyl-CoA per mole of fatty acid. Propionyl-CoA is further metabolized to succinate.

3.1. Mitochondrial uptake of fatty acids

Fatty acyl-CoA thioesters that are formed at the outer mitochondrial membrane cannot directly enter the mitochondrial matrix, where the enzymes of β -oxidation are located, because the inner mitochondrial membrane is impermeable to CoA and its derivatives. Instead, carnitine carries the acyl residues of acyl-CoA thioesters across the inner mitochondrial membrane. The carnitine-dependent translocation of fatty acids across the inner mitochondrial membrane is schematically shown in Fig. 1 [8]. The reversible transfer of fatty acyl residues from CoA to carnitine is catalyzed by carnitine palmitoyltransferase I (CPT I), which is an enzyme of the outer mitochondrial membrane. The resultant acylcarnitines cross the inner mitochondrial membrane via the carnitine:acylcarnitine translocase. This carrier protein catalyzes a rapid mole for mole exchange of acylcarnitine for carnitine, carnitine for carnitine, and acylcarnitine for acylcarnitine. This exchange, especially of acylcarnitine for carnitine, is essential for the translocation of long chain fatty acids from the cytosol into mitochondria. In addition, the translocase facilitates a slow unidirectional flux of carnitine across the inner mitochondrial membrane. This unidirectional flux of carnitine may be important

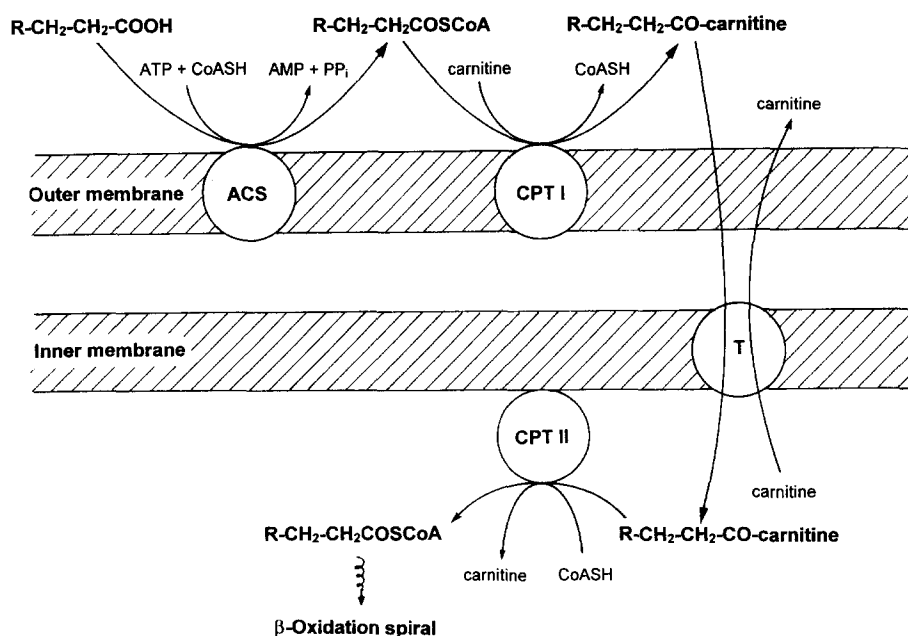


Fig. 1. Carnitine-dependent transfer of acyl groups across the inner mitochondrial membrane. ACS, acyl-CoA synthetase; CPT I and CPT II, carnitine palmitoyltransferase I and II, respectively; T, carnitine : acylcarnitine translocase.

for mitochondria of organs other than liver to acquire carnitine, which is synthesized in the liver. The rat liver translocase, which has a subunit molecular mass of 32.5 kDa, has been purified and its cDNA has been cloned. The protein has also been functionally reconstituted into proteoliposomes. In the mitochondrial matrix, carnitine palmitoyltransferase II (CPT II), an enzyme that is associated with the inner mitochondrial membrane, catalyzes the transfer of acyl residues from carnitine to CoA to form acyl-CoA thioesters that then enter the β -oxidation spiral. CPT II has been purified from mitochondria of bovine heart and rat liver. The purified enzyme has a subunit molecular mass of approximately 70 kDa and catalyzes the reversible transfer of acyl residues with 10–18 carbon atoms between CoA and carnitine. The cDNAs of rat and human CPT II have been cloned and sequenced. The predicted amino acid sequences of the corresponding proteins show a greater than 80% homology with each other. CPT I, in contrast to CPT II, is reversibly inhibited by malonyl-CoA, its natural regulator, and is covalently modified and inactivated by CoA derivatives of certain alkyl glycidic acids [8]. The latter property was utilized to label this protein for generating sequence information that permitted the molecular cloning of CPT I. Human and rat cDNAs code for 88-kDa proteins that are highly homologous (88%) to each other and also are very similar (50%) to CPT II. An isoform of liver CPT I (L-CPT I) is present in skeletal muscle (M-CPT I) while both isoforms are expressed in heart mitochondria. L-CPT I and M-CPT I are products of different genes and have different kinetic properties. M-CPT I compared to L-CPT I is much more sensitive toward malonyl-CoA ($K_i \approx 20$

nM vs. 2 μ M) but has a lower affinity for carnitine ($K_m \approx 500 \mu\text{M}$ vs. 30 μM). CPT I is anchored in the outer mitochondrial membrane via two transmembrane segments so that the 46 N-terminal residues and the large C-terminal catalytic domain remain on the cytosolic side of the membrane. CPT I together with CPT II and acyl-CoA synthetase seem to be concentrated at contact sites between the inner and outer mitochondrial membrane. The effective expression of active CPT I in the yeast *Pichia pastoris*, which is devoid of this enzyme, made it possible to study the structure–function relationship of this enzyme. The general conclusion of these studies is that the cytosolic N-terminal region of CPT I harbors positive and negative regulatory elements that determine the sensitivity of L-CPT I toward malonyl-CoA and the affinity of M-CPT I for carnitine [9].

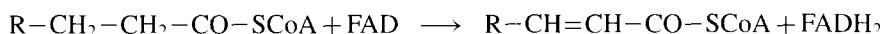
In addition to CPT I and CPT II, mitochondria contain a carnitine acetyltransferase, which has been purified and its cDNA has been cloned. The enzyme from bovine heart has an estimated molecular mass of 60 kDa and is composed of a single polypeptide chain. It catalyzes the transfer of acyl groups with 2–10 carbon atoms between CoA and carnitine. The function of this enzyme has not been established conclusively. Perhaps, the enzyme regenerates free CoA in the mitochondrial matrix by transferring acetyl groups and other short-chain or medium-chain acyl residues from CoA to carnitine. The resultant acylcarnitines can leave mitochondria via the carnitine:acylcarnitine translocase and can be metabolized by the same or other tissues, or can be excreted in urine. In addition, carnitine acetyltransferase together with CPT II may convert acylcarnitines that were formed by the partial β -oxidation of fatty acids in peroxisomes to acyl-CoAs for further oxidation in mitochondria.

Short-chain and medium-chain fatty acids with less than 10 carbon atoms can enter mitochondria as free acids independent of carnitine. They are activated by short-chain and medium-chain acyl-CoA synthetases that are present in the mitochondrial matrix.

3.2. Enzymes of β -oxidation in mitochondria

The enzymes of β -oxidation either are associated with the inner mitochondrial membrane or are located in the mitochondrial matrix. The reactions catalyzed by these enzymes are shown schematically in Fig. 2, which also provides a hypothetical view of the physical and functional organization of these enzymes.

In the first of four reactions that constitute one cycle of the β -oxidation spiral acyl-CoA is dehydrogenated to 2-*trans*-enoyl-CoA according to the following equation.



Four acyl-CoA dehydrogenases with different but overlapping chain length specificities cooperate to assure the complete degradation of all fatty acids that can be metabolized by mitochondrial β -oxidation. The names of the four dehydrogenases, short-chain, medium-chain, long-chain, and very-long-chain acyl-CoA dehydrogenases, reflect their chain-length specificities. Purification of these enzymes has permitted detailed studies of their molecular and mechanistic properties [10,11]. The first three dehydrogenases are soluble matrix enzymes with similar molecular masses between 170 and 190 kDa. They are composed of four identical subunits, each of which carries a tightly, but non-covalently bound, flavin adenine dinucleotide (FAD). Their cDNAs

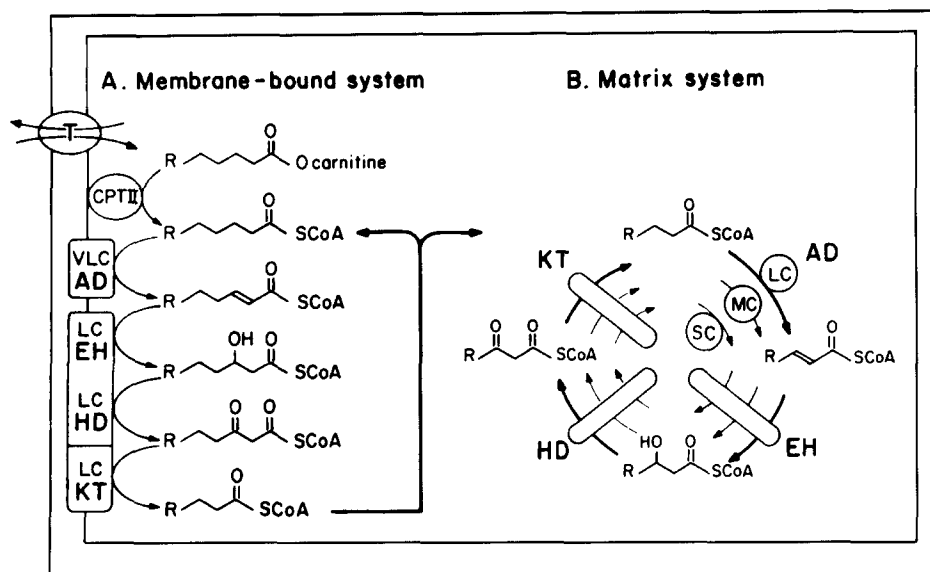
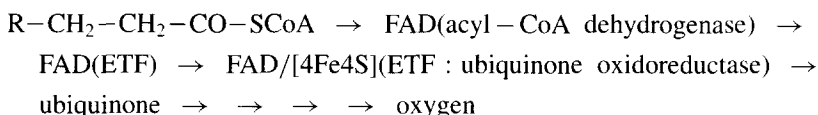


Fig. 2. Model of the functional and physical organization of β -oxidation enzymes in mitochondria. (A) β -Oxidation system active with long-chain (LC) acyl-CoAs; (B) β -Oxidation system active with medium-chain (MC) and short-chain (SC) acyl-CoAs. Abbreviations: T, carnitine: acylcarnitine translocase; CPT II, carnitine palmitoyltransferase II; AD, acyl-CoA dehydrogenase; EH, enoyl-CoA hydratase; HD, 1,3-hydroxyacyl-CoA dehydrogenase; KT, 3-ketoacyl-CoA thiolase; VLC, very long-chain.

have been cloned and sequenced. High degrees of homology (close to 90%) have been observed for the same type of enzyme from man and rat and significant homologies (30–35%) are apparent when different enzymes from one source are compared. The crystal structure of medium-chain acyl-CoA dehydrogenases at 2.4 Å resolution confirmed the homotetrameric structure of the enzyme with one FAD bound per subunit in an extended conformation [12]. Very-long-chain acyl-CoA dehydrogenase, in contrast to the three other dehydrogenases, is a protein of the inner mitochondrial membrane. Purification of this enzyme and its molecular cloning established that it is a 133-kDa homodimer with one FAD bound per subunit. The four dehydrogenases differ with respect to their specificities for substrates of various chain lengths. Short-chain acyl-CoA dehydrogenase only acts on short-chain substrates like butyryl-CoA and hexanoyl-CoA. Medium-chain acyl-CoA dehydrogenase is most active with substrates from hexanoyl-CoA to dodecanoyl-CoA, whereas long-chain acyl-CoA dehydrogenase preferentially acts on octanoyl-CoA and longer-chain substrates. Very-long-chain acyl-CoA dehydrogenase extends the activity spectrum to longer-chain substrates, including those having acyl chains with 22 and 24 carbon atoms. However, long-chain acyl-CoA dehydrogenase may have a specific function in the β -oxidation of unsaturated fatty acids because this enzyme, in contrast to very long-chain acyl-CoA dehydrogenase, effectively dehydrogenates 4-enoyl-CoAs and 5-enoyl-CoAs, which are β -oxidation intermediates of unsaturated fatty acids. This hypothesis is supported by the observed accumulation of

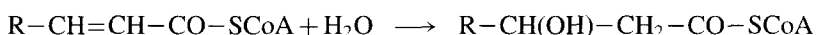
intermediates of unsaturated but not of saturated fatty acids in knock-out mice lacking long-chain acyl-CoA dehydrogenase. The dehydrogenation of acyl-CoA thioesters involves the removal of a proton from the α -carbon of the substrate and hydride transfer from the β -carbon to the FAD cofactor of the enzyme to yield 2-*trans*-enoyl-CoA and enzyme-bound FADH₂ [12]. Studies based on X-ray crystallography, chemical modifications, and site-specific mutagenesis established that glutamate 376 is the base responsible for the α -proton abstraction in medium-chain acyl-CoA dehydrogenase. Other acyl-CoA dehydrogenases follow a similar mechanism. Reoxidation of FADH₂ occurs by two successive single-electron transfers from the dehydrogenase to the FAD prosthetic group of a second flavoprotein named electron-transferring flavoprotein (ETF), which donates electrons to an iron-sulfur flavoprotein named ETF:ubiquinone oxidoreductase. The latter enzyme, a component of the inner mitochondrial membrane, feeds electrons into the mitochondrial electron transport chain via ubiquinone. The flow of electrons from acyl-CoA to oxygen is schematically shown below.



ETF is a soluble matrix protein with a molecular mass of close to 60 kDa. It is composed of two non-identical subunits of similar molecular masses with one FAD per protein dimer. The crystal structure of ETF revealed the location of FAD in a cleft between the two subunits.

In addition to the four acyl-CoA dehydrogenases involved in fatty acid oxidation, two acyl-CoA dehydrogenases specific for metabolites of branched-chain amino acids have been isolated and purified. They are isovaleryl-CoA dehydrogenase and 2-methyl-branched chain acyl-CoA dehydrogenase.

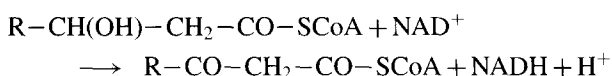
In the second step of β -oxidation 2-*trans*-enoyl-CoA is reversibly hydrated by enoyl-CoA hydratase to L-3-hydroxyacyl-CoA as shown below.



Two enoyl-CoA hydratases have been identified in mitochondria [4]. The better characterized of the two enzymes is enoyl-CoA hydratase or crotonase, which is a 161-kDa homohexamer. The best substrate of crotonase is crotonyl-CoA ($\text{CH}_3\text{-CH=CH-CO-SCoA}$), which is hydrated to form L(*S*)-3-hydroxybutyryl-CoA. The activity of the enzyme decreases with increasing chain length of the substrate so that the activity with 2-*trans*-hexadecenoyl-CoA is only 1–2% of the activity achieved with crotonyl-CoA. Crotonase also hydrates 2-*cis*-enoyl-CoA to D-3-hydroxyacyl-CoA and exhibits very low Δ^3, Δ^2 -enoyl-CoA isomerase activity. The crystal structure of crotonase revealed that it is a member of the hydratase/isomerase superfamily with two active site glutamate residues that function as general acid and general base in the syn addition of water to crotonyl-CoA. The second enoyl-CoA hydratase, referred to as long-chain enoyl-CoA hydratase, is virtually inactive with crotonyl-CoA, but effectively hydrates medium-chain and long-chain substrates. The activities of crotonase and long-chain enoyl-CoA hydratase complement each other thereby assuring high rates of hydration

of all enoyl-CoA intermediates. Long-chain enoyl-CoA hydratase is a component enzyme of the trifunctional β -oxidation complex, which additionally exhibits long-chain activities of L-3-hydroxyacyl-CoA dehydrogenase and 3-ketoacyl-CoA thiolase [13]. This β -oxidation complex is a protein of the inner mitochondrial membrane. It consists of equimolar amounts of a large α -subunit with a molecular mass of close to 80 kDa and of a small β -subunit with a molecular mass of approximately 48 kDa. Cloning and sequencing of the cDNAs that code for this complex revealed significant homologies of the amino-terminal and central regions of the large subunit with enoyl-CoA hydratase and L-3-hydroxyacyl-CoA dehydrogenase, respectively, and of the small subunit with 3-ketoacyl-CoA thiolase. These homologies are indicative of the locations of the component enzymes on the complex.

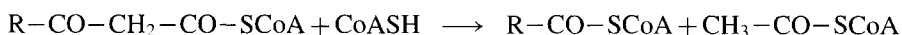
The third reaction in the β -oxidation cycle is the reversible dehydrogenation of L-3-hydroxyacyl-CoA to 3-ketoacyl-CoA catalyzed by L-3-hydroxyacyl-CoA dehydrogenase as shown in the following equation.



Four L-3-hydroxyacyl-CoA dehydrogenases have been identified in mitochondria.

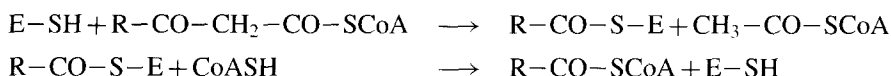
L-3-Hydroxyacyl-CoA dehydrogenase is a soluble matrix enzyme, which has a molecular mass of approximately 65 kDa and is composed of two identical subunits [4]. The enzyme and its cDNA have been sequenced. The crystal structure of the pig heart enzyme revealed a bilobal structure with the NAD^+ binding site in the N-terminal region and the substrate binding site in the cleft between the C-terminal and N-terminal domains. The enzyme is specific for NAD^+ as a coenzyme. It acts on L-3-hydroxyacyl-CoAs of various chain lengths but is most active with medium-chain and short-chain substrates. A second L-3-hydroxyacyl-CoA dehydrogenase was recently isolated from bovine liver. This enzyme is a homotetramer with a subunit molecular mass of 27 kDa. Its substrate specificity resembles that of the L-3-hydroxyacyl-CoA dehydrogenase described above. However, the enzyme's primary function may be in androgen metabolism and not in β -oxidation because it exhibits significant 17 β -hydroxysteroid dehydrogenase activity and is absent or almost absent from some tissues with high β -oxidation activity. Long-chain L-3-hydroxyacyl-CoA dehydrogenase is a component enzyme of the trifunctional β -oxidation complex or trifunctional protein [13]. This dehydrogenase is active with medium- and long-chain substrates, but not with 3-hydroxybutyryl-CoA, and hence complements the soluble dehydrogenase to assure high rates of dehydrogenation over the whole spectrum of β -oxidation intermediates. A soluble short-chain L-3-hydroxy-2-methylacyl-CoA dehydrogenase is also present in the mitochondrial matrix. This enzyme, which acts on short-chain substrates with or without 2-methyl substituents, is believed to function only in isoleucine metabolism.

In the last reaction of the β -oxidation cycle 3-ketoacyl-CoA is cleaved by thiolase as shown below.



The products of the reaction are acetyl-CoA and an acyl-CoA shortened by two carbon atoms. The equilibrium of the reaction is far to the side of the thiolytic cleavage

products thereby driving β -oxidation to completion. All thiolases that have been studied in detail contain an essential sulfhydryl group, which participates directly in the carbon-carbon bond cleavage as outlined in the following equations where E-SH represents thiolase.



According to this mechanism, 3-ketoacyl-CoA binds to the enzyme and is cleaved between its α and β carbon atoms. An acyl residue, which is two carbons shorter than the substrate, is transiently bound to the enzyme via a thioester bond, while acetyl-CoA is released from the enzyme. Finally, the acyl residue is transferred from the sulfhydryl group of the enzyme to CoA to yield acyl-CoA.

Several types of thiolases have been identified, some of which exist in multiple forms [4]. Mitochondria contain three classes of thiolases: (1) acetoacetyl-CoA thiolase or acetyl-CoA acetyltransferase, which is specific for acetoacetyl-CoA (C_4) as a substrate; (2) 3-ketoacyl-CoA thiolase or acetyl-CoA acyltransferase, which acts on 3-ketoacyl-CoA thioesters of various chain lengths (C_4 – C_{16}) and (3) long-chain 3-ketoacyl-CoA thiolase, which acts on medium-chain and long-chain 3-ketoacyl-CoA thioesters, but not on acetoacetyl-CoA. The latter two enzymes are essential for fatty acid β -oxidation, whereas acetoacetyl-CoA thiolase most likely functions only in ketone body and isoleucine metabolism. Long-chain 3-ketoacyl-CoA thiolase is a component enzyme of the membrane-bound trifunctional β -oxidation complex [13], whereas the other two thiolases are soluble matrix enzymes. All mitochondrial thiolases have been purified and their cDNAs have been cloned and sequenced. A comparison of amino acid sequences proved all mitochondrial thiolases to be different, but homologous, enzymes. 3-Ketoacyl-CoA thiolase is composed of four identical subunits with a molecular mass of close to 42 kDa. This enzyme acts equally well on all substrates tested except for acetoacetyl-CoA, which is cleaved at half the maximal rate observed with longer chain substrates.

The absence or near absence of intermediates of β -oxidation from mitochondria prompted the idea of intermediate channeling due to the existence of a multienzyme complex of β -oxidation enzymes in intact mitochondria. The identification and characterization of at least two isozymes for each of the four reactions of the β -oxidation spiral led to the presentation of a model for their physical and functional organization as shown in Fig. 2 [14]. By this model, the membrane-bound, long-chain specific β -oxidation system, consisting of very-long-chain acyl-CoA dehydrogenase and the trifunctional β -oxidation complex, converts long-chain to medium-chain fatty acyl-CoAs, which are completely degraded by the matrix system of soluble enzymes that have a preference for medium-chain and short-chain substrates. An assumption underlying this model is that all enzymes thought to function in fatty acid β -oxidation are essential for this process. So far this assumption has proven to be correct. The characterization of inherited disorders of fat metabolism in humans has revealed that each of the many β -oxidation enzymes found to be deficient in a patient is essential for the normal degradation of fatty acids (for more detail see Section 5).

3.3. β -Oxidation of unsaturated fatty acids

Unsaturated fatty acids, which usually contain *cis* double bonds, also are degraded by β -oxidation. However, additional (auxiliary) enzymes are required to act on the pre-existing double bonds once they are close to the thioester group as a result of chain-shortening [15]. All double bonds present in unsaturated and polyunsaturated fatty acids can be classified either as odd-numbered double bonds, like the 9-*cis* double bond of oleic acid and linoleic acid or as even-numbered double bonds like the 12-*cis* double bond of linoleic acid. Since both classes of double bonds are present in linoleic acid, its degradation illustrates the breakdown of all unsaturated fatty acids. A summary of the β -oxidation of linoleic acid is presented in Fig. 3. Linoleic acid, after conversion to its CoA thioester(I), undergoes three cycles of β -oxidation to yield 3-*cis*,6-*cis*-dodecadienoyl-CoA(II) which is isomerized to 2-*trans*,6-*cis*-dodecadienoyl-CoA(III) by Δ^3, Δ^2 -*trans*-enoyl-CoA isomerase, an auxiliary enzyme of β -oxidation. 2-*trans*,6-*cis*-Dodecadienoyl-CoA(III) is a substrate of β -oxidation and can complete one cycle to yield 4-*cis*-decenoyl-CoA(IV), which is dehydrogenated to 2-*trans*,4-*cis*-decadienoyl-CoA(V) by medium-chain acyl-CoA dehydrogenase. 2-*trans*,4-*cis*-Decadienoyl-CoA(V) cannot continue on its course through the β -oxidation spiral, but instead is reduced by NADPH in a reaction catalyzed by 2,4-dienoyl-CoA reductase. The product of this reduction, 3-*trans*-decenoyl-CoA(VI), is isomerized by Δ^3, Δ^2 -enoyl-CoA isomerase to 2-*trans*-decenoyl-CoA(VII), which can be completely degraded by completing four cycles of β -oxidation. Altogether, the degradation of unsaturated fatty acids in mitochondria involves at least Δ^3, Δ^2 -enoyl-CoA isomerase and 2,4-dienoyl-CoA reductase as auxiliary enzymes in addition to the enzymes of the β -oxidation spiral.

More recent is the demonstration that odd-numbered double bonds can be reduced at the stage of 5-enoyl-CoA intermediates formed during the β -oxidation of unsaturated fatty acids. Shown in Fig. 4 is the sequence of reactions that explains the NADPH-dependent reduction of 5-*cis*-enoyl-CoA(I) [16]. After introduction of a 2-*trans* double bond by acyl-CoA dehydrogenase, the resultant 2,5-dienoyl-CoA(II) is converted to 3,5-dienoyl-CoA(III) by Δ^3, Δ^2 -enoyl-CoA isomerase. A novel enzyme, $\Delta^{3,5} \Delta^{2,4}$ -dienoyl-CoA isomerase, converts 3,5-dienoyl-CoA(III) to 2-*trans*,4-*trans*-dienoyl-CoA(IV) by a concerted shift of both double bonds. Finally, 2,4-dienoyl-CoA reductase catalyzes the NADPH-dependent reduction of one double bond to produce 3-*trans*-enoyl-CoA(V), which, after isomerization to 2-*trans*-enoyl-CoA(VI) by Δ^3, Δ^2 -enoyl-CoA isomerase, can reenter the β -oxidation spiral. Although the significance of this new pathway has not been fully explored, it seems likely that it provides an avenue for the metabolism of 3,5-dienoyl-CoAs that may be formed fortuitously by Δ^3, Δ^2 -enoyl-CoA isomerase acting on 2,5-dienoyl-CoA intermediates.

Two Δ^3, Δ^2 -enoyl-CoA isomerases exist in rat mitochondria. One is mitochondrial Δ^3, Δ^2 -enoyl-CoA isomerase that has been purified and its cDNA has been cloned [17]. This enzyme is a multimer of one type of subunit with a molecular mass of 30 kDa. In addition to converting the CoA derivatives of 3-*cis*-enoic acids and 3-*trans*-enoic acids with 6–16 carbon atoms to the corresponding 2-*trans*-enoyl-CoAs, the enzyme catalyzes the conversion of 2,5-dienoyl-CoA to 3,5-dienoyl-CoA and of

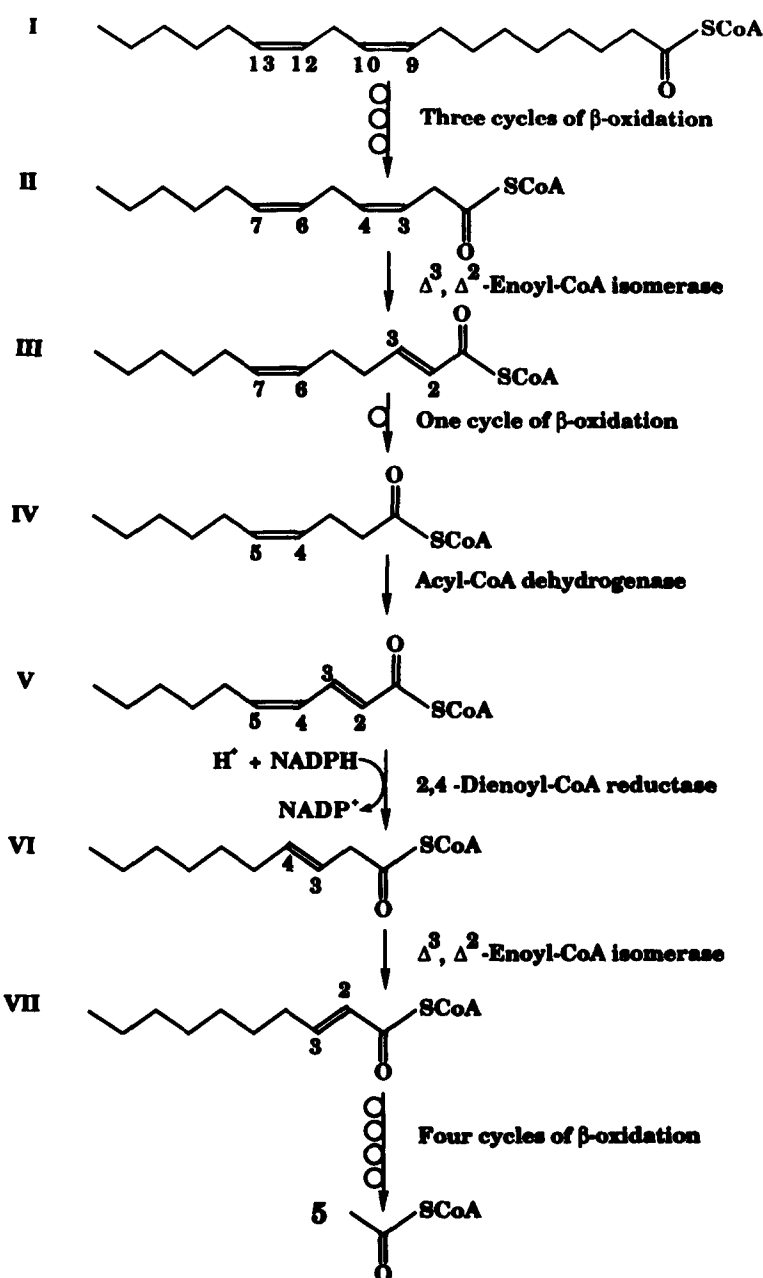


Fig. 3. β -Oxidation of linoleoyl-CoA.

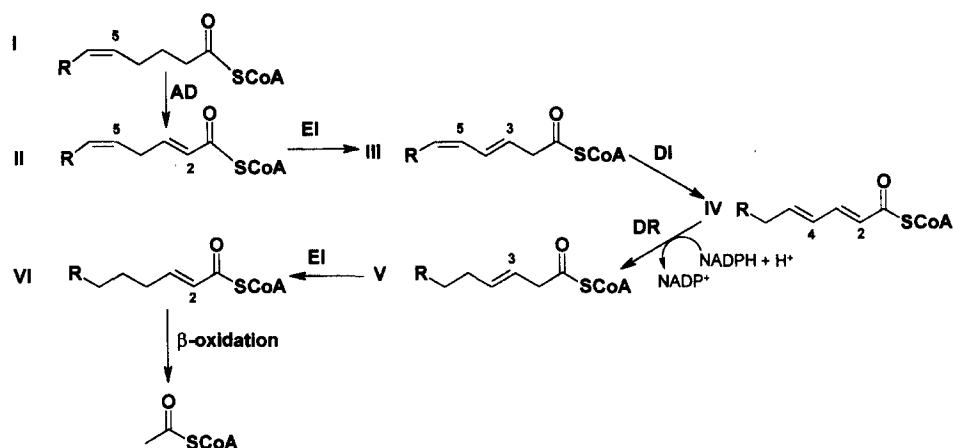


Fig. 4. β -Oxidation of 5-*cis*-enoyl-CoA. AD, acyl-CoA dehydrogenase; EI, Δ^3, Δ^2 -enoyl-CoA isomerase; DI, $\Delta^{3,5}, \Delta^{2,4}$ -dienoyl-CoA isomerase; DR, 2,4-dienoyl-CoA reductase.

3-ynoyl-CoA to 2,3-dienoyl-CoA. A second Δ^3, Δ^2 -enoyl-CoA isomerase has been identified in mitochondria. This isomerase is identical with peroxisomal Δ^3, Δ^2 -enoyl-CoA isomerase and therefore has a dual subcellular localization. It is more active with long-chain than medium-chain substrates and has a preference for 3-*trans*-enoyl-CoA as compared to 3-*cis*-enoyl-CoAs.

2,4-Dienoyl-CoA reductase has been purified and its cDNA has been cloned. This enzyme is a homotetramer with a native molecular mass of 124 kDa. The reductase has a specific requirement for NADPH. A second mitochondrial 2,4-dienoyl-CoA reductase has been observed but has not yet been characterized.

$\Delta^{3,5}, \Delta^{2,4}$ -Dienoyl-CoA isomerase is a member of the hydratase/isomerase superfamily with a 32-kDa subunit in a homohexameric arrangement. It is present in both mitochondria and peroxisomes because of mitochondrial and peroxisomal targeting signals at its N-terminus and C-terminus, respectively. The crystal structure of this isomerase revealed the presence of one active site glutamate and aspartate residue each, which catalyze simultaneous proton transfers that facilitate the 3,5 $\rightarrow$ 2,4 double bond isomerization with substrates having acyl chains with 8–20 carbon atoms. It also catalyzes the isomerization of 3,5,7-trienoyl-CoA to 2,4,6-trienoyl-CoA. This reaction is a likely step in the β -oxidation of conjugated linoleic acid like 9-*cis*, 11-*trans*-octadecadienoic acid.

3.4. Regulation of fatty acid oxidation in mitochondria

The rate of fatty acid oxidation is a function of the plasma concentration of unesterified fatty acids. Unesterified or free fatty acids are generated by lipoprotein lipase or are released from adipose tissue into the circulatory system, which carries them to other tissues or organs. Hormones like glucagon and insulin regulate the lipolysis of triacylglycerols in adipose tissue (see Chapter 10). The utilization of fatty acids for either oxidation or lipid synthesis depends on the nutritional state of the animal,

more specifically on the availability of carbohydrates. Because of the close relationship among lipid metabolism, carbohydrate metabolism, and ketogenesis, the regulation of fatty acid oxidation in liver differs from that in tissues like heart and skeletal muscle, which have an overwhelming catabolic function. For this reason, the regulation of fatty acid oxidation in liver and heart will be discussed separately.

The direction of fatty acid metabolism in liver depends on the nutritional state of the animal. In the fed animal, the liver converts carbohydrates to fatty acids, while in fasted animals fatty acid oxidation, ketogenesis, and gluconeogenesis are the more active processes. Clearly, there exists a reciprocal relationship between fatty acid synthesis and fatty acid oxidation. Although it is well established that lipid and carbohydrate metabolism are under hormonal control, it has been more difficult to identify the mechanism that regulates fatty acid synthesis and oxidation. McGarry and Foster [18] have proposed that the concentration of malonyl-CoA, the first committed intermediate in fatty acid biosynthesis, determines the rate of fatty acid oxidation. The essential features of their hypothesis are presented in Fig. 5. In the fed animal, where glucose is actively converted to fatty acids, the concentration of malonyl-CoA is elevated. Malonyl-CoA at micromolar concentrations inhibits hepatic CPT I thereby decreasing the transfer of fatty acyl residues from CoA to carnitine and their translocation into mitochondria. Consequently, β -oxidation is depressed. When the animal changes from the fed to the fasted state, hepatic metabolism shifts from glucose breakdown to gluconeogenesis with a resultant decrease in fatty acid synthesis. The concentration of malonyl-CoA decreases, and the inhibition of CPT I is relieved. Furthermore, starvation causes an increase in the total CPT I activity and a decrease in the sensitivity of CPT I toward malonyl-CoA. Altogether, during starvation acylcarnitines are more rapidly formed and translocated into mitochondria thereby stimulating β -oxidation and ketogenesis.

It appears that the cellular concentration of malonyl-CoA is directly related to the activity of acetyl-CoA carboxylase, which is hormonally regulated. The short-term regulation of acetyl-CoA carboxylase involves the phosphorylation and dephosphorylation of the enzyme (see Chapter 6). In the fasting animal, a high [glucagon]/[insulin] ratio causes the phosphorylation and inactivation of acetyl-CoA carboxylase. As a consequence, the concentration of malonyl-CoA and the rate of fatty acid synthesis decrease, while the rate of β -oxidation increases. A decrease of the [glucagon]/[insulin] ratio reverses these effects. Thus, both fatty acid synthesis and fatty acid oxidation are regulated by the ratio of [glucagon]/[insulin].

It has been suggested that malonyl-CoA also regulates fatty acid oxidation in non-hepatic tissues like heart and skeletal muscle [19,20]. The formation of malonyl-CoA in these tissues is catalyzed by a 280-kDa isoform (ACC 2) of the 265-kDa acetyl-CoA carboxylase (ACC 1) that is the predominant form of lipogenic tissues. The disposal of malonyl-CoA is thought to be catalyzed by cytosolic malonyl-CoA decarboxylase. If so, the tissue concentration of malonyl-CoA is determined by the activities of both the carboxylase and decarboxylase. Both enzymes seem to be regulated. ACC 2 is phosphorylated and inactivated by AMP-dependent kinase in response to stress caused by ischemia/hypoxia and exercise and is activated allosterically by citrate. A concern about this model for the regulation of fatty acid oxidation in heart and skeletal muscle is the discrepancy between the micromolar tissue concentration of malonyl-CoA and

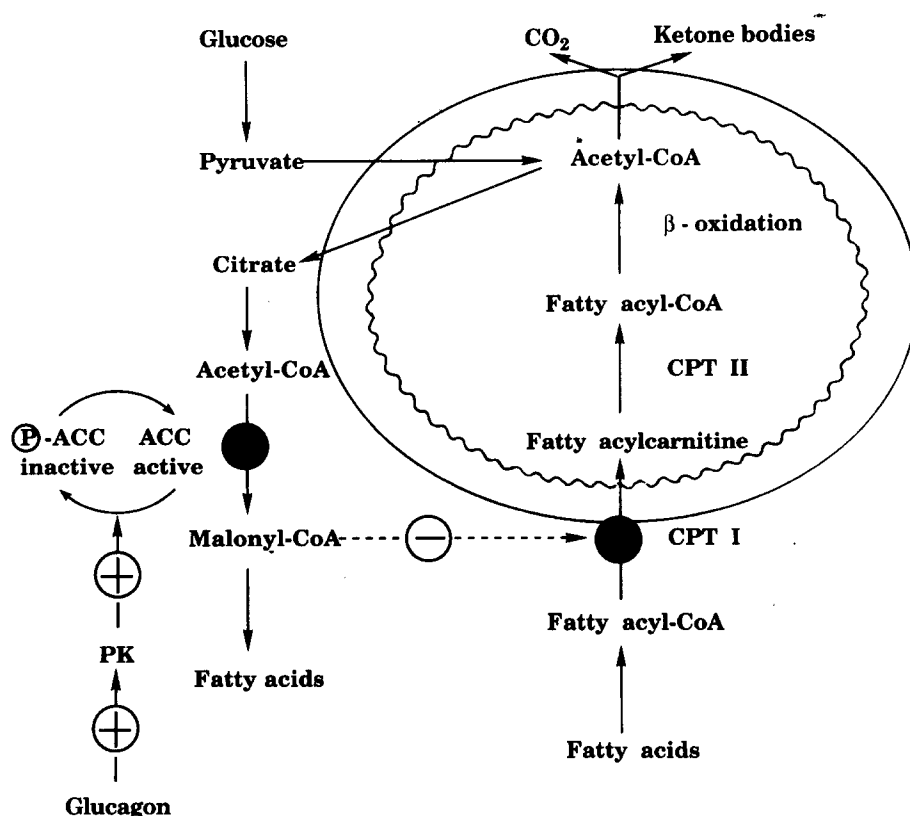
FATTY ACID SYNTHESIS**FATTY ACID OXIDATION**

Fig. 5. Proposed regulation of fatty acid oxidation in liver. $\oplus$, Stimulation; $\ominus$, inhibition; $\bullet$, enzymes subject to regulation. ACC, acetyl-CoA carboxylase; CPT, carnitine palmitoyltransferase; PK, protein kinase.

the nanomolar K_i of muscle CPT I (M-CPT I) for malonyl-CoA. This enzyme should be completely inhibited at all times unless the effective malonyl-CoA concentration is lower due to binding to other proteins or due to intracellular compartmentation.

In heart, and possibly in other tissues, the rate of fatty acid oxidation is tuned to the cellular energy demand in addition to being dependent on the concentration of plasma free fatty acids [21]. At sufficiently high concentrations (>0.6 mM) of free fatty acids the rate of fatty acid oxidation is only a function of the cellular energy demand. Studies with perfused hearts and isolated heart mitochondria have shown that a decrease in the energy demand results in elevated concentrations of acetyl-CoA and NADH and in lower concentrations of CoA and NAD⁺. The resultant increases in the ratios of [acetyl-CoA]/[CoA] and [NADH]/[NAD⁺] in the mitochondrial matrix may be the cause for the reduced rate of β -oxidation. Experiments with isolated heart mitochondria have provided support for this view. Moreover, these experiments support the conclusion that the ratio of [acetyl-CoA]/[CoA], and not of [NADH]/[NAD⁺],

controls the rate of β -oxidation. Although the site of this regulation has not been identified unequivocally, it is possible that the [acetyl-CoA]/[CoA] ratio regulates the activity of 3-ketoacyl-CoA thiolase and thereby controls the flux of fatty acids through the β -oxidation spiral.

4. Fatty acid oxidation in peroxisomes

Peroxisomes and glyoxysomes, collectively referred to as microbodies, are subcellular organelles capable of respiration. They do not have an energy-coupled electron transport system like mitochondria, but instead contain flavin oxidases, which catalyze the substrate-dependent reduction of oxygen to H_2O_2 . Because catalase is present in these organelles, H_2O_2 is rapidly reduced to water. Thus, peroxisomes and glyoxysomes are organelles with a primitive respiratory chain where energy released during the reduction of oxygen is lost as heat. Glyoxysomes are peroxisomes that contain the enzymes of the glyoxylate pathway in addition to flavin oxidases and catalase. Peroxisomes or glyoxysomes are found in all major groups of eukaryotic organisms including yeasts, fungi, protozoa, plants and animals.

An extramitochondrial system capable of oxidizing fatty acids was first detected in glyoxysomes of germinating seeds. When rat liver cells were shown to contain a β -oxidation system in peroxisomes [22], the interest in the peroxisomal pathway was greatly stimulated and one set of β -oxidation enzymes was soon identified and characterized [23,24]. It should be noted that peroxisomal β -oxidation occurs in all eukaryotic organisms, whereas mitochondrial β -oxidation seems to be restricted to animals. Studies of peroxisomal β -oxidation were aided by the use of certain drugs, e.g. clofibrate and di(-2-ethylhexyl)phthalate, which induce the expression of the enzymes of peroxisomal β -oxidation and in addition cause the proliferation of peroxisomes in rodents. The induction of peroxisomal β -oxidation by xenobiotic proliferators or fatty acids involves the peroxisomal proliferator-activated receptors (PPARs), which are members of the nuclear hormone receptor family and which recognize peroxisomal proliferator response elements upstream of the affected structural genes [4]. Although rat liver peroxisomes are capable of chain-shortening regular long-chain fatty acids, their main function seems to be the partial β -oxidation of very-long-chain fatty acids, methyl-branched carboxylic acids like pristanic acid, prostaglandins, dicarboxylic acids, xenobiotic compounds like phenyl fatty acids, and hydroxylated 5- β -cholestanoic acids, formed during the conversion of cholesterol to cholic acid.

4.1. Fatty acid uptake by peroxisomes

The mechanism of fatty acid uptake by peroxisomes is poorly understood. Although small molecules like substrates and cofactors can freely cross the membrane of isolated peroxisomes from animals, it seems that in vivo the peroxisomal membrane constitutes a permeability barrier that would require transporters to facilitate the uptake of substrates and cofactors [25]. In animal and yeast cells, long-chain fatty acids are activated outside of the peroxisomal membrane. Long-chain acyl-CoAs are thought to enter peroxisomes

in a facilitated process involving half ABC transporters like ALDP, ALDRP, PMP70, and PMP69 in animal cells and Pxa1p and Pxa2p in yeast. In contrast, very long-chain fatty acids in animal cells and medium-chain fatty acids in yeast can be activated in the peroxisomal matrix. The β -oxidation of medium-chain fatty acids in yeast requires two membrane proteins that are assumed to facilitate the uptake of medium-chain fatty acids and ATP, respectively.

4.2. Pathways and enzymology of peroxisomal α -oxidation and β -oxidation

The first step in peroxisomal β -oxidation (see Fig. 6) is the dehydrogenation of acyl-CoA to 2-*trans*-enoyl-CoA catalyzed by acyl-CoA oxidase. This enzyme, in contrast to the mitochondrial dehydrogenases, transfers two hydrogens from the substrate to its FAD cofactor and then to O_2 , which is reduced to H_2O_2 . Rat liver contains three acyl-CoA oxidases with different substrate specificities. Their names, palmitoyl-CoA oxidase, pristanoyl-CoA oxidase, and trihydroxycoprostanoyl-CoA oxidase are indicative of their preferred substrates [26]. Interestingly, human liver contains only one branched-chain acyl-CoA oxidase besides palmitoyl-CoA oxidase. Cloning and sequencing of the gene and cDNAs coding for rat acyl-CoA oxidase revealed two

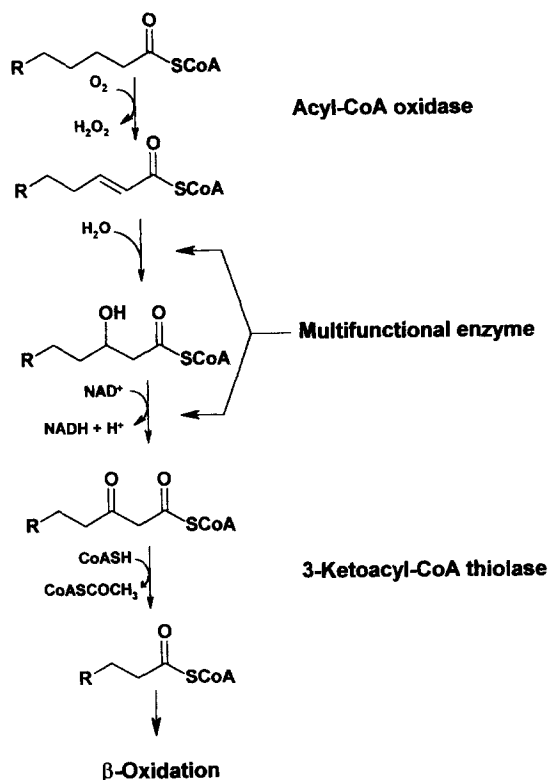


Fig. 6. Pathway of β -oxidation in peroxisomes.

isoforms of this enzyme as the result of the alternative use of two exons [24]. The rat liver palmitoyl-CoA oxidase is a homodimer with a molecular mass of close to 150 kDa. Ligands of PPARs induce the expression of this enzyme but not of the other two acyl-CoA oxidases. Palmitoyl-CoA oxidase is inactive with butyryl-CoA and hexanoyl-CoA as substrates, but dehydrogenates all longer chain substrates with similar maximal velocities. Acyl-CoA oxidases from organisms other than mammals are either active with substrates of all chain lengths or the organisms express more than one acyl-CoA oxidase with complementing chain length specificities. Consequently, fatty acids can be completely degraded in yeasts, plants, and other lower eukaryotic organisms, but not in mammals.

The next two reactions of β -oxidation, the hydration of 2-enoyl-CoA to 3-hydroxyacyl-CoA and the NAD^+ -dependent dehydrogenation of 3-hydroxyacyl-CoA to 3-ketoacyl-CoA, are catalyzed in rat liver peroxisomes by multifunctional enzymes 1 (MFE 1) and 2 (MFE 2) [25,27]. Both multifunctional enzymes harbor enoyl-CoA hydratase and 3-hydroxyacyl-CoA dehydrogenase activities. MFE 1 additionally exhibits Δ^3, Δ^2 -enoyl-CoA isomerase activity. 3-Hydroxyacyl-CoA intermediates formed and acted on by MFE 1 have the L-configuration, whereas the hydroxy intermediates produced by MFE 2 have the D-configuration. Both MFE 1 and MFE 2 convert medium-chain and long-chain 2-*trans*-enoyl-CoAs to 3-ketoacyl-CoAs but show little or no activity with short-chain substrates. MFE 2 is slightly more active than MFE 1 with longer-chain substrates. However, MFE 2 alone acts on substrates with 2-methyl branches like those formed during the β -oxidation of pristanic acid and hydroxylated 5 β -cholestanoic acid. MFE 1 consists of a single 80-kDa polypeptide while MFE 2 is a homodimer with a molecular mass of approximately 150 kDa. Yeast and fungi contain only one multifunctional enzyme each with D-specific enoyl-CoA hydratase and 3-hydroxyacyl-CoA dehydrogenase activities.

The last reaction of β -oxidation, the CoA-dependent cleavage of 3-ketoacyl-CoA, is catalyzed by 3-ketoacyl-CoA thiolase. Two rat 3-ketoacyl-CoA thiolases coded for by different genes have been detected. One is constitutively expressed, whereas the expression of the other is highly induced in response to peroxisomal proliferators [25,27]. Both enzymes are homodimers with molecular masses of close to 80 kDa. Both exhibit little activity toward acetoacetyl-CoA, but are active with all longer chain substrates except for 3-keto-2-methylacyl-CoA intermediates formed during the β -oxidation of pristanic acid and hydroxylated 5 β -cholestanoic acids. However, these intermediates are acted upon by another 3-ketoacyl-CoA thiolase (SCP₃-thiolase) that was identified in peroxisomes during a study of the 58-kDa precursor of sterol carrier protein-2. The C-terminal segment of this 58-kDa protein is identical with sterol carrier protein-2, whereas the N-terminal domain harbors the thiolase, which is most active with medium-chain substrates. The crystal structure of the peroxisomal 3-ketoacyl-CoA thiolase from *Saccharomyces cerevisiae* at 2.8 Å resolution shows two cysteine residues in close proximity at the presumed active site.

Because peroxisomes contain at least two enzymes for each step of β -oxidation, specific functions for these enzymes were inferred from their substrate specificities. Most of the predictions were verified by analyzing fatty acids that accumulate in patients and/or knock-out mice deficient for individual enzymes. Together, these data

support the proposal that branched-chain acyl-CoA oxidase, MFE 2, and SCP_N-thiolase are essential for the degradation of pristanic acid and hydroxylated 5- β -cholestanic acid. The β -oxidation of very-long-chain fatty acids requires the involvement of acyl-CoA oxidase and MFE 2. 3-Ketoacyl-CoA thiolase is believed, but has not been proven to function in the breakdown of straight-chain fatty acids. Surprisingly the knock-out mouse for MFE 1 does not exhibit an obvious phenotype, thus leaving the function of this enzyme in doubt.

Unsaturated fatty acids are degraded in peroxisomes by the pathways outlined in Figs. 3 and 4. Two Δ^3, Δ^2 -enoyl-CoA isomerases are present in rat liver peroxisomes. One is the monofunctional Δ^3, Δ^2 -enoyl-CoA isomerase that is also present in mitochondria, the other is the Δ^3, Δ^2 -enoyl-CoA isomerase activity of multifunctional enzyme 1. The monofunctional Δ^3, Δ^2 -enoyl-CoA isomerase has a preference for long-chain substrates and may play the major role in the partial β -oxidation of long-chain unsaturated fatty acids. So far this isomerase has only been obtained by cloning and heterologous expression based on its homology to the sole Δ^3, Δ^2 -enoyl-CoA isomerase of yeast. The crystal structure of yeast Δ^3, Δ^2 -enoyl-CoA isomerase revealed the presence of a single glutamate residue at the active site, which catalyzes a 1,3-proton transfer that results in the shift of the double bond. A 2,4-dienoyl-CoA reductase distinct from the mitochondrial reductase but homologous with the yeast 2,4-dienoyl-CoA reductase has been identified in mammalian peroxisomes by a cloning and expression approach. This enzyme acts on a wide spectrum of 2,4-dienoyl-CoAs but is most active with medium-chain substrates. $\Delta^{3,5} \Delta^{2,4}$ -Dienoyl-CoA isomerase of mammalian peroxisomes is the same enzyme that is also present in mitochondria (see Section 3.3).

The products of peroxisomal β -oxidation in animals are chain-shortened acyl-CoAs, acetyl-CoA, and NADH. The β -oxidation of chain-shortened acyl-CoAs is completed in mitochondria. For this purpose, acyl-CoAs, including acetyl-CoA, are thought to leave peroxisomes as acylcarnitines, which can be formed by peroxisomal carnitine octanoyltransferase and/or carnitine acetyltransferase [25]. These reactions, as well as the observed hydrolysis of acetyl-CoA to acetate, would regenerate CoA in peroxisomes. The transporters that facilitate the exit of the β -oxidation products from peroxisomes have not yet been identified.

Phytanic acid (3,7,11,15-tetramethylhexadecanoic acid), a component of the human diet that is derived from phytol, a constituent of chlorophyll, is not degraded by β -oxidation because its 3-methyl group interferes with this process. Instead it is chain shortened by α -oxidation in peroxisomes as outlined in Fig. 7 [25,27]. Activation of phytanic acid(I) to phytanoyl-CoA(II) by long-chain or very long-chain acyl-CoA synthetase converts it to a substrate of hydroxyphytanoyl-CoA hydroxylase. Cleavage of the resultant 2-hydroxyphytanoyl-CoA(III) by 2-hydroxyphytanoyl-CoA lyase yields pristanal(IV) and formyl-CoA that are oxidized to pristanic acid(V) and CO₂, respectively. Pristanic acid after activation to pristanoyl-CoA is a substrate of β -oxidation because the 2-methyl group does not interfere with the process as long as the 2-methyl group has the *S* configuration. A 2-methylacyl-CoA racemase that is present in both peroxisomes and mitochondria epimerizes (2*R*)-pristanoyl-CoA and (2*S*)-trihydroxycholestanoyl-CoA to their *S* isomers that are substrates of peroxisomal β -oxidation.

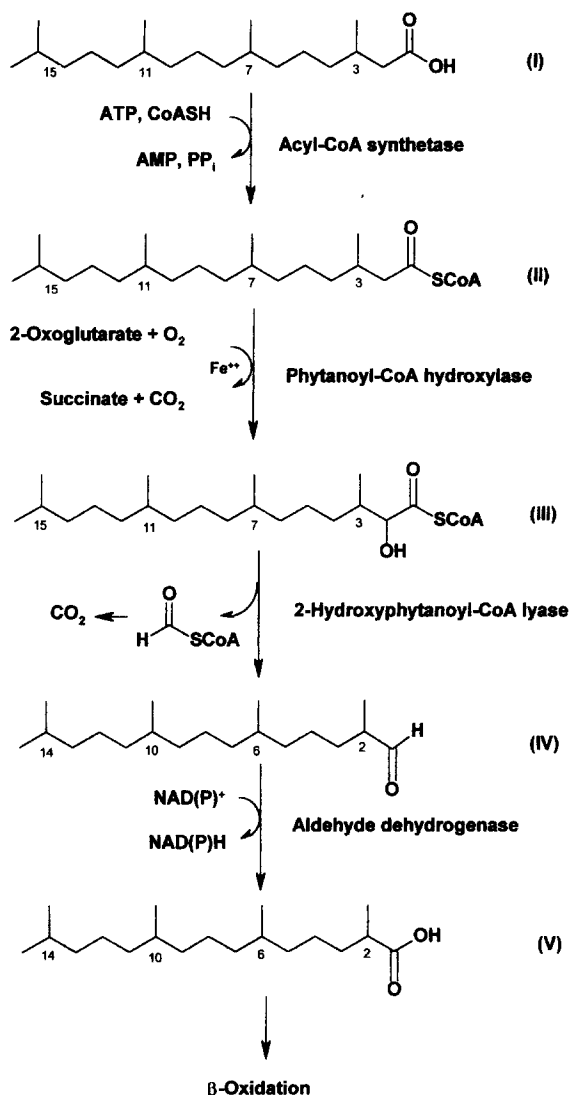


Fig. 7. α -Oxidation of phytanic acid.

5. Inherited diseases of fatty acid oxidation

Disorders of fatty acid oxidation were first described in 1973 when deficiencies of carnitine and carnitine palmitoyltransferase (CPT II) were identified as causes of muscle weakness [28]. Patients with low levels (5–45% of normal) of CPT II have recurrent episodes of muscle weakness and myoglobinuria, often precipitated by prolonged exercise and/or fasting. Almost a decade later, a deficiency of medium-chain dehydrogenase was identified in patients with a disorder of fasting adaptation [28]. This

relatively common disorder is characterized by episodes of non-ketotic hypoglycemia provoked by fasting during the first 2 years of life. Between episodes, patients with medium-chain acyl-CoA dehydrogenase deficiency appear normal. Therapy is aimed at preventing fasting, if necessary by the intravenous administration of glucose, and includes carnitine supplementation. The molecular basis of medium-chain acyl-CoA dehydrogenase deficiency is an A $\rightarrow$ G base transition in 90% of the disease causing alleles. This mutation results in the replacement of lysine-329 by a glutamate residue, which impairs the assembly of subunits into the functional tetrameric enzyme. In the years following the identification of medium-chain acyl-CoA dehydrogenase deficiency, fatty acid oxidation disorders due to the following enzymes deficiencies have been described: short-chain acyl-CoA dehydrogenase, very-long-chain acyl-CoA dehydrogenase, electron-transferring flavoprotein (ETF), ETF:ubiquinone oxidoreductase, 3-hydroxyacyl-CoA dehydrogenase, long-chain 3-hydroxyacyl-CoA dehydrogenase, trifunctional β -oxidation complex, 2,4-dienoyl-CoA reductase, carnitine palmitoyltransferase I, and carnitine:acylcarnitine translocase [28,29]. A deficiency of mitochondrial acetoacetyl-CoA thiolase impairs isoleucine and ketone body metabolism, but not fatty acid oxidation.

Many of these disorders are associated with the urinary excretion of acylcarnitines, acyl conjugates of glycine, and dicarboxylic acids that are characteristic of the metabolic block. A general conclusion derived from studies of these disorders is that an impairment of β -oxidation makes fatty acids available for microsomal ω -oxidation by which fatty acids are oxidized at their terminal (ω) methyl group or at their penultimate ($\omega - 1$) carbon atom. Molecular oxygen is required for this oxidation and the hydroxylated fatty acids are further oxidized to dicarboxylic acids. Long-chain dicarboxylic acids can be chain-shortened by peroxisomal β -oxidation to medium-chain dicarboxylic acids, which are excreted in urine.

Several disorders associated with an impairment of peroxisomal β -oxidation have been described [30]. Of these, Zellweger syndrome and neonatal adrenoleukodystrophy are characterized by the absence, or low levels, of peroxisomes due to the defective biogenesis of this organelle. As a result of this deficiency, compounds that are normally metabolized in peroxisomes, for example very long-chain fatty acids, dicarboxylic acids, hydroxylated 5- β -cholestanoic acids, and also phytanic acid, accumulate in plasma [25,30]. Infants with Zellweger syndrome rarely survive longer than a few months due to hypotonia, seizures and frequently cardiac defects. In addition to disorders of peroxisome biogenesis, defects of each of the three enzymes of the peroxisomal β -oxidation spiral and of the peroxisomal very long-chain acyl-CoA synthetase (X-linked adrenoleukodystrophy) have been reported [25,30]. Most of these patients were hypotonic, developed seizures, failed to make psychomotor gains, and died in early childhood. The importance of α -oxidation in humans has been established as a result of studying Refsum's disease, a rare and inherited neurological disorder. Patients afflicted with this disease accumulate large amounts of phytanic acid due to a deficiency of phytanoyl-CoA hydroxylase [25,30].

6. Future directions

Fatty acid oxidation has been studied for almost a century with the result that a fairly detailed view of this process has emerged. The molecular characterization of most β -oxidation enzymes has yielded a wealth of structural information while the dynamics of this pathway remain less well understood. This is in part due to an absence of information about the organization of the β -oxidation enzymes and the impact such organization has on the control of the process. Also a number of questions about the regulation of this process remain unanswered, especially about its regulation in extrahepatic tissues. Even the extensively studied regulation of hepatic fatty acid oxidation by malonyl-CoA continues to be further investigated to provide an understanding of the regulatory mechanism at the molecular level. In spite of impressive progress in the area of peroxisomal β -oxidation, aspects of this process remain unresolved. For example, it is unclear how fatty acids enter peroxisomes and how products exit from this organelle. Also, the transcriptional regulation of this process has not been fully explored. Moreover, the cooperation between peroxisomes and mitochondria in fatty acid oxidation remains to be studied. Not all of the reactions of the β -oxidation spiral have been verified experimentally and hence some may not take place as envisioned. Finally, the complete characterization of known disorders of β -oxidation in humans and the identification of new disorders will raise questions about some accepted features of this process and will prompt re-investigations of issues thought to be resolved.

Abbreviations

ACC	acetyl-CoA carboxylase
ACS	acyl-CoA synthetase
CPT	carnitine palmitoyltransferase
ETF	electron-transferring flavoprotein
FABP	fatty acid binding protein
MFE	multifunctional enzyme
PPAR	peroxisomal proliferator-activated receptor
SCP	sterol carrier protein

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Fatty acid synthesis in eukaryotes

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1. Introduction

Fatty acids fulfill several crucial roles in animals. They represent a major storage form of energy, they provide an essential structural component of membranes, through direct covalent linkage they are used to modify and regulate the properties of many proteins and, as components of certain lipid signaling molecules, they perform important roles in metabolic regulation. This chapter will focus primarily on the structure, mechanism of action and regulation of the enzymes responsible for the biosynthesis of long-chain saturated fatty acids, *de novo*. All of the carbon atoms of fatty acids are derived from the two-carbon precursor, acetyl-CoA and until the demonstration in 1958 that CO₂ was required for the biosynthesis of fatty acids *de novo*, it had been assumed that the pathway utilized the enzymes of β -oxidation operating in the reverse direction (S.J. Wakil, 1958). Studies during the late 1950s and early 1960s, primarily in the laboratories of S.J. Wakil, P.R. Vagelos and F. Lynen, established clearly that the thermodynamic barrier posed by condensation of two acetyl-CoA molecules is circumvented by the introduction of an energy-dependent carboxylation step that generates malonyl-CoA as a co-substrate. Much of the early progress in identifying the individual enzymes involved uniquely in the biosynthetic route was made using the *Escherichia coli* system and it was not until the mid-1970s that it became clear that in eukaryotes, the enzymes are covalently linked in 'multifunctional' polypeptides. In prokaryotes, exemplified by *E. coli*, more than 10 individual proteins are involved in the biosynthesis of long-chain saturated fatty acids from acetyl-CoA (Chapter 3). This system, in which each enzyme is present on a single polypeptide, became known as the 'type II' FAS system and the multifunctional polypeptide system, exemplified by yeast and animals, became known as the 'type I' FAS system. Remarkably, in animals, the catalytic components required for the entire fatty acid biosynthetic pathway are integrated into two multifunctional polypeptides, acetyl-CoA carboxylase (ACC) and fatty acid synthase (FAS) [1]. Both proteins are posttranslationally modified by the covalent attachment of vitamin derivatives that play essential roles as 'swinging arms' in the translocation of intermediates between different catalytic sites. In ACC, a biotin moiety, attached to the ϵ -amino group of a lysine residue, serves as a carboxyl carrier between the carboxylase and transcarboxylase catalytic domains and in the FAS, a phosphopantetheine moiety, attached to a serine hydroxyl, serves to transport substrates and the growing acyl chain through the various catalytic centers of the complex. Expression of both enzymes is regulated at the transcriptional level, in a tissue-specific manner, in response to various developmental, nutritional, and hormonal signals [2]. In addition, ACC is subject to short-term regulation by allosteric and phosphorylation

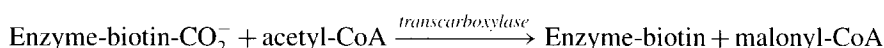
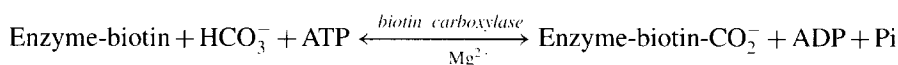
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mechanisms and is commonly regarded as the pace setting enzyme for fatty acid synthesis [3]. The only free intermediate in the entire biosynthetic pathway is malonyl-CoA; all other intermediates exist only as covalently bound acyl-enzyme complexes [1]. Recently, this metabolite has been recognized as playing a critical role in a fuel sensing and signaling mechanism that regulates food intake and energy metabolism [4,5].

2. Acetyl-CoA carboxylase

2.1. The reaction sequence

The formation of malonyl-CoA from acetyl-CoA is a two-step reaction involving first, the ATP-dependent carboxylation of the biotinyl moiety, followed by transfer of the carboxyl to an acetyl-CoA acceptor.



2.2. Domain organization

The amino acid sequence and domain structure of ACC is highly conserved in eukaryotes, such that antibodies raised against animal and fungal forms cross-react with each other. The two catalytic domains, biotin carboxylase and transcarboxylase, are located in the amino- and carboxyterminal halves of the polypeptide, respectively (Fig. 1). Between these domains lies the conserved biotin-binding site motif Met–Lys–Met within the biotin

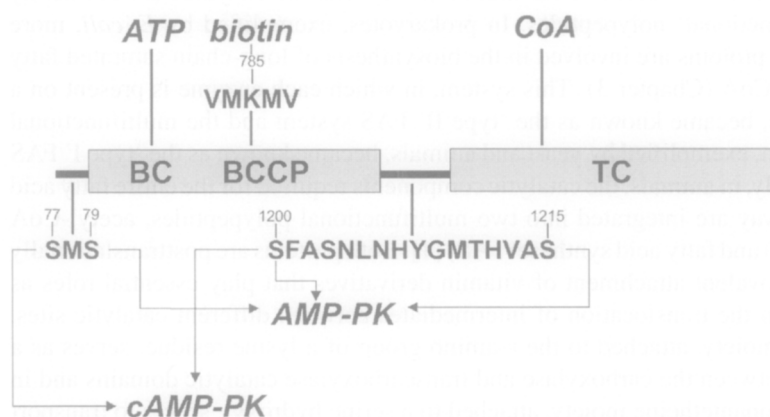


Fig. 1. Domain organization of the animal ACC α -isoform. The ACC contains 2346 residues. The precise locations of boundaries between the biotin carboxylase (BC), biotin carboxyl carrier protein (BCCP) and transcarboxylase (TC) domains has not been established; none of the domains has been expressed as individual proteins. The locations of ATP, biotin and CoA-binding sites are shown above the map and the location of phosphorylation sites shown to influence activity are shown below the map. Adapted from Corton and Hardie [6] and Kim [7].

carboxyl carrier protein domain. The presumed ATP and HCO_3^- binding sites within the biotin carboxylase domain and the acetyl-CoA-binding site, within the transcarboxylase domain, have been identified by sequence analysis but have yet to be confirmed by mutagenesis. The major differences between animal and yeast ACC are in the first 100 or so aminoterminal residues and near the center of the polypeptides, both regions that are implicated in regulation of activity of the animal form by phosphorylation.

The minimal functional unit (protomer) is a homodimer. In the presence of citrate, the inactive protomer rapidly assumes a catalytically active conformation and undergoes a slower, reversible polymerization to a long filamentous polymeric structure, ~ 100 Å wide and up to 5000 Å long, containing as many as 20 protomers [8]. The polymeric structure may stabilize the enzyme protomers in an active conformation. Depolymerization and inactivation of the enzyme is promoted by long-chain acyl-CoA thioesters, which are direct competitors with citrate, and malonyl-CoA.

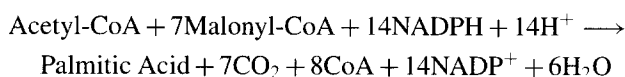
2.3. *Isoforms*

Two major isoforms of ACC, α and β , have been described in animals (K.-H. Kim, 1988, 1996; S.J. Wakil, 1995) that, paradoxically, have distinctly different roles in metabolism [4]. The 265 kDa α -isoform is the predominant form expressed in the soluble cytoplasm of lipogenic tissues such as adipose, liver and mammary gland, where it plays an essential role in providing malonyl-CoA as a carbon source for fatty acid synthesis. The 280 kDa β -isoform, on the other hand, is expressed mainly in heart and skeletal muscle, tissues with very low lipogenic capacity, but also to some extent in liver, and is associated with the outer mitochondrial membrane. In these tissues, malonyl-CoA produced by the β -isoform functions primarily as a negative regulator of carnitine palmitoyltransferase I and, in so doing, controls the flux of fatty acids into the mitochondria for β -oxidation (Chapter 5). The two isoforms exhibit extensive sequence similarity, the major difference being the presence of ~ 120 additional residues at the aminoterminal of the β -form that serve to anchor the enzyme in the outer mitochondrial membrane [7]. The presence of a third isoform, containing eight additional amino acids, four residues upstream from Ser1200, has been inferred by detection of a mRNA species containing an additional 24 nucleotides that may result from alternative splicing of the gene for the α -isoform [9]. The mRNA encoding the longer version of the α -isoform dominates in liver and adipose, whereas that encoding the shorter one dominates in lactating mammary gland. Activity of the two α -isoforms appears to be regulated by different phosphorylation mechanisms (Section 4.2).

3. *Fatty acid synthase*

3.1. *The reaction sequence*

The overall reaction catalyzed by the animal FAS can be summarized by the equation:



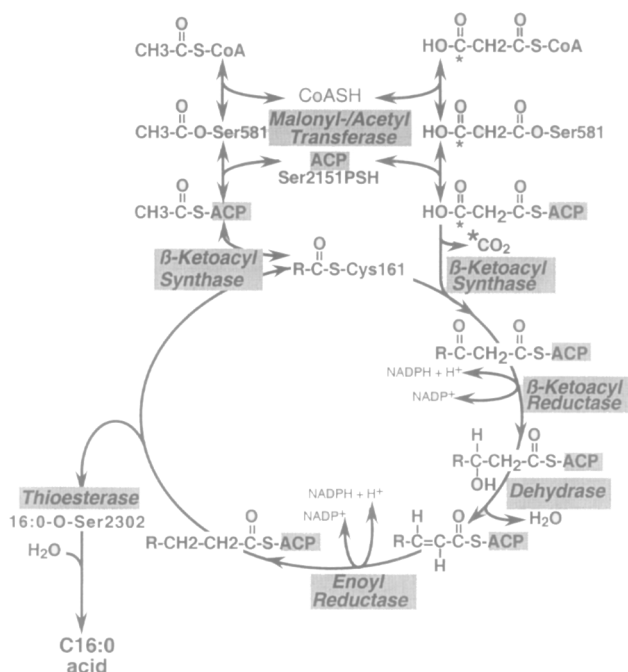


Fig. 2. Reaction sequence catalyzed by the fatty acid synthase. The location of all covalent acyl-enzyme intermediates is shown. PSH, phosphopantetheine moiety. The condensation reaction involves the stereochemical inversion of the C2 of the malonyl moiety. Reduction of the 3-ketoacyl moiety to a (3*R*)-hydroxyacyl moiety involves transfer of the prochiral 4*S* hydrogen of NADPH. Dehydration to the *trans*-enoyl moiety occurs through the syn-elimination of the prochiral 2*S* hydrogen and the 3*R* hydroxyl group. Reduction of the *trans*-enoyl moiety proceeds by the transfer of the prochiral 4*R* hydrogen of NADPH to the prochiral 2*S* position [11]. Recent studies indicate that the malonyl C-3 is released as HCO₃⁻, rather than CO₂ in the decarboxylation reaction (S. Smith, 2002).

The pathway can be visualized as a cyclic process in which the acetyl primer undergoes a series of Claisen condensation reactions with seven malonyl extender molecules and, following each condensation, the β-carbon of the β-ketoacyl moiety formed is completely reduced by a three-step ketoreduction–dehydration–enoylreduction process [10]. The saturated acyl chain product of one cycle becomes the primer substrate for the following cycle, so that two saturated carbon atoms are added to the primer with each turn of the cycle (Fig. 2). The final product is released as a free fatty acid by the animal FAS and as a CoA ester by the yeast complex.

3.2. The catalytic components

Several catalytic elements are required for the biosynthetic process: acyltransferases that load the primer and extender substrates onto the FAS complex; a posttranslationally phosphopantetheinylated acyl carrier protein, which translocates the various intermediates between catalytic sites; a β-ketoacyl synthase, which performs the condensation reaction; the β-ketoacylreductase, dehydrase and enoylreductase enzymes responsible

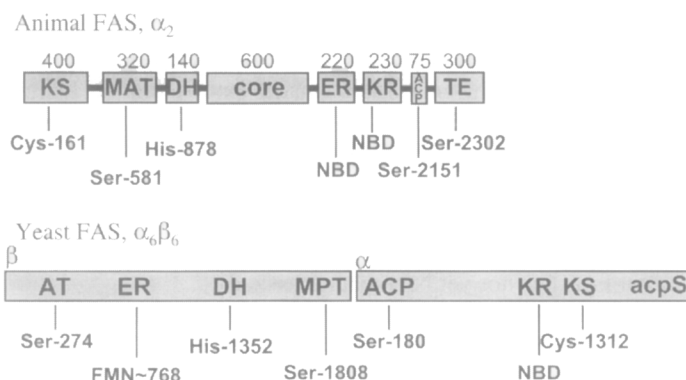


Fig. 3. Domain maps of the animal and yeast fatty acid synthases. The rat fatty acid synthase contains 2505 residues. KS, β -ketoacyl synthase; MAT, malonyl/acetyltransferase; DH, dehydrase; ER, enoylreductase; KR, ketoreductase; ACP, acyl carrier protein; TE, thioesterase; NBD, nucleotide-binding domain. The approximate number of residues in each domain is indicated above the map. The MAT, ACP and thioesterase domains have been isolated, by limited proteolysis, and expressed as independent proteins. The central core region has no known catalytic function, but is thought to contain a dimerization domain that promotes interaction between the subunits. The location of essential active-site residues, and the nucleotide-binding region of the two reductases are shown below the map. The yeast β subunit contains 1887 residues and the α subunit 1845. AT, acetyltransferase; MPT, malonyl/palmitoyltransferase; acpS, holo-ACP synthetase. The boundaries between component domains of the yeast protein are less clearly defined; none of these domains has been isolated or expressed as independent proteins. Adapted from Smith [13] and Fichtlscherer et al. [14].

for the β -carbon processing reactions and a chain terminating enzyme responsible for product release. Although the animal and yeast FASs employ essentially the same overall reaction scheme, they exhibit minor differences in the enzymatic details and major differences in their macromolecular architecture. (1) Whereas the animal FAS utilizes the same acyltransferase for loading of both the acetyl and malonyl substrates (S. Smith, 1996, 1997), the yeast FAS uses two different enzymes, one specific for acetyl moieties, the other able to load acetyl or malonyl [12]. (2) The animal FAS releases the product as a free fatty acid, through the action of a thioesterase, which severs the bond between fatty acid and phosphopantetheine thiol (S. Smith, 1978) whereas the yeast FAS uses the same broad specificity acyltransferase responsible for substrate loading to transfer the product to a CoA acceptor, so that the product is an acyl-CoA thioester [12]. (3) The enoylreductase of the animal FAS utilizes only NADPH, but the yeast enoylreductase requires both NADPH and FMN as cofactors [1].

3.3. Domain organization

In eukaryotes, the enzymes required for fatty acid synthesis *de novo* are integrated into large multifunctional polypeptides that are located in the cytosol. However, the domain organization and overall molecular architecture of the animal and yeast complexes are quite different (Fig. 3). Thus, the animal FAS is a dimer of identical, 272 kDa polypeptides [10,11,13], whereas the yeast FAS contains six copies each of two different subunits, α and β , of molecular masses 208 kDa and 220 kDa, respectively (E. Schweizer, 1973).

Remarkably, the ordering of the catalytic domains in the yeast multifunctional polypeptides is quite dissimilar to that of the animal and the two multifunctional proteins appear to have evolved along quite different gene fusion pathways. A unique feature of the yeast FAS is that phosphopantetheinylation is catalyzed by a discrete domain located at the carboxyterminus of the α subunit (E. Schweizer, 2000). This endogenous phosphopantetheinyl transferase cooperates with the ACP domain of the second α subunit within the $\alpha_2\beta_2$ protomer. Phosphopantetheinylation of the animal, prokaryotic and plant ACPs is catalyzed by separate, discrete enzymes. The phosphopantetheinyl transferase acting on the animal FAS has not yet been characterized.

Both eukaryotic forms of FAS require the oligomeric structure for activity: the animal α_2 complex has two equivalent sites for fatty acid synthesis (S.J. Wakil, 1984; S. Smith, 1995) and the yeast $\alpha_6\beta_6$ complex has six (S.J. Wakil, 1980). Thus far the complexes have been refractory to crystallographic analysis and structural analysis has been limited to lower-resolution methods. Electron microscopy and small-angle neutron scattering studies (S.J. Wakil, 1987; A. Ikai, 1988) reveal the animal FAS as an ellipsoid structure, $\sim 216 \times 144 \times 72$ Å, containing two cavities, one at each end of the molecule. The two cavities may represent the two sites for fatty acid synthesis in the dimeric structure. Electron micrographic images of the complex labeled with Fab fragments derived from antibodies raised against the thioesterase domain indicate that the two carboxylterminal domains are located one at each pole of the ellipsoid. Imaging of complexes labeled by Fab fragments from antibodies raised against the region between the dehydrase and enoylreductase domains, indicate that this non-catalytic core is located near the center of the ellipsoid structure. This core region contains dimerization domains that may serve to stabilize interactions between the two subunits [13].

A more detailed 25 Å resolution structure of the yeast FAS has been computed from galleries of electron microscopic images (J. Stoops, 1996). The complex appears as a barrel-shaped structure, $\sim 245 \times 220$ Å, formed by three zig-zag-shaped, α subunit pairs, each with a pair of over- and under-lying arch-shaped β subunits that cap the ends of the barrel. Thus the protomeric unit is an $\alpha_2\beta_2$ structure. The barrel contains six cavities that likely constitute the six equivalent sites for fatty acid synthesis. Each cavity has two funnel-shaped openings of ~ 20 Å that may allow diffusion of substrates into, and products out of, the catalytic centers.

3.4. Chain initiation

Since the animal FASs utilize a common loading site for both the primer and chain-extender substrates, acetyl- and malonyl-CoA are mutually competitive inhibitors of each other. These FASs cannot order the sequential loading of one acetyl moiety and seven malonyl moieties but instead rely on a self-editing process, in which both substrates are rapidly exchanged between CoA ester and enzyme-bound forms [11,13]. Only when the appropriate pair of substrates is bound, that is an acetyl, or longer saturated acyl moiety, at the cysteine active site and a malonyl moiety at the phosphopantetheine site, does condensation take place. Nonproductive binding, for example when the malonyl moiety binds before the acetyl moiety, or when two acetyl moieties are bound sequentially, results in rapid translocation of the substrates back to

their CoA ester form. Paradoxically then, for the system to function efficiently, free CoA, which appears on the product side of the equation, must be available at all times (S. Smith, 1982).

The yeast FAS employs two different acyltransferases for loading of acetyl and malonyl substrates and does not rely on a self-editing mechanism to facilitate translocation of the appropriate substrates to the condensation sites [12]. Nevertheless, since these complexes terminate acyl-chain growth by transfer of the acyl moiety to a CoA acceptor, free CoA is also required for efficient operation of the pathway. Both the animal α_2 and yeast $\alpha_6\beta_6$ FASs exhibit the expected stoichiometry for acyl chain assembly when all substrates and cofactors are present (i.e. two fatty acyl chains per α_2 , six fatty acyl chains per $\alpha_6\beta_6$). However, neither the animal (S. Smith, 1985) nor the yeast [12] FAS can be fully saturated with substrate, when only malonyl-CoA or acetyl-CoA is present. This substoichiometric binding of substrates by the yeast FAS has been referred to by Schweizer and colleagues as a negative cooperativity that ensures that the complex is not overloaded with substrates that would otherwise compete for sites required for processing the various acyl-enzyme intermediates present on the enzyme [12].

3.5. Chain termination and product specificity

The eukaryotic FASs synthesize predominantly the 16-carbon saturated product with smaller amounts of 14- and 18-carbon products. The enzymatic basis of product specificity of the animal FAS has been studied in some detail in Gordon Hammes', and in our, laboratory. The β -ketoacyl synthase has a relatively broad chain-length specificity and is able to transfer efficiently saturated acyl moieties with 2 to 14 carbon atoms from the phosphopantetheine thiol to the active-site cysteine thiol (Fig. 4). However, longer chain-length acyl moieties are transferred between thiols with increasing difficulty

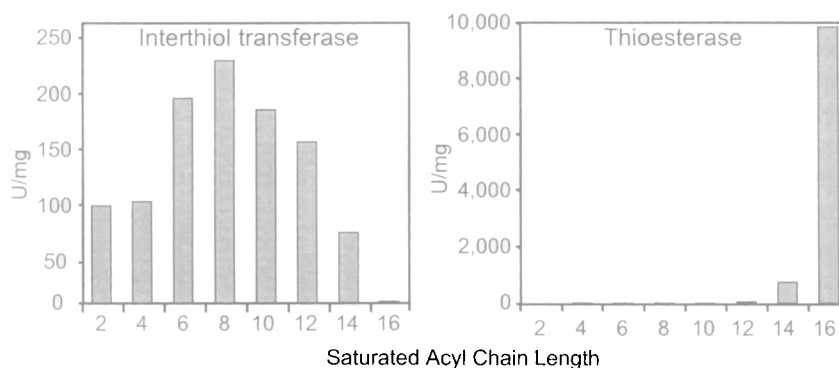


Fig. 4. Chain-length specificities of the interthiol transferase activity associated with the 3-ketoacyl synthase and the chain-terminating thioesterase. The interthiol transferase activity was assayed on a recombinant form of FAS in which functionality of the malonyl/acetyl transferase, ACP and thioesterase domains had been inactivated by mutation; acyl-CoAs were used as model acyl donors and pantetheine as a model acceptor [15]. Thioesterase activity was assayed on a recombinant form of the isolated thioesterase domain and acyl-CoAs were used as model substrates (S. Smith, 1978).

[11]. In contrast, the chain-terminating thioesterase has very limited ability to remove acyl moieties with less than 16 carbon atoms from the phosphopantetheine thiol (S. Smith, 1978, 1991). Thus the specificities of the chain-elongating and chain-terminating enzymes complement each other perfectly, ensuring that the 16-carbon fatty acid is the major product (Fig. 4). The β -ketoacyl synthase cannot transfer incompletely reduced intermediates (β -keto-, β -hydroxy or enoyl) from the phosphopantetheine thiol to the active-site cysteine, so only the saturated intermediates are elongated (S. Smith, 1997).

Some specialized animal tissues, such as the lactating mammary gland of mammals and the preen glands of birds are able to use the FAS to produce the shorter chain-length fatty acids characteristic of milk fat and the oily secretion used for waterproofing feathers, respectively. In the mammary glands of nonruminants (J. Knudsen, 1981; S. Smith, 1981) and the preen glands of birds (P.E. Kolattukudy, 1981), formation of these unusual products is attributable to the expression of a separate, discrete 29 kDa thioesterase that is able to access and release saturated acyl chain intermediates directly from the phosphopantetheine of the FAS complex. Ruminants, on the other hand, possess a FAS with unusual properties. In these species, the acyltransferase responsible for substrate loading on the FAS has a very broad acyl chain-length specificity so that saturated C4, C6, C8 and C10 intermediates can equilibrate between enzyme-bound and CoA ester forms (J. Knudsen, 1981). In the lactating mammary gland, this pool of short chain-length acyl-CoAs can be utilized for triglyceride synthesis and milk fat production. The FASs of some specialized sebaceous glands, such as the harderian, meibomian and preen glands, produce significant amounts of odd carbon-number and methyl-branched fatty acids, primarily as a result of the availability of significant amounts of propionyl-CoA and methylmalonyl-CoA as primer and chain-extender substrates (P.E. Kolattukudy, 1978, 1987).

3.6. *Interdomain communication*

The 20-Å-long, phosphopantetheine 'swinging arm' of the animal and fungal FASs has long been accepted as a key factor in allowing each of the six catalytic centers access to the ACP domains. However, distances between some of the catalytic sites of the animal FAS have been estimated, by fluorescence energy transfer, as being considerably greater than 20 Å; for example, 48 Å between the phosphopantetheine thiol and the thioesterase active site and 40 Å between the phosphopantetheine thiol and the two nucleotide-binding domains (P.E. Kolattukudy, 1985; G.G. Hammes, 1986). These observations suggest that cooperation between the acyl carrier domain and the catalytic domains during fatty acid biosynthesis may involve extensive conformational changes within the complex. Such conformational changes may be mediated in part through the presence of flexible linker regions between the catalytic domains.

Although the dimeric form of the animal FAS contains two sites for fatty acid synthesis, the monomeric form is incapable of transferring substrates from the CoA ester form to the phosphopantetheine thiol and cannot catalyze the condensation reaction. In 1991, the demonstration by J. Stoops and S.J. Wakil that the cysteine active-site thiol could be cross-linked to the phosphopantetheine thiol of the opposite subunit by dibromopropanone inspired the proposal that the two subunits of the animal

FAS are positioned in a fully extended antiparallel orientation, such that two centers for fatty acid synthesis are formed at the subunit interface. The arrangement of domains in the multifunctional polypeptide is consistent with a model in which each center for fatty acid synthesis requires cooperation of the three aminoterminal domains of one subunit (β -ketoacyl synthase, malonyl/acetyl transferase and dehydrase) with the four carboxyterminal domains of the other subunit (enoylreductase, β -ketoacylreductase, acyl carrier protein and thioesterase) [10,11,13]. The development, in our laboratory, of procedures for the production of recombinant animal FAS dimers containing different mutations on each subunit has afforded a unique opportunity to test this model (S. Smith, 1998). Several of our findings cannot be explained adequately by the prevailing model [16]. First, *in vitro* mutant complementation analyses have revealed that the β -ketoacyl synthase and malonyl/acetyltransferase can cooperate functionally with the acyl carrier protein domains of *both* subunits, although the intrasubunit functional interaction is less efficient than is the intersubunit interaction. Second, mutant complementation analysis also indicates that the β -hydroxy dehydration reaction is catalyzed exclusively by cooperation between the dehydrase and acyl carrier protein domains associated with the *same* subunit.

Third, the results of a thorough reinvestigation of the specificity of dibromopropanone interaction with the animal FAS are inconsistent with the original interpretation that cross-linking of active-site cysteine and phosphopantetheine thiols by this reagent occurs exclusively intersubunit. Thus, treatment of animal FAS dimers with dibromopropanone generates three new molecular species with decreased electrophoretic mobilities; none of these species is formed by fatty acid synthase mutant dimers lacking either the active-site cysteine of the β -ketoacyl synthase domain (Cys161Ala FAS mutant) or the phosphopantetheine thiol of the acyl carrier protein domain (Ser2151Ala FAS mutant). When dimers carrying one or both mutations on one or both subunits were treated with dibromopropanone and analyzed by a combination of sodium dodecyl sulfate/polyacrylamide gel electrophoresis, Western blotting, gel filtration and matrix-assisted laser desorption mass spectrometry, the two slowest moving of the three cross-linked species were identified as doubly and singly cross-linked dimers, respectively, whereas the fastest moving species was identified as originating from internally cross-linked subunits. This internally cross-linked species accounted for as much as 35% of the total cross-linked species [17].

On the other hand, certain features of the original model are supported by results of the mutant complementation studies. Thus, the substrate loading and condensation reactions are catalyzed most efficiently by cooperation of the β -ketoacyl synthase and malonyl/acetyltransferase domains with the ACP domain of the opposite subunit and the β -carbon processing and chain-terminating reactions are catalyzed by cooperation of the phosphopantetheine moiety with catalytic domains associated with the same subunit [16].

Perhaps the most important implication of these recent findings is that the structural organization of the animal FAS must permit head-to-tail interactions between domains located on opposite subunits and must allow for functional interactions between domains located distantly on the same subunit. These requirements could be met by a modified head-to-tail model in which the two ACP domains, at each of the subunit 'tails', have

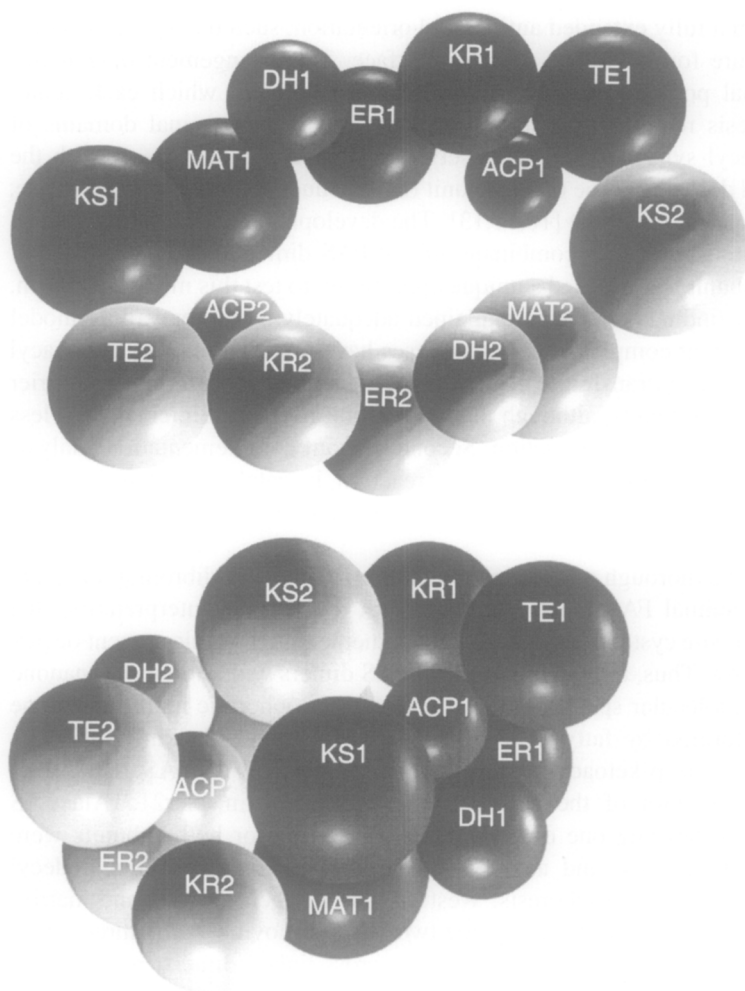


Fig. 5. Alternative functional models for the animal fatty acid synthase. (Top) The classical 'head-to-tail' model [10,11,13,16]. (Bottom) The alternative model [16]. The two subunits are distinguished by dark and light shading and by the numbers 1 or 2 on each domain. KS, β -ketoacyl synthase; MAT, malonyl/acetyltransferase; DH, dehydrase; ER, enoyl reductase; KR, β -ketoacyl reductase; ACP, acyl carrier protein, TE, thioesterase. For simplicity, only those domains for which catalytic functions have been established are shown in the model; thus the non-catalytic core region has been omitted.

access to the β -ketoacyl synthase and malonyl/acetyltransferase domains at the 'heads' of either subunit (Fig. 5, top). Although such a mechanism would require a remarkable degree of flexibility within the FAS polypeptides, to ensure productive interactions both inter- and intrasubunit, it cannot be formally excluded at this time.

An alternative model has been envisaged that requires much less conformational flexibility in the subunits (Fig. 5, bottom). In this model, the β -ketoacyl synthase and malonyl/acetyltransferase domains of both subunits are located near the center of the

dimer, where they can freely access the ACP domains of both subunits, and the domains responsible for the β -carbon processing reactions are located at opposite ends of the dimer and therefore have access to only one ACP domain. Thus the model retains the original concept of head-to-tail oriented subunits, but also allows for head-to-tail interactions between domains of the same subunit.

4. Short-term regulation of fatty acid synthesis

Fatty acids, esterified in triglycerides, represent the major form of stored energy in animals. A relatively small proportion of excess caloric intake in the form of carbohydrate is stored as glycogen, most is converted to fat via the *de novo* lipogenic pathway. Thus the flux of substrate through the pathway is high in animals that have ingested high amounts of carbohydrate and is low during periods of fasting. In most animals, the liver and adipose tissue are the major sites of fatty acid biosynthesis, although the pathway is also of vital importance in other tissues during certain stages of development. In animals having access to a regular daily supply of food, fatty acid synthesis in the liver varies dramatically during the diurnal cycle. In the laboratory rat fed *ad libitum*, the peak in hepatic lipogenesis occurs during the feeding period, in the dark phase of the daily cycle, and is paralleled by changes in the activity of ACC (N. Iritani, 1985, 1990). Untreated diabetic rats do not exhibit this diurnal rhythm in lipogenic activity and ACC activity remains low throughout each cycle; administration of insulin restores the normal diurnal rhythm. However, most investigations into the role of hormones and nutrients in the regulation of lipogenesis have not studied these diurnal changes, but instead have focused on models in which the animals are first subjected to the stress of a prolonged fast of one or two days, prior to reintroduction of a food supply.

4.1. Regulation of substrate supply for fatty acid synthesis

Following a period of fasting, the initial stimulus for activation of the lipogenic pathway appears to be the influx of dietary carbohydrate. Glucose stimulates secretion of insulin from pancreatic β -cells and, as the ratio of insulin to glucagon increases in the blood, the activities of several enzymes in the glycolytic and lipogenic pathways are elevated, while the activities of key gluconeogenic enzymes are decreased (Fig. 6). These early changes in activity are brought about primarily by changes in the catalytic efficiency of key enzymes, whereas the longer-term responses can also involve changes in enzyme concentration (Section 5.4).

The resulting increased flux of pyruvate into the mitochondria and the subsequent activation of pyruvate dehydrogenase causes a rapid increase in the production and export of citrate via the tricarboxylate anion carrier. The elevated concentration of cytosolic citrate can simultaneously provide acetyl-CoA, the carbon source for fatty acid synthesis, through the action of ATP: citrate lyase, and stimulate the conversion of acetyl-CoA to malonyl-CoA, by activation of ACC. The reducing equivalents required for fatty acid synthesis are provided largely through the action of malic enzyme and

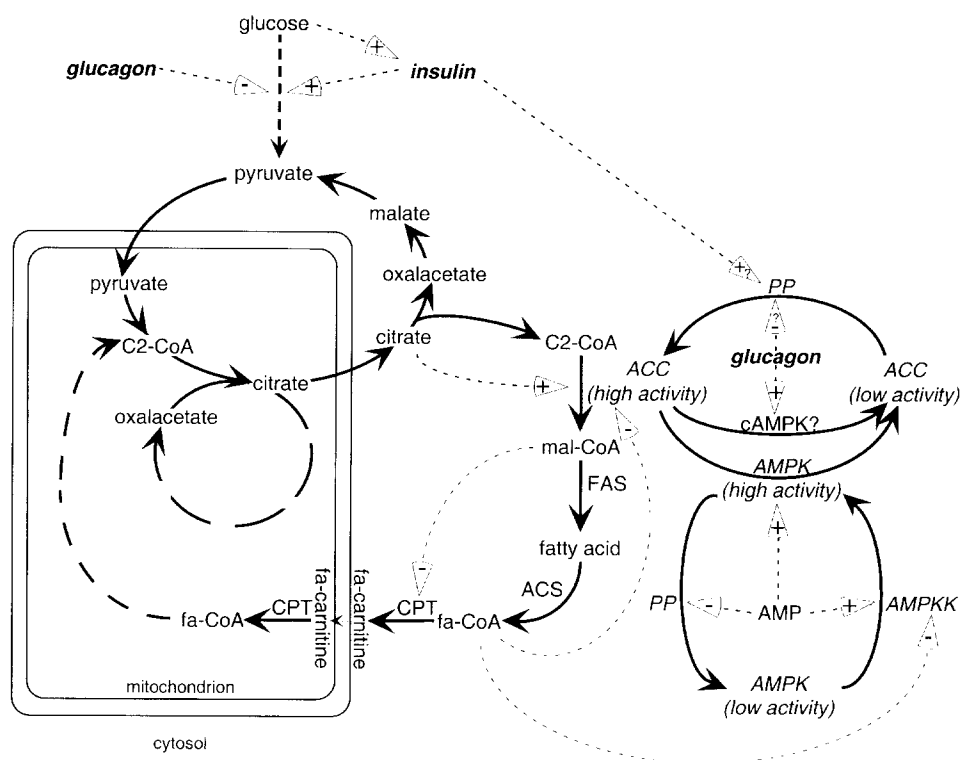


Fig. 6. Schematic representation of the pathways involved in the conversion of glucose to fatty acid. Dashed arrows with + or - signs, indicate points of positive or negative regulation. ACS, acyl-CoA synthetase; PP, protein phosphatase; AMPKK, AMPK kinase.

the two pentose phosphate cycle enzymes, glucose-6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase. The supply of NADPH by these enzymes appears to be largely dependent on its rate of utilization and is unlikely to be limiting in the lipogenic pathway.

The two pancreatic hormones, insulin and glucagon exert essentially opposite effects on energy metabolism. Thus, during the intervals between feeding and during prolonged periods of fasting, when the ratio of insulin to glucagon decreases in the blood, the activities of key enzymes in the glycolytic and lipogenic pathways are decreased, while the activities of key gluconeogenic enzymes are increased resulting in a reversal of the process described above.

4.2. Regulation of acetyl-CoA carboxylase α activity by reversible phosphorylation

ACC preparations purified from lipogenic tissues contain up to eight phosphate residues per subunit. Six of the phosphorylation sites are located within the first 100 aminoterminal residues, the other two near the center of the polypeptide (Fig. 2). Mild, limited proteolysis of ACC results in cleavage of the highly phosphorylated aminoterminal

domain and full activation of the enzyme in the absence of citrate. Unless special precautions are observed to prevent activation of protein kinases during tissue disruption, ACC preparations typically are highly phosphorylated and require addition of supraphysiological concentrations of citrate (1–5 mM) for full activity, regardless of the nutritional status of the animal (N.B. Madsen, 1987; S.J. Wakil, 1988 and D.G. Hardie, 1992). However, when livers of fed animals are freeze-clamped prior to isolation of the enzyme, preparations are obtained that have low phosphate content and can be fully activated by physiological concentrations of citrate (~ 0.2 mM). On the other hand, the enzyme purified from freeze-clamped livers of fasted animals has a high phosphate content and requires high added citrate concentrations for full activity. A plethora of kinases has been implicated in the phosphorylation of the enzyme (Fig. 2) but, despite the flurry of activity that occurred in this field in the 1980s and early 1990s, there is still no clear consensus as to which of them is physiologically the most important in regulating ACC activity. Two kinases, one dependent on cAMP for activity (cAMPK) the other dependent on 5'-AMP (AMPK) can inactivate ACC by phosphorylation in vitro and have been considered the strongest candidates [6,7]. The cAMPK phosphorylates ACC at Ser77 and Ser1200 and causes primarily an increase in the K_a for citrate and a slight reduction in V_{max} , whereas the AMPK phosphorylates mainly at Ser79, Ser1200 and, to a lesser extent, at Ser1215 and produces a large decrease in V_{max} . The AMPK has been considered as a particularly attractive candidate, since it is also able to phosphorylate and inactivate 3-hydroxy-3-methylglutaryl coenzyme-A reductase, and so potentially is in a position to coordinately regulate both the fatty acid and sterol biosynthetic pathways [6]. Treatment in vitro with cAMPK and AMPK of recombinant ACCs, mutated in one or more of the potential phosphorylation sites, indicated that when only Ser79 is available as a phosphorylation target, AMPK is able to inactivate ACC and when only Ser1200 is available, cAMPK, but not AMPK is able to inactivate ACC [7]. Thus, the critical target for cAMPK appears to be Ser1200 and for AMPK, Ser79. Strong arguments have been made supporting roles for both cAMPK and AMPK as regulators of ACC activity. During fasting, glucagon secretion raises the intracellular concentration of cAMP, which potentially can result in phosphorylation and inactivation of ACC by cAMPK, and indeed glucagon treatment of hepatocytes increases phosphorylation at Ser1200 (D.G. Hardie, 1988). Similarly, during fasting, as ATP production falls into deficit and AMP concentrations rise, AMPK could be activated, resulting in phosphorylation and inactivation of ACC. Indeed not only is AMPK activated directly by AMP, but so is an AMPK kinase that activates AMPK by phosphorylation, whereas a AMPK phosphatase that inactivates AMPK is inhibited by AMP (Fig. 6). Nevertheless, some puzzling observations remain unresolved. First, a report from Ki-Han Kim's laboratory in 1990 concluded that the α -isoform of ACC containing eight additional residues near Ser1200 is not phosphorylated at this site by the cAMPK. However, the same report documented that it is precisely this putative cAMPK-resistant, longer isoform that dominates in liver, the major site of lipogenesis and a tissue that is responsive to glucagon; this finding casts serious doubt on the role of the cAMPK in regulation of ACC activity. Second, although the diurnal rhythm in the activation state of ACC correlates nicely with variations in the phosphate content at position Ser79, AMPK activity remains at the same level throughout the light and dark cycles [6]. Furthermore,

no reduction in AMPK activity is observed when either hepatocytes or pancreatic β -cell lines are exposed to high levels of glucose. The possibility that it is in fact a phosphatase with specificity for Ser79 of ACC that is the regulated enzyme has yet to be resolved.

Collectively, these observations indicate that the catalytic activity of ACC is determined by a complex interplay between citrate-induced conformational changes and phosphorylation events. However, the question as to exactly how the covalent modification and conformational changes in ACC bring about changes in activity has not yet been addressed at the structural level.

4.3. Malonyl-CoA, fuel sensing and appetite control

Within the last several years there has grown an increasing awareness of the important role played by malonyl-CoA as a fuel sensing and signaling molecule [4]. Of 28 intermediates formed between acetyl-CoA and palmitic acid, malonyl-CoA is the only true free intermediate and, as a regulator of CPTI, is uniquely positioned to control both the rate of biosynthesis and degradation of fatty acids.

Malonyl-CoA formed in liver is utilized primarily as a substrate for fatty acid synthesis, but at the same time, as an inhibitor of carnitine palmitoyltransferase I, it controls entry of fatty acids into the mitochondria for oxidation (J.D. McGarry, 1980) (Fig. 6). Thus, following feeding, when excess dietary carbohydrate is being converted to fat and tissue malonyl-CoA levels are elevated, activity of carnitine palmitoyltransferase I is inhibited and flux of fatty acids into the mitochondria is prevented [4]. This control mechanism ensures that the fatty acid biosynthetic and oxidation pathways are not simultaneously activated in the same tissue, thus avoiding the establishment of a futile cycle. Conversely, during periods of fasting hepatic malonyl-CoA levels are lowered, flux of substrate through the fatty acid biosynthetic pathway is halted and the block in entry of fatty acids into the mitochondria is removed.

Recent studies have revealed a previously unsuspected correlation between intracellular concentration of malonyl-CoA and appetite control [5]. As anticipated, mice treated with cerulenin, an inhibitor of the β -ketoacyl synthase activity of the FAS (Chapter 3), exhibit decreased rates of hepatic lipogenesis and elevated intracellular malonyl-CoA. However, these animals also cease feeding, maintain normal activity, exhibit a dramatic loss of adipose tissue mass and suffer a major reduction in body weight. Treatment with 5-(tetradecyloxy)-2-furoic acid, an inhibitor of ACC, also results in decreased fatty acid synthesis but has no effect on feeding behavior. Indeed, administration of 5-(tetradecyloxy)-2-furoic acid restores normal feeding behavior in mice treated with a FAS inhibitor, indicating that it is the elevated intracellular concentration of malonyl-CoA that triggers loss of appetite. The attenuation effect is observed when the compounds are administered intracerebroventricularly, implicating the central nervous system as the site of action of the inhibitors. This conclusion is further supported by the observation that expression of the hypothalamic neuropeptide Y, an appetite stimulant, is dramatically reduced in mice treated with an inhibitor of FAS. Since both ACC and FAS are expressed in hypothalamic neurons, it appears likely that the malonyl-CoA concentration sensing mechanism may be directly 'hard-wired' to the appetite control center in the hypothalamus. Under normal conditions, the rise in tissue malonyl-CoA

concentration following feeding may provide a 'satiety signal' indicating that excess caloric intake is now being channeled into fat stores, so that neuropeptide Y production is turned off and feeding ceases. Thus the condition induced by treatment with FAS inhibitors likely mimics the normal fed state so that the animal is deceived into a prolonged period of fasting for the duration of the treatment.

Perhaps it would appear paradoxical that in mice treated with FAS inhibitors, where food intake is dramatically reduced and intracellular malonyl-CoA levels rise in the lipogenic tissues, fatty acid oxidation provides the major source of energy and adipose mass is reduced. The answer to this puzzle likely is attributable to unique metabolic control mechanisms operative in muscle, the main site of fatty acid oxidation during physical activity [4]. The activity of the ACC β -form and the concentration of malonyl-CoA in skeletal muscle decreases within seconds of the initiation of exercise. The mechanism by which activity of the β -form is regulated in muscle appears to involve both allosteric regulation by citrate and reversible phosphorylation, since activity of AMPK is elevated during muscle contraction. The mechanism of disposal of malonyl-CoA produced in muscle by ACC β is not well understood. Most likely malonyl-CoA is utilized by a cytosolic malonyl-CoA decarboxylase, since genetic defects in this enzyme produce a phenotype characteristic of mitochondrial fatty acid oxidation disorders (S.J. Gould, 1999). Almost certainly, in view of the scarcity of FAS in muscle, malonyl-CoA is not utilized for fatty acid synthesis, so that administration of the FAS inhibitors is unlikely to impact significantly the concentration of malonyl-CoA in muscle. The regulation of malonyl-CoA levels by different mechanisms in muscle and lipogenic tissues allows for use of fatty acids as metabolic fuel in muscle, even when malonyl-CoA levels may be high in liver, as in the case of animals treated with FAS inhibitors.

5. Regulation of the intracellular concentration of lipogenic enzymes

5.1. Strategies and methodology

The intracellular concentrations of lipogenic enzymes potentially can be altered through changes in their rate of synthesis or degradation. Thus changes in the rates of either transcription or translation, as well as changes in the stability of specific mRNAs and their encoded proteins need to be examined when evaluating factors that may regulate changes in enzyme concentration. In reality, the decrease in concentration of ACC and FAS that occurs during relatively long-term fasting appears to be due to both a cessation in transcription of their genes and a decrease in the half-life of the proteins themselves. On refeeding of animals that have been subjected to long-term fasting, transcription of ACC and FAS is activated and the half-life of the proteins is lengthened until a new steady state concentration of enzymes is established. During differentiation of preadipocytes into mature adipocytes, *in vitro*, the increase in the intracellular concentration of lipogenic enzymes can be attributed to both stabilization of their mRNAs and increased rate of transcription of their genes. In recent years investigators have focused primarily on factors that influence the rate of gene transcription, so that relatively few details are known as to how changes in the stability of the mRNAs and proteins are brought about.

Cell culture systems are widely used to study the regulation of gene transcription, as these cells can usually be induced to take up and express heterologous reporter gene constructs and can be manipulated to mimic changes in the hormonal and nutritional environment that occur in intact animals. The most commonly used cell culture systems are primary hepatocytes and hepatoma cell lines, as models for studying the effects of hormones and nutrients on expression of lipogenic enzymes in the liver, and 3T3-L1 and 30A5 cells, as adipocyte models. These adipocyte lines are derived from fibroblasts which, at confluence, can be modulated to differentiate and accumulate fat droplets. Upon differentiation of these preadipocytes, lipogenic enzymes accumulate to levels similar to that characteristic of normal adipose tissue, so that these cell lines are commonly used to study gene transcription in both differentiating and mature adipocytes. However, the differentiating preadipocyte culture system does not mimic all of the characteristics of intact adipose tissue and the responsiveness of hepatoma cells to hormones and nutrients is typically muted, compared to that of hepatocytes or the liver of whole animals. Therefore, caution needs to be exercised in interpreting results obtained with these model systems, particularly when corroborating evidence from *in vivo* experiments is lacking. The use of transgenic animals affords a powerful model system for evaluating the roles of putative *cis*-regulatory and *trans*-acting proteins in mediating tissue-specific gene expression in response to nutritional status.

Regulation of gene transcription involves interaction of DNA regulatory sequences, termed *cis*-acting elements, that are usually found in the 5'-flanking regions of genes, with nuclear proteins, termed *trans*-acting factors. Understanding the mechanism of transcriptional regulation of a gene, therefore, requires the identification of both the *cis*-acting elements in the gene and the *trans*-acting factors in the nuclei that bind to them as well as elucidation of the signaling events that lead to changes in the interaction between the *cis* elements and *trans* factors.

5.2. The acetyl-CoA carboxylase promoter

The two major isoforms of ACC, α and β , which exhibit about 80% amino acid sequence similarity, are encoded by separate genes that map to chromosomes 17q21 and 12q23.1, respectively. The α -isoform is expressed predominantly in tissues that exhibit high rates of fatty acid synthesis, such as liver and adipose tissue, whereas the β -isoform is expressed mainly in heart and skeletal muscle, and to a lesser extent in liver. Transcription of the α gene is under the control of two promoters, designated as PI and PII that are separated by 12.3 kbp. Five different species of mRNA, all of which contain the same base sequence in the coding region but differ in the 5'-untranslated region, are generated by the alternative splicing of the first four exons [18].

The usage of α gene promoters is tissue-specific (Fig. 7). Thus, only promoter PI is active in adipose tissue and promoter PII in mammary gland, whereas both promoters are active in liver. PII, which is active in all tissues at least at a low level, is sometimes described as a 'housekeeping promoter', although it contains a strong enhancer element and several regulatory *cis*-elements that confer responsiveness to external stimuli. A third ACC α promoter, PIII, has been described in sheep that gives rise to an N-terminal variant of ACC α that exhibits tissue-restricted mode of expression (M.C. Barber, 2001).

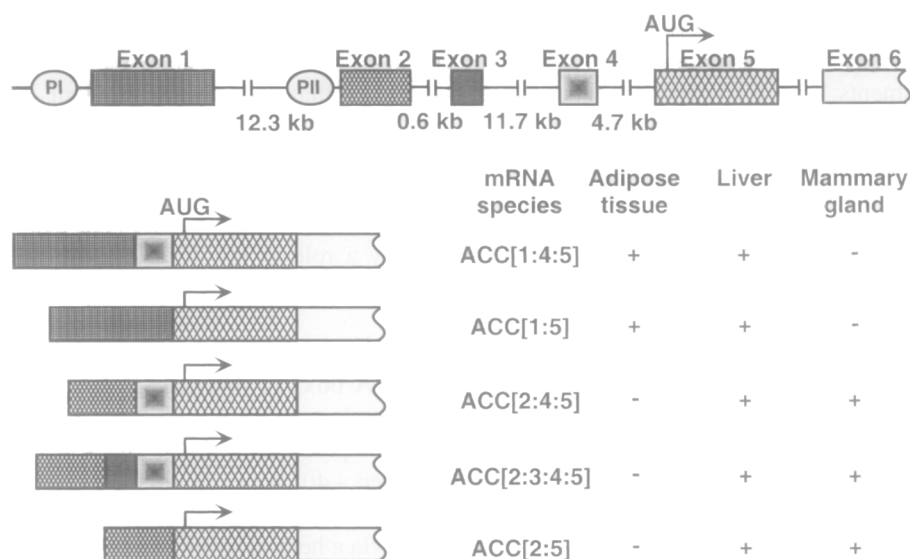


Fig. 7. Structure of the 5'-flanking region of the rat ACC α gene and tissue-specific distribution of the various mRNA species transcribed. The upper section of the figure shows a genomic map of the first six exons of the gene and includes the location and sizes of the intronic sequences. The circles labeled PI and PII indicate the positions of the two ACC gene promoters. The AUG codon, indicated by a right-angled arrow in exon 5, represents the translation initiation site. The lower section of the figure depicts the structural organization of the 5'-flanking region of the various mRNA transcripts. The mRNAs are named based on their exon content. The presence (+) or absence (-) of particular mRNA species in three different rat tissues is indicated by + and - signs, respectively. This figure is adapted from Kim [7].

The choice of gene promoter usage also seems to vary with animal species and developmental status. Thus, when animals are fasted and refed a high-carbohydrate, low-fat diet, transcripts from both PI and PII promoters are elevated in the liver of 12 days old chicks, whereas mainly the transcript from the PI promoter-driven transcript is induced in adult rats; similarly, changes in the thyroid hormone level affect both PI and PII promoter activity in chickens but only PI promoter activity in adult rats. The ACC β gene is also transcribed by two promoters, PI β and PII β , that are located immediately upstream of the first and second exons, respectively. However, the promoter regions of the α and β genes exhibit little sequence similarity as the ACC β promoter region contains *cis*-acting elements that bind muscle-specific transcription factors (K.-S. Kim, 2001).

5.3. The fatty acid synthase promoter

FAS genomic clones containing the 5'-flanking DNA have been isolated and partially characterized from rat, goose, chicken and human. The FAS gene reportedly is transcribed from two promoters in humans [19], but from only one in other species. Promoter I, which contains recognizable TATA and CAAT boxes, controls transcription from an initiation site near the beginning of exon 1, whereas promoter II, which contains

neither a TATA nor CAAT box, controls transcription from two sites, one in intron 1, the other 49 nt upstream of the ATG start codon in exon 2. Transient transfection experiments with FAS-reporter gene chimeras indicate that promoter I is about 15 times stronger than promoter II and that the presence of promoter II attenuates the transcriptional activity of promoter I. Based on these results, a model has been proposed (S.J. Wakil, 1996) in which intronic promoter II exerts a roadblock for RNA polymerase II that has initiated transcription from promoter I. Promoter II of the human FAS gene, which has not yet been well characterized, may play a role in low-level constitutive expression of FAS, whereas promoter I may be preferred for rapid transcription, under demand for high lipogenic rates.

The rat and goose genes exhibit 90 and 61% identity, respectively, with the human FAS gene in the promoter region containing the TATA box, an inverted CCAAT box and a functional Sp1-binding site. In common with the human FAS gene, the rat gene contains a long first intron that exerts a negative effect on promoter activity. The negative regulatory element has been mapped to a region between +405 to +1083 in the rat gene. Sequence elements within this region, which are capable of binding a variety of nuclear proteins, also confer a negative effect on transcription via a heterologous promoter, in an orientation-dependent manner. However, the results of run-on assays with nuclei derived from tissues that express FAS at either high or low levels indicate that the different rates of transcription can be accounted for by differences in the extent of initiation, rather than by operation of a transcriptional pausing mechanism, as proposed for the human FAS gene (S. Smith, 1997). The sequence of the first intronic region is poorly conserved between rat and human genes and the physiological significance, if any, of the use of an intronic promoter in the human, but not in the rat, FAS gene remains to be demonstrated.

Table 1 summarizes various *cis*-acting elements that have been identified in the FAS and ACC promoters and the *trans*-acting factors that bind to them.

5.4. Transcriptional regulation of lipogenic gene expression in response to dietary carbohydrate

The adaptive response of gene expression to nutrient availability is vital to the survival of all organisms. In animals, the response to availability of dietary carbohydrate is mediated both by effects of glucose metabolism itself and by secondary effects on hormone secretion. Thus, glucose entering the blood stream provides the initial stimulus for the synthesis and secretion of insulin, which in turn promotes the entry of glucose into cells and its subsequent metabolism. Insulin release at the onset of feeding also stimulates the conversion of thyroxine into the more potent hormone 3,5,3'-triiodothyronine (T3) and may increase the level of T3 receptors in the liver. Thus the feeding stimulus activates both insulin and T3-mediated signaling pathways, both of which have the effect of increasing lipogenesis [20].

The initial, rapid hepatic response to carbohydrate intake involves activation of key glycolytic and lipogenic enzyme activities (Fig. 6). The long-term response is accompanied by an increased enzyme production, primarily as a result of increased gene transcription. Conversely, glucagon production is elevated during fasting, resulting in down-regulation of glycolytic and lipogenic enzymes. The involvement of insulin,

Table 1
Cis-elements and trans-factors involved in the regulation of FAS and ACC genes

Consensus cis-acting motif	Trans-acting factor	Effect on transcription	Location in rat FAS promoter	Location in ACC α promoter
TATA box (TTTAAAT)	TBP	Initiates transcription	-33/-26	Not present
E-Box (CANNTG)	USF and other bHLH-LZ family members	Mediates the insulin response in FAS and ACC α PI promoter	-60/-65, -326/-332	-931/-936 (Rat, PII); -109/-114 (Sheep, PI)
Inverted CAAT box (ATTGGCC)	NF-Y	Activates the basal transcription; required for conferring sterol response to FAS promoter; mediates cAMP response in FAS promoter	-98/-92, -516/-498	Not identified
GC-Box (G/TG/AAGGCG/TG/AG/AG/T)	Sp1 family members	Activates transcription; participates in conferring sterol response to ACC and FAS promoters; mediates glucose response in ACC α PII promoter	-91/-83, -168/-160, -226/-218, -242/-284, -482/-474 and -557/-549	-254/-242, -329/-317 (Rat, PII)
Sterol response element (TCACNCCAC)	SREBP family members	Mediates the sterol response; mediates the nutritional response	-73/-43, -150/-141	-273/-256, -977/-967 (Rat, PII)
Thyroid hormone response element (A/GGGT/AC/AAN ₁ /GGGT/AC/AA)	TR/RXR and/or LXR/RXR	Mediates the thyroid hormone response	-120/-125, -669/-650	-108/-82 (Chicken, PII)
Carbohydrate response element (CATGN ₇ CGTG)	ChoRF/USF	Mediates the carbohydrate response in FAS and ACC α PI promoter	-7214/-7190	-126/-102 (Rat, PI)

TBP, TATA box binding protein; USF and bHLH-LZ, upstream stimulatory factor and basic helix-loop-helix leucine zipper; NF-Y, nuclear factor Y; Sp1, stimulatory protein 1; SREBP, sterol-response element binding protein; TR/RXR, a heterodimer consisting of thyroid hormone receptor and retinoid receptor; LXR/RXR, a heterodimer consisting of liver receptor X and retinoid receptor; ChoRF, carbohydrate-response element binding factor. Consensus cis-motifs are given in 5' to 3' orientation and N represents any nucleotide. It should be noted that many variations of the consensus motif are functional in various genes.

glucagon and T3 in the regulation of lipogenic gene expression has been appreciated for some time, but only recently have carbohydrates themselves been implicated in this process. The identity of the intracellular signaling molecule produced by the metabolism of glucose has not yet been determined. One candidate is glucose-6-phosphate, since intracellular concentration of this metabolite varies in parallel with ACC and FAS mRNA concentrations in liver and adipose tissue, in response to dietary carbohydrate (P. Ferre, 1997). Another is xylulose-5-phosphate, an intermediate in the pentose phosphate cycle, since this metabolite can mimic the effect of hyperglycemia on glucose-regulated genes, without causing an elevation of glucose-6-phosphate [21].

5.4.1. The role of SREBPs

The effects of glucose, insulin and glucagon on transcription of lipogenic enzymes involves recruitment of a number of transcriptional activators, including SREBP, USF, NF-Y and Sp1. Pioneering work in the Goldstein and Brown laboratories established that SREBPs belong to the basic helix-loop-helix leucine zipper family of transcription factors and are involved in the regulation of the biosynthesis of both cholesterol and fatty acids [22] (Chapter 15). These DNA-binding proteins form dimers that can recognize both the direct repeat sterol regulatory element 5'-TCACNCCAC-3' and the inverted repeat E-box, 5'-CANNTG-3' [23].

The SREBPs are synthesized as ~1150 amino acid precursor proteins, bound to the endoplasmic reticulum and nuclear membrane, that undergo a sequential two-step proteolytic cleavage under low cellular sterol concentrations. This proteolytic cleavage results in the release of the aminoterminal section of SREBP which enters the nucleus and activates the transcription of genes involved in cholesterol and fatty acid synthesis, by binding to regulatory elements within their promoter regions. There are three major SREBP isoforms encoded by two different genes. The single SREBP-2 isoform produced by the SREBP-2 gene is believed to function primarily in the maintenance of cholesterol homeostasis [23]. The SREBP-1 gene can be transcribed from two different promoters, generating two isoforms, SREBP-1c and SREBP-1a, the latter having 29 additional amino acids at the aminotermus. The additional aminoterminal residues of SREBP-1a carry a binding site for NF-Y (also known as CBP) and imparts a strong promoter enhancing effect to this isoform; SREBP-1c, on the other hand, is a relatively weaker transcriptional activator, unless additional sequences are present in the promoter region that can recruit co-activators such as NF-Y and Sp1. The relative levels of SREBP-1a and 1c mRNAs vary significantly in different tissues. In tissues active in lipogenesis, such as liver and adipose tissue, the SREBP-1c mRNA is more abundant, whereas in spleen and most cultured cell lines, the SREBP-1a mRNA predominates.

The results of various *in vivo* and *in vitro* studies indicate that SREBPs play a major role in the regulation of lipogenic gene expression. For example, transcription of SREBP-1c, ACC and FAS genes is shut off during fasting of normal animals and turned on following refeeding a high-carbohydrate diet. However, transgenic animals carrying a disrupted SREBP-1 gene fail to modulate transcription of the lipogenic genes in response to these changes in nutritional status (H. Shimano, 1999). Over-expression of SREBP-1a or 1c in cultured preadipocytes or in transgenic animal liver activates transcription of lipogenic genes [24]. Transgenic animals over-expressing SREBP-1c in

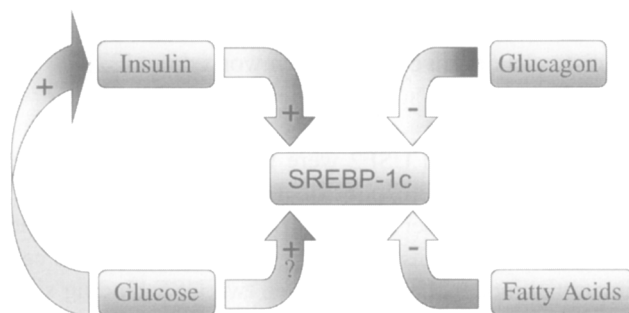


Fig. 8. Central role of SREBP-1c in the regulation of lipogenesis. Influx of glucose leads to increased secretion of insulin by pancreatic β -cells. Insulin and glucose induce lipogenic gene expression by increasing synthesis of SREBP-1c, whereas glucagon and fatty acids exert negative effect by decreasing SREBP-1c synthesis. Arrows carrying + and – symbol indicate positive and negative effect, respectively, on expression of SREBP-1c by corresponding external stimuli. The question mark indicates that direct activation of SREBP-1c expression by glucose has not been demonstrated definitively.

adipose cells develop insulin-resistant hyperglycemia, accumulate triglycerides in the liver and serum, and fail to down-regulate hepatic lipogenic gene expression in response to fasting. SREBP-1c also has been shown to be involved in the insulin-induced expression of the glucokinase gene [25,26]. The observation that insulin activates the expression of the SREBP-1c gene in primary hepatocytes whereas glucagon and cAMP have the opposite effect suggests that SREBP could mediate some of the effects of insulin on lipogenic gene expression in the liver, indirectly, by increasing glucose flux through glucokinase, and/or directly, by activation of lipogenic gene expression (Fig. 8).

5.4.2. The role of SREBP co-activators

As relatively weak transcriptional activators, SREBPs typically operate synergistically with other general transcriptional activators such as Sp1, NF-Y and cAMP-response element binding protein (T.F. Osborne, 2000). However, between different promoters, there are significant differences in the identity of these co-regulators and in the position of their binding sites relative to the position and number of SREBP-binding sites. These differences may be attributable, at least in part, to the differential recruitment of SREBP isoforms having different co-regulatory factor requirements. Sp1 may function synergistically with SREBP in mediating the glucose/insulin activation of both the FAS and ACC α genes. Increased binding of Sp1 to GC-rich regions between nucleotides –329/–317 and –254/–242 of the ACC α II promoter has been observed in 30A5 adipocytes exposed to glucose. Disruption of either Sp1-binding site by mutation eliminates glucose responsiveness of the ACC α II promoter. The increased binding of Sp1 has been attributed to a glucose-mediated induction of protein phosphatase I in the nucleus, which dephosphorylates Sp1, generating a form that binds more effectively to the promoter (K.-H. Kim, 1996).

NF-Y, a heterotrimeric transcription factor that binds to the inverted CCAAT-box motif of the FAS promoter, also appears to play a role as an activator of basal transcription of this gene. The cAMP-mediated down-regulation of FAS transcription

that occurs on fasting appears to involve the interaction of NF-Y with the inverted CCAAT-box, although the mechanistic details have not yet been worked out [27].

5.4.3. *The role of USFs*

The ubiquitous DNA-binding proteins USF1 and USF2 were initially proposed as the key mediators of the glucose/insulin-response in lipogenic genes. USF1 and USF2, members of a basic helix–loop–helix leucine zipper family of transcription factors, bind to the E-box sequence motif, CANNTG. Both transient transfection experiments in cell culture and the use of transgenic animal models have implicated two USF-binding sites in the glucose/insulin responsiveness of the FAS gene, one at –65, the other at –332 [28]. The USF-binding site at position –65 in the rat FAS promoter is overlapped by an SREBP-binding site. Although this region of the gene can bind USF and SREBP independently, studies with transgenic mice suggest that it is the binding of USF at this site that is more important in mediating the fasting/refeeding response. Thus, it appears that it is SREBP binding at the second site, –150 to –141, that is essential in modulating transcription of the FAS gene. The nucleotide sequence in this region is a motif that is highly conserved in rat, human and goose FAS promoters. An E-box located at –114 of the ACC α I promoter that binds USF1 and 2, but not SREBP-1, has also been implicated in the insulin responsiveness of the ovine gene [29]. As yet, it is unclear as to whether, or how, SREBPs and USFs interact in modulating transcription of the FAS and ACC genes in response to fasting and refeeding. Quite possibly, the USF proteins act synergistically with SREBP, as do other co-activators such as Sp1 and NF-Y.

5.4.4. *The role of carbohydrate-response elements*

Additional sequence elements have been identified within the ACC and FAS genes that play a role in modulation of transcription in response to glucose, independent of insulin. These sequences, termed glucose-response elements, or carbohydrate-response elements, consist of two E-box half-sites related to the sequence 5'-CACG, in either direct or inverted orientation and separated by 7 or 9 nucleotides, and are similar to those identified as carbohydrate-response elements in the pyruvate kinase and S14 genes (H.C. Towle, 2000). The carbohydrate-response elements, which are located at –126 to –102 in the rat ACC α I promoter, and at –7240 to –7190 in the rat FAS gene, bind both USF and an additional, as yet unidentified *trans*-acting factor [30,31]. This factor, termed a carbohydrate-response factor, presumably is linked to the metabolism of glucose by an as yet unidentified signaling pathway.

5.4.5. *Signaling pathways*

Although it is well established that genes involved in glucose catabolism and lipid biosynthesis are up-regulated by high glucose and insulin concentrations in both liver and adipose tissues, the signaling pathway from glucose/insulin to the transcriptional machinery is not clearly understood. The binding of insulin to its receptor activates the associated tyrosine kinase and results in autophosphorylation of the receptor. The signal is then propagated by two main routes: the insulin receptor substrates/phosphatidylinositol 3-kinase pathway and the Ras/mitogen-activated protein kinase pathway. It is the phosphatidylinositol 3-kinase signaling pathway that has been implicated in

the regulation of FAS transcription by insulin, although the identity of the participants remains to be established. The AMPK which is activated under various stress conditions due to a rise in the cellular AMP to ATP ratio, plays a significant role in the regulation of glucose-activated genes including FAS and ACC. Glucose-mediated transcriptional activation of FAS and ACC genes is inhibited by activated AMPK possibly due to inhibition of a yet to be identified activator or activation of an unidentified repressor. However, AMPK plays no role in the induction of glucose-activated gene expression since basal AMPK activity is not reduced by high levels of glucose in the cell [32]. A similar situation may be operative in the regulation of ACC activity by reversible phosphorylation. Here too, clear evidence implicates AMPK in the down-regulation (phosphorylation), but not in the up-regulation (dephosphorylation) of ACC activity.

5.4.6. The role of thyroid hormone

Thyroid hormone plays a significant role in regulation of expression of lipogenic enzymes in response to changes in nutritional status. T3 stimulates transcription of ACC and FAS in a tissue-specific manner, exerting its strongest effect in liver. In chick embryo hepatocytes, the rate of lipogenesis and the levels of lipogenic enzymes are low until the newly hatched chicks are fed and transcription of the lipogenic enzymes is activated. The induction of lipogenic enzymes can be simulated in vitro by treatment of chick embryo hepatocytes with T3. The mechanism of thyroid hormone-induced gene expression involves activation of the thyroid hormone receptor to form a heterodimer with the retinoid X receptor (RXR) or liver X receptor (LXR), which in turn binds to the consensus thyroid hormone-response element (Table 1) and activates the transcription. Treatment of chick embryo hepatocytes with T3 results in the stimulation of the promoter II-driven transcript of the ACC α gene, whereas in Hep G2 cells, T3 increases transcription from the human FAS promoter I, but has no effect on promoter II (S.J. Wakil, 1998; F.B. Hillgartner, 2001). The T3 response is mediated by the consensus thyroid hormone-response element half-site sequence, arranged as direct repeats on the noncoding strand of ACC α promoter II and the coding strand of FAS promoter I. The FAS promoter I also harbors an additional thyroid hormone-response element motif having only one half-site of consensus sequence. The amount of T3-receptor/RXR heterodimers that binds to these thyroid hormone-response elements is increased in the presence of T3. As yet, the details of the mechanism by which T3 regulates ACC α and FAS transcription remain unclear. F.B. Hillgartner and colleagues (2001) have proposed that T3 regulates ACC α transcription by a novel mechanism involving changes in the composition of nuclear receptor complexes bound to thyroid hormone-response element. Heterodimeric complexes containing LXR/RXR ensure basal level of ACC α expression whereas the complexes containing LXR/RXR and T3-receptor/RXR mediate T3-induced ACC α expression. Insulin also contributes to the T3-mediated activation of lipogenic gene expression by facilitating the conversion of thyroxine to T3 and increasing the level of nuclear T3 receptor.

5.4.7. Down-regulation during fasting

The down-regulation of transcription of the ACC and FAS genes that accompanies fasting is most likely initiated by a change in the relative amounts of glucagon and

insulin produced by the pancreas that results in decreased transcription of SREBP-1, phosphorylation of Sp1, and withdrawal of the carbohydrate-response element-mediated stimulatory effect and down-regulation of T3 production. The elevation of intracellular cAMP concentration that results from increased glucagon production also plays a role in the down-regulation process. However, the *cis*-acting elements and *trans*-acting factors involved appear to be different for ACC and FAS. NF-Y binding at an inverted CCAAT-box has been implicated in the case of the FAS promoter and AP-2 binding at variants of well-characterized cAMP-response element in the ACC α PII promoter. Phosphorylation of AP-2 by cAMPK facilitates the binding of AP-2 to the cAMP-response element [33]. Further investigation is required to determine the precise mechanism whereby cAMP exerts its effects on these *trans*-acting factors. The *cis*-elements and the *trans*-factors that confer cAMP responsiveness to the ACC α PI promoter still remain to be identified.

5.4.8. Summary

In summary, a tentative model for the transcriptional regulation of the ACC and FAS genes, in response to fasting and refeeding a carbohydrate-rich diet, can be envisaged in which the carbohydrate-response factor, together with SREBP, play central roles and Sp1, NF-Y and USF function as co-activators of SREBP. However, experimental validation of this model is incomplete in that it remains unclear as to exactly how functionality of the putative co-activators NF-Y and USF is modulated. For example, binding of NF-Y to the inverted CCAAT-box of the FAS gene is not altered either in cultured hepatoma cells treated with cAMP, or in the livers of fasted animals, although FAS transcription is down-regulated in both situations. Similarly, binding of USF to the E-box at -68/-52 of the rat FAS gene is unaltered in the livers of refed animals as well as in cultured hepatoma or adipose cells treated with insulin/glucose, although this interaction appears essential for mediating the fasting/refeeding response. Possibly, covalent modification of these co-activators of SREBP may alter their functionality within the transcriptional initiation complex, but experimental evidence is lacking at present.

5.5. Transcriptional regulation of lipogenic gene expression in response to dietary polyunsaturated fatty acids

Dietary fish oils containing n-3 fatty acids, such as eicosapentaenoic acid and docosahexaenoic acid, decrease blood triglyceride concentrations in hypertriglyceremic patients and are considered to have protective effects against cardiovascular diseases. Moreover, polyunsaturated fatty acids, when fed to rodents inhibit hepatic de novo lipogenesis, triglyceride secretion, and increase activities of enzymes involved in fatty acid oxidation. These effects on lipid metabolism are due to activation or suppression of transcription of key metabolic enzymes. Thus, polyunsaturated fatty acids inhibit the transcription of the FAS and ACC genes and activate the transcription of genes encoding acyl-CoA oxidase and cytochrome P450 4A2, enzymes that are involved in peroxisomal and mitochondrial fatty acid oxidation, respectively (S.D. Clarke, 1994). The transcriptional activation of genes encoding fatty acid oxidation enzymes is mediated by a nuclear peroxisome proliferator-activated receptor (Chapter 5). However, these receptors are not involved in

transcriptional regulation of the genes encoding lipogenic enzymes. The mechanism of down-regulation of lipogenic enzyme expression appears to be mediated, at least in part, through SREBP-1, since feeding of dietary fish oil to mice results in a decrease of the hepatic mRNA for SREBP-1, but not in that for SREBP-2 [34,35].

5.6. Transcriptional regulation during development

Fatty acids are important components of membrane phospholipids and are essential for phosphatidylcholine biosynthesis in proliferating cells. Nevertheless, transcription of ACC and FAS genes is not regulated by cell-cycling, but is regulated by certain growth factors such as colony stimulating factor 1 (S. Jackowski, 2000). The transcription of ACC and FAS genes is also regulated during the differentiation and development of several tissues. For example, in mammary glands, expression of lipogenic enzymes is increased during late pregnancy and early lactation, in preparation for the production of milk fat. In brain, FAS expression is elevated during the myelination phase of development, but is low in mature animals (P.R. Vagelos, 1972, 1973). In the lung, FAS activity increases during late gestation, in preparation for the synthesis of the palmitoyl component of lung surfactant that is required for air breathing (L.W. Gonzales, 1999). Our knowledge of the tissue-specific mechanisms regulating most of these processes is, as yet, scant. Perhaps the most extensively studied differentiation model is that of the preadipocyte, in which the maturation process is characterized by the coordinate induction of the lipogenic enzymes and the accumulation of intracellular triglycerides (Chapter 10). Paradoxically, cAMP, which is implicated in the down-regulation of lipogenic enzyme gene expression during fasting, is required as a positive effector in the transcriptional activation of lipogenic enzymes in both the developing lung and the differentiating preadipocyte. In culture, preadipocytes must first be sensitized by exposure to cAMP before they will respond to insulin and turn on the expression of lipogenic enzymes (K.-H. Kim, 1991). In the case of ACC, this effect appears to be mediated primarily through the PII promoter. However, it still remains to be established whether this synergistic interaction between insulin and cAMP occurs in vivo or is specific to differentiation of 30A5 cells in culture.

6. Future directions

A major obstacle in understanding how the enzymes of the multifunctional polypeptide system cooperate to effect the biosynthesis of fatty acids is the absence of a detailed three-dimensional structure for either ACC or FAS. Crystal structures of many of the individual type II FAS enzymes have been obtained in recent years and have provided valuable information about the various catalytic mechanisms that likely operate in their type I counterparts. However, the most interesting aspects of the type I FASs clearly concern the overall architecture of the complex. The intriguing question as to how an ACP domain of the animal FAS is able to make functional contacts with eight catalytic domains, six on the same subunit and two on the companion subunit, can only be answered by generating high-quality images of the entire 540 kDa dimer.

either by X-ray crystallography or high-resolution electron microscopy. Similarly, a detailed structural analysis of the animal ACC is required in order to understand how reversible phosphorylation and the conformational changes induced by citrate modulate the activity of this enzyme.

Substantial progress has been made during the last 5 years in identifying the various *trans*-acting factors that regulate transcription of the ACC and FAS genes in response to nutritional stimuli and we are likely to see many of the remaining details of these processes elaborated in the near future.

Arguably, the most exciting recent discovery in the field of fatty acid biosynthesis is that both ACC and fatty acid synthase may constitute legitimate targets for the development of therapeutic agents for the treatment of obesity and cancer. The potential usefulness of inhibitors of the two enzyme complexes rests largely on their ability to modulate intracellular malonyl-CoA concentrations. Cerulenin (2,3-epoxy, 4-oxo, 7,10-dienoyl, dodecylamide), the fatty acid synthase inhibitor that, when administered to mice, causes loss of appetite and reduction in adipose mass, is too unstable for therapeutic use, but a chemically stable inhibitor has been synthesized (C75; 3-carboxy-4-octyl-2-methylenebutyrolactone) that is equally effective in suppressing appetite and inducing weight loss [36]. Both cerulenin and C75 are also potent inhibitors of DNA replication in many types of tumors that characteristically overexpress fatty acid synthase. The toxicity of these compounds appears to be due to their ability to inhibit fatty acid synthesis and cause an elevation in intracellular malonyl-CoA concentration that ultimately results in apoptosis of the tumor cell. Tumor cells exposed to these inhibitors are unable to synthesize or oxidize fatty acids, since the elevated malonyl-CoA concentration inhibits CPT1 and blocks entry of fatty acids into the mitochondria. The effects of cerulenin can be mimicked by the combined administration of 5-(tetradecyloxy)-2-furoic acid and etomoxir, inhibitors of ACC and carnitine palmitoyltransferase I, respectively, supporting the theory that it is the inhibition of both fatty acid synthesis and fatty acid oxidation that triggers the apoptotic process (F.P. Kuhajda, 2001). The exact mechanism by which apoptosis is initiated by the disruption in fatty acid metabolism is not yet completely understood. One possibility is that fatty acids are diverted away from mitochondrial oxidation into ceramide synthesis, a situation that is known to induce apoptosis. Alternatively, the process could involve interaction of carnitine palmitoyltransferase I with Bcl-2, a mitochondrial membrane protein that regulates programmed cell death.

In another ground-breaking study, ACC β -isoform knockout mice were generated and found to have a higher-than-normal rate of fatty acid oxidation and lower amounts of adipose fat (S.J. Wakil, 2001). The absence of malonyl-CoA in muscle tissue appears to free the carnitine palmitoyltransferase from its normal down-regulation control mechanism and lead to efficient oxidation of fatty acids that have been mobilized from adipose tissue. Most surprisingly, the knockout mice accumulate 50% less adipose fat, despite consuming 20% more food than wild-type mice!

Although further study is needed to carefully evaluate the long-term consequences of administration of inhibitors of ACC and fatty acid synthase, undoubtedly these findings will stimulate searches for new inhibitors that could offer novel therapies for the treatment of both obesity, its associated disorders, and of cancer.

Abbreviations

ACC	acetyl-CoA carboxylase
FAS	fatty acid synthase
ACP	acyl carrier protein
cAMPK	cAMP-dependent protein kinase
AMPK	AMP-dependent protein kinase
T3	3,5,3'-triiodothyronine
SREBP	sterol-response element binding protein
USF	upstream stimulatory factor
NF-Y	nuclear factor Y
Sp1	stimulatory protein 1
RXR	retinoid X receptor
LXR	liver X receptor

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Fatty acid desaturation and chain elongation in eukaryotes

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1. Introduction

Although the contribution of lipid molecules to the hydrophobic character of membranes was recognized late in the nineteenth century, the nutritional importance of specific lipid molecules was first revealed through the pioneering work of Burr and Burr in 1929. They fed rats a fat-free diet and observed that retarded growth, scaly skin, tail necrosis and eventual death were reversed by feeding specific fats. Linoleic acid was recognized as the active agent and the term ‘essential fatty acid’ was coined.

During the 1950s, the advent of chromatographic techniques (gas–liquid and thin-layer chromatography) and greater availability of appropriate substrates and precursors labelled with radioisotopes contributed in a major way to current understanding of fatty acid metabolism. In the 1960s, studies focused on *in vitro* assays of specific enzymatic steps [1]. $\Delta 9$ desaturase was measured in yeast, rat liver microsomes and plants. It was determined that polyunsaturated fatty acid (PUFA) formation in animal tissues involved $\Delta 6$ and $\Delta 5$ desaturation. Chain elongation of long chain fatty acids was found in the endoplasmic reticulum (ER) and mitochondria [2,3]. Relationships between essential fatty acids and prostaglandins (Chapter 13) were elucidated. Subsequent work led to an understanding of primary, alternate and competitive pathways of desaturation and chain elongation and of factors that regulate fatty acid metabolism.

The 1990s and 21st century have seen the advent of molecular and genetic systems that support in-depth analysis of the regulation of fatty acid elongation and desaturation pathways [4–9]. More complete understanding of the roles of particular unsaturated fatty acids in the regulation of cell biology and human health and disease continues to evolve.

1.1. Physical consequences of fatty acyl chain desaturation and elongation

Among the forces holding lipid molecules within membranes, London–Van der Waals interactions have a major impact along the fatty acyl chains. These relatively weak forces are additive, proportional to the number of overlapping methylene groups, and inversely proportional to the distance between them. Thus, for long fatty acyl chains, total London–Van der Waals bonding strength is greater than electrostatic and hydrogen bonding of polar head groups. Accordingly, length of a fatty acyl chain is a crucial

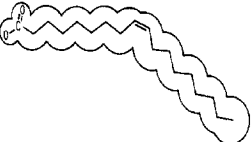
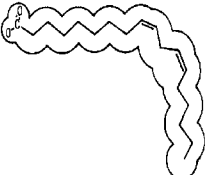
FATTY ACID	ABBREVIATION	MELTING POINT	SPATIAL WIDTH(nm)	POSSIBLE CONFIGURATION
Stearic Acid	18:0	70°	0.25	
Oleic Acid	c-18:1n-9	16°	0.72	
Elaidic Acid	t-18:1n-9	43°	0.31	
Linoleic Acid	c,c-18:2n-6	-5°	1.13	

Fig. 1. Physical characteristics of fatty acids. See Table 1 for nomenclature of the fatty acids. Single bonds have a length of 0.154 nm and an angle of approximately 111°; double bonds have a length of 0.133 nm and an angle of approximately 123°. *c*, *cis*; *t*, *trans*.

parameter in its contribution to membrane structure and stability. Restricted solubility in an aqueous environment is governed by the tendency of acyl chains to remain associated with one another (oil does not dissolve well in water). Similarly, response to temperature is modulated by the extent to which thermal influences dissociate acyl chains. Thus, as acyl chain length increases, solubility decreases and melting point increases. Where more loosely packed membrane structures are advantageous, rigidity of lengthy saturated acyl chains can be countered by double bonds. A double bond of *cis* geometric configuration results in a 30° angle change from the linearity of the saturated chain (Fig. 1). Double bonds also are non-rotating, restrict chain movement and increases polarity through charge concentration.

Double bonds cause bends in fatty acyl chains within biological membranes (potentially, but probably not in a membrane milieu, an acyl chain with six double bonds could assume nearly circular shape) and decrease rigidity. Accordingly, within membranes, acyl chain length and the number and position of double bonds markedly influence fluidity, permeability and stability of biological membranes.

1.2. Fatty acyl chains and biology

Many (but not all) required fatty acids also can be biosynthesized by mammalian cells. Altering fatty acyl composition of membrane lipids influences physical properties and thus changes the function of integral proteins. Release of specific fatty acids from

membrane stores regulates cellular signal transduction and gene transcription with specificity determined by differences in the unsaturation and chain length [10–13].

Fatty acids with several double bonds are known as PUFA. Some PUFA can only be synthesized from dietary fats as animal cells do not have the ability to make double bonds at some positions along the fatty acyl chain. Required dietary fatty acids are known as ‘essential fatty acids’. PUFA serve as precursors of biologically active prostaglandins and leukotrienes (Chapter 13) that are required to mount immune and inflammatory responses to infection and pain. In the nucleus, PUFA also control gene transcription through separate classes of receptors called peroxisomal proliferator activated receptors (PPARs) [14] and sterol response elements binding proteins (SREBPs) [15] (also see Chapters 10 and 15). PUFA released from membrane phospholipids by agonist-induced stimulation of phospholipases (Chapter 11) can be involved in signal transduction through modulation of cellular proteins including isoforms of protein kinase C or cyclases, and in translocation of specific enzymes to membranes [12]. Many of these diverse functions are modulated by a specific type of fatty acid. Thus, a large variety of fatty acyl chains are required in the lipids of biological membranes and storage depots. Types obtained through dietary intake and *de novo* synthesis are insufficient to meet the varied demands of cells, so there is substantial metabolism and rearrangement as development, growth, and aging proceed. Knowledge of how the array of fatty acyl chains is derived and modified, and what regulates metabolism of fatty acyl chains by desaturation and chain elongation are described in sections that follow.

2. Chain elongation of long chain fatty acids

De novo synthesis of fatty acids (Chapter 6) produces mainly 16-carbon palmitic acid, with minor amounts of 18-carbon stearic acid (Table 1). Quantitatively, these chain lengths are major components of many membrane lipids; qualitatively, they are related to optimal width of the membrane lipid bilayer. Many major chains are longer than 16 or 18 carbons, constituting more than half of the total acyl chains of many tissues. For example, in myelin surrounding axonal processes of neuronal cells, fatty acyl chains of 18 carbons or greater are more than 60% of the total, and in sphingolipids, 24-carbon acyl chains are prominent. Chain lengths of 28–36 carbons have been reported in phospholipids of retinal photoreceptors.

Many eukaryotic cells have capacity for 2-carbon chain elongation of both endogenously synthesized and dietary fatty acids (Fig. 2). In liver, brain and other tissues, two primary systems — one in the endoplasmic reticulum (ER) and the other in mitochondria — provide chain elongation. The ER system predominates quantitatively. Mitochondrial elongation is distinct from reversal of fatty acid catabolism through β -oxidation (Chapter 5). Peroxisomes also contain an acetyl-CoA dependent elongation system that is enhanced markedly after treatment with specific fatty acids and drug mimics known collectively as peroxisomal proliferators (Chapter 10).

Table 1
Nomenclature and bond positions of major long chain fatty acids

Common name	Systematic name ^a	Abbreviations	Δ Bond positions
Palmitic acid	hexadecanoic acid	16:0	
Palmitoleic acid	9-hexadecenoic acid	16:1 n-7	9
	6-hexadecenoic acid	16:1 n-10	6
Stearic acid	octadecanoic acid	18:0	
Oleic acid	9-octadecenoic acid	18:1 n-9	9
Vaccenic acid	11-octadecenoic acid	18:1 n-7	11
Petroselenic acid	6-octadecenoic acid	18:1 n-12	6
Elaidic acid	<i>t</i> -9-octadecenoic acid	<i>t</i> -18:1 n-9	9
Linoleic acid	9,12-octadecadienoic acid	18:2 n-6	9,12
Linoelaidic acid	<i>t,t</i> -9,12-octadecadienoic acid	<i>t,t</i> -18:2 n-6	9,12
α-Linolenic acid	9,12,15-octadecatrienoic acid	18:3 n-3	9,12,15
γ-Linolenic acid	6,9,12-octadecatrienoic acid	18:3 n-6	6,9,12
Stearidonic acid	6,9,12,15-octadecatetraenoic acid	18:4 n-3	6,9,12,15
Arachidic acid	eicosanoic acid	20:0	
Gadoleic acid	9-eicosenoic acid	20:1 n-11	9
Gondoic acid	11-eicosenoic acid	20:1 n-9	11
Dihomo-γ-linolenic acid	8,11,14-eicosatrienoic acid	20:3 n-6	8,11,14
Mead acid	5,8,11-eicosatrienoic acid	20:3 n-9	5,8,11
Arachidonic acid	5,8,11,14-eicosatetraenoic acid	20:4 n-6	5,8,11,14
Timnodonic acid	5,8,11,14,17-eicosapentaenoic acid	20:5 n-3	5,8,11,14,17
Behenic acid	docosanoic acid	22:0	
Cetoleic acid	11-docosenoic acid	22:1 n-11	11
Erucic acid	13-docosenoic acid	22:1 n-9	13
Adrenic acid	7,10,13,16-docosatetraenoic acid	22:4 n-6	7,10,13,16
Docosapentaenoic acid	4,7,10,13,16-docosapentaenoic acid	22:5 n-6	4,7,10,13,16
Docosapentaenoic acid	7,10,13,16,19-docosapentaenoic acid	22:5 n-3	7,10,13,16,19
Clupanodonic acid	4,7,10,13,16,19-docosahexaenoic acid	22:6 n-3	4,7,10,13,16,19
Lignoceric acid	tetracosanoic acid	24:0	
Nervonic acid	15-tetracosenoic acid	24:1 n-9	15
Cerotic acid	hexacosanoic acid	26:0	
Ximenic acid	17-hexacosenoic acid	26:1 n-9	17

^a All double bonds are of *cis* geometric configuration except where *t* indicates a *trans* double bond.

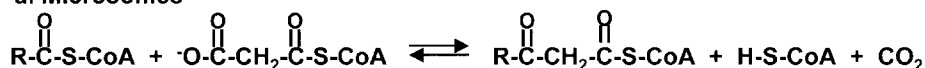
2.1. Endoplasmic reticulum elongation system

Fatty acyl chain elongation associated with the ER is highly active. Fatty acids must be activated to CoA derivatives; the CoA moiety provides energy for attachment of the donor group to a growing fatty acyl chain. Several fatty acyl-CoA synthetases with specificities for chain length and degree of unsaturation have been purified and their cDNAs cloned. Describing their precise subcellular locations is a needed step in determining links to the synthesis of particular lipids.

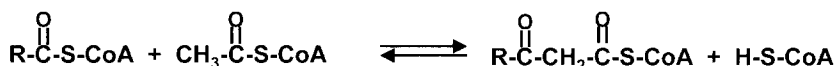
Distinctive to the ER elongation system, the 2-carbon elongation unit is donated from 3-carbon malonyl-CoA; interestingly, malonyl-CoA itself is not synthesized in the ER. Microsomal chain elongation is active with both saturated and unsaturated fatty acids, with γ-linolenate, 18:3 n-6, being the most effective substrate. In a manner analogous

1. Condensation

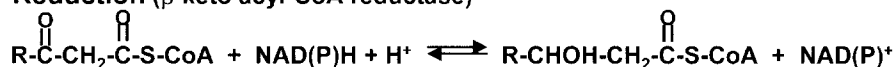
a. Microsomes



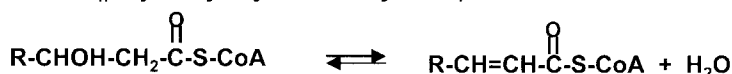
b. Mitochondria



2. Reduction (β -keto acyl-CoA reductase)



3. Dehydration (β -hydroxy acyl-CoA dehydrase)



4. Reduction (2-*trans* enoyl-CoA reductase)

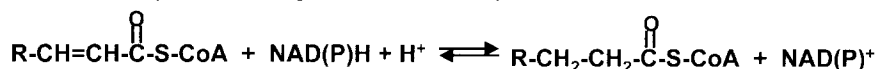


Fig. 2. Reactions in 2-carbon chain elongation of long chain fatty acids.

to de novo synthesis of 16-carbon acyl chains, a four-component reaction occurs in the 2-carbon elongation process of long chain fatty acyl-CoA (Fig. 2). Condensation of fatty acyl-CoA and malonyl-CoA to β -ketoacyl-CoA is the initial step. It is rate-limiting, determines fatty acyl specificity, and results in addition of the 2-carbon moiety. The second reaction is catalyzed by a reductase that utilizes NADPH (in preference to NADH) to form β -hydroxy acyl-CoA. It is not known if the ability to use both electron donors is due to two separate reductases. Flow of reducing equivalents from NADPH or NADH to β -keto reductase appears to involve cytochrome *b5* and cytochrome *P*-450 reductase. The third reaction in chain elongation involves dehydration to enoyl-CoA, and the final reaction is a second reductase step catalyzed by 2-*trans*-enoyl-CoA reductase that requires NADPH. With possible exception of the dehydrase, active sites of all components of the ER elongation system have a cytosolic orientation. Partial purification of components of the elongation system suggest that the enzymes are discrete entities (in contrast to a single polypeptide for de novo fatty acid synthase of animal tissues); acyl-CoA derivatives can be isolated at each intermediate step. Despite such evidence, some investigations suggest covalent linkage of acyl-CoA to a multi-functional complex of the enzymes after condensation.

Regulation of the ER chain elongation system is not well understood. The rate-limiting condensation enzyme, but not the reductases or dehydrase, can be influenced by diet. Fasting depresses elongation and refeeding a carbohydrate diet increases overall chain elongation, whereas refeeding a high protein diet does not. Dietary effects seem

similar for saturated and PUFA substrates. In rat liver, elongation activity peaks at two weeks after birth and declines before another increase after weaning; old adult rats have lower elongation activities than young adult rats. Elongation of 16:0 by rat brain microsomes declines with age but elongation of PUFA increases and remains high for several months. Little is known about developmental changes of elongation activities in human tissues. Inhibition of elongation by induction of a diabetic state and reversal by insulin is greater for 16:0 than for 18:3 n-3 suggesting divergent regulation of elongation of acyl chain classes. Xenobiotics, including plasticizers or hypolipidemic agents such as fibrates, markedly alter chain elongation, primarily at the condensation step. Studies with brain indicate different elongation enzymes reacting with dissimilar acyl chain lengths. In Quaking mice (a genetic mutant with defective myelination), chain elongation of 20:0 to 22:0 and 24:0 was reduced by 70%, whereas elongation of 16:0 and 18:0 was unaltered relative to control mice. Thus, more than one elongation system with specificity based on length and unsaturation of acyl chains appears to be operative in the ER.

2.2. Mitochondrial elongation system

Although less active than the microsomal system, mitochondrial chain elongation has been extensively investigated, particularly in liver and brain. The 2-carbon condensing donor in mitochondria is acetyl-CoA (Fig. 2; reaction 1b). Generally, monounsaturated fatty acyl-CoA is more active than saturated-CoA and both support higher activity than PUFA, particularly in brain. Maximal mitochondrial elongation in liver, brain, kidney, and adipose tissue seems to require both NADPH and NADH, whereas heart, aorta and muscle require only NADH.

During the 1970s, mechanisms of mitochondrial chain elongation were elucidated [3]. Although β -oxidation (Chapter 5) and mitochondrial chain elongation have the same organelle location, reversal of β -oxidation is not feasible; the FAD-dependent acyl-CoA dehydrogenase of β -oxidation is substituted by a more thermodynamically favorable enzyme, enoyl-CoA reductase (Fig. 2; reaction 4), to produce overall negative free energy for the sequence. Enoyl-CoA reductase from liver mitochondria is distinct from the ER reductase, based on pH optima and specificities for saturated and unsaturated acyl-CoAs. Kinetic studies suggest that enoyl-CoA reductase is rate-limiting in mitochondrial elongation.

2.3. Functions of elongation systems

Chain elongation in the ER appears to be the most important source of acyl chains greater than 16 carbons for membrane phospholipids during growth and maturation, when long chain acids may not be supplied adequately in the diet. 18- to 24-carbon saturates and monoenes and 20- and 22-carbon polyunsaturates are required for neural growth and myelination, regardless of dietary fluctuations.

The function of the mitochondrial elongation system is less clear but it may participate in biogenesis of mitochondrial membranes or in transfer of reducing equivalent between carbohydrates and lipids. However, in view of the relatively low activity toward